Radiative Transfer

Course A of two · A graduated course for pre-master students: theory, worked examples, illustrations, and exam questions & answers

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Part A — Short answers (2 marks each)
Part B — Problems
Part C — Formation of the RTE, polarization, history (4 marks each)
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Formula Sheet — Course A: Radiative Transfer

Radiation field (Lecture RT2)
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Moments, diffusion, scattering (Lecture RT8)
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Historical approximations (Lecture RT10)
Polarization (Lecture RT9)

Lectures RT1–RT10, final exam (RT11) and formula sheet (RT12). Course B, *Stellar Atmospheres*, applies this theory to stars.

Based on: A670 *Advanced Astrophysics* Lecture 1 · W. Hu, *Astro 305 Notes, Set 1* · MIT 8.901 (2019) Lecture 7 · G. W. Collins II, *Fundamentals of Stellar Astrophysics*, Ch. 9 · S. Chandrasekhar, *Radiative Transfer* (1950)

Code: all figures and numbers are reproducible with the Python scripts in `scripts/`. The interactive web app is in `app/` and the spreadsheet is `Radiative_Transfer_Lab.xlsx`.

Lecture RT1 — Introduction: How Is the Radiative Transfer Equation Used in Astrophysics?

> **Level:** pre-master → Astrophysics master entry. **Sources:** A670 *Advanced Astrophysics* Lecture RT1 (course outline); W. Hu, *Astro 305 Lecture Notes, Set 1*; MIT 8.901 (2019) Lecture RT7; G. W. Collins II, *The Fundamentals of Stellar Astrophysics*, Ch. 9; S. Chandrasekhar, *Radiative Transfer* (1950), Ch. I.

Learning goals

By the end of this lecture you should be able to:

1. State in one sentence what radiative transfer (RT) is and why almost all of astrophysics depends on it.
2. Write the one-dimensional radiative transfer equation (RTE) and name every symbol in it.
3. Distinguish **thermodynamic equilibrium (TE)**, **local thermodynamic equilibrium (LTE)** and **non-LTE (NLTE)** in words.
4. List the main astrophysical applications of the RTE.

Key concepts in plain words

Radiative transfer is photon bookkeeping. Imagine following one narrow beam of light as it travels from a star to your telescope. At every step along the way, some photons are removed from the beam (absorbed, or knocked sideways by scattering), and some new photons are added to it (emitted by the gas, or scattered into the beam from other directions). Radiative transfer is simply careful accounting of these gains and losses. The radiative transfer equation (RTE) is that account written as a differential equation: *change of the beam = gains − losses*.

Why astronomers care. We cannot visit stars or gas clouds; we only receive their light. The light we receive is not the light that was originally produced: it has been processed by every layer it crossed. RT is the "translation dictionary" between the physical conditions (temperature, density, composition, motion) and the spectrum we record. Without RT we could not measure the temperature of the Sun's surface, the chemical composition of a distant star, or the density of an interstellar cloud.

Specific intensity — the brightness of one beam. The basic quantity is the *specific intensity* I_ν: the energy carried by light of one frequency, travelling in one direction, through a unit area, per unit time. It is what one pixel of a camera measures when it looks at an extended object. All other radiation quantities (flux, energy density, pressure) are built from it by averaging over directions.

Optical depth — distance measured in "fog units". Optical depth τ_ν counts how many photon mean free paths lie along a path. On a foggy day, visibility might be 50 m: an object 50 m away is at $\tau \approx 1$, and one 500 m away is at $\tau \approx 10$, completely hidden. In the same way, $\tau \ll 1$ means "transparent" (optically thin) and $\tau \gg 1$ means "opaque" (optically thick). Optical depth, not geometrical distance, decides what we can see.

Source function — what the gas would like the beam to be. The ratio $S_\nu = j_\nu / \alpha_\nu$ (emission divided by absorption) is the brightness the gas tends to impose on any beam passing through it. If the beam is brighter than S_ν it is dimmed; if it is fainter it is brightened. A thick layer always ends up "painting" the beam with its own S_ν.

TE, LTE and NLTE — three levels of equilibrium. In *thermodynamic equilibrium* (TE) everything, matter and light, shares one temperature, as inside a closed hot oven. In *local thermodynamic equilibrium* (LTE) the **matter** at each place behaves as if it were in an oven at the local temperature, because collisions between particles are frequent; the **light**, however, can stream freely and need not be in equilibrium. In *non-LTE* (NLTE) collisions are too rare, and the radiation field itself controls how atoms are excited and ionised. Most of this course works in LTE, because it is the first and most important model for stellar atmospheres.

1.1 What is radiative transfer?

Almost everything we know about the Universe beyond the Solar System comes to us as **photons**. Before they reach our telescopes these photons are **emitted**, **absorbed** and **scattered** by atoms, ions, molecules, free electrons and dust grains. *Radiative transfer* is the theory of how a beam of radiation changes as it passes through matter. It is the link between:

- the **physical state** of the gas (temperature T, density ρ, chemical composition, velocity field), and
- the **observable** quantity: the spectrum I_ν that we record.

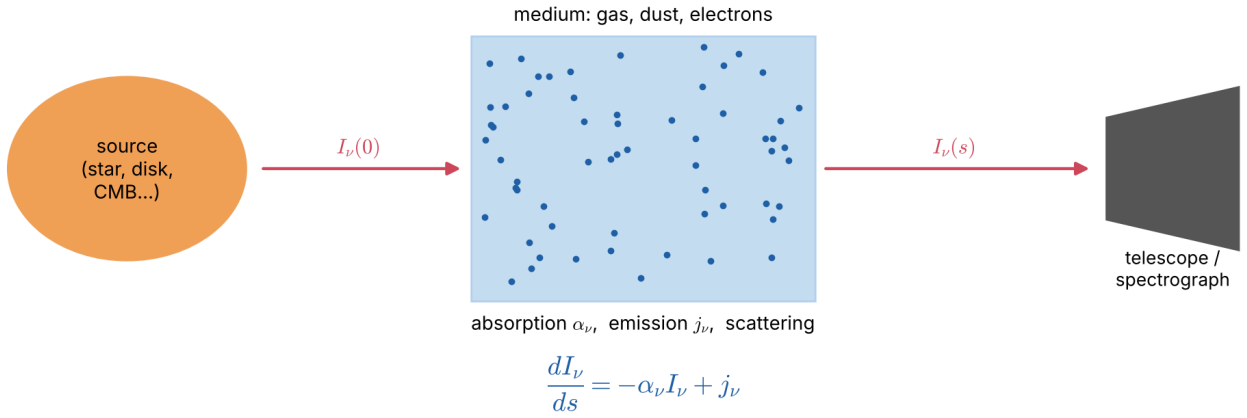


Figure 1.1 — The radiative transfer problem. A beam with intensity $I_\nu(0)$ enters a medium. Absorption (α_ν) removes energy from the beam, emission (j_ν) adds energy, and scattering does both by redirecting photons. We observe $I_\nu(s)$.

Two complementary problems:

Problem	Given	Find	Example
Forward (modelling)	$T(z)$, $\rho(z)$, composition	emergent spectrum I_ν	model stellar atmosphere, synthetic spectrum
Inverse (diagnostics)	observed I_ν	T, ρ, abundances	abundance analysis, limb-darkening inversion

1.2 The fundamental equation (preview)

Along a ray, with path length s:

$$\boxed{\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu}$$

Symbol	Name	SI units
I_ν	specific intensity	W m^{-2} Hz^{-1} sr^{-1}
$\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu$	absorption (extinction) coefficient	m^{-1}
κ_ν	opacity (mass absorption coefficient)	m^2 kg^{-1}
j_ν	emission coefficient (emissivity)	W m^{-3} Hz^{-1} sr^{-1}

Physical reading. The first term is a *loss* proportional to how much light there already is, so absorption removes a fixed *fraction* per unit length. The second term is a *gain* that does not depend on I_ν; matter emits whether or not light is present.

We define the dimensionless **optical depth** $d\tau_\nu = \alpha_\nu\, ds$ and the **source function** $S_\nu = j_\nu/\alpha_\nu$:

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + S_\nu.$$

This compact form is the one we will use throughout the course. Lectures RT2–RT5 derive it carefully.

1.3 A short history

Year	Who	Contribution
1859	Kirchhoff	j_ν/α_ν is a universal function of T for matter in equilibrium
1900	Planck	the universal function: $B_\nu(T)$
1906	K. Schwarzschild	RTE for the solar atmosphere; radiative equilibrium explains solar limb darkening
1916–1926	Eddington	radiative equilibrium of stars; the **Eddington approximation** $K = J/3$
1917	Einstein	A and B coefficients; stimulated emission
1930–34	Hopf, Milne	exact solution of the gray atmosphere (Hopf function $q(\tau)$)
1950	Chandrasekhar	*Radiative Transfer*: discrete ordinates, H-functions, polarisation
1960s–1978	Mihalas, Auer & Mihalas	NLTE model atmospheres, numerical methods
today	many groups	3D, time-dependent, Monte-Carlo, NLTE, polarised RT on supercomputers

1.4 LTE versus non-LTE: the central idea of this course

> **Correction to a common statement.** The A670 Lecture RT1 notes state that "in LTE the radiative transfer equation simplifies to $I_\nu = B_\nu(T)$". That is the condition for **full thermodynamic equilibrium (TE)**, which holds deep inside stars. In **LTE** the correct statement is
>
> $$S_\nu = B_\nu(T_{\rm local}) \qquad \text{but in general} \qquad I_\nu \neq B_\nu.$$
>
> The *matter* is in equilibrium at the local temperature, so it emits like a black body per unit absorption. The *radiation field* is **not** Planckian, because photons near the surface come from regions of different temperature and can escape. If I_ν were equal to B_ν everywhere, there would be no net flux and stars would not shine. Lectures SA1 and SA5 return to this point in detail.

	TE (stellar interior)	LTE (most photospheres)	NLTE (upper atmospheres, nebulae, ISM)
Particle velocities	Maxwell at T	Maxwell at local T	Maxwell (usually)
Excitation / ionisation	Boltzmann + Saha at T	Boltzmann + Saha at local T	from **statistical equilibrium** (rate equations)
Source function	$S_\nu = B_\nu$	$S_\nu = B_\nu(T)$	S_ν depends on J_ν (non-local)
Radiation field	$I_\nu = B_\nu$ (isotropic)	$I_\nu \neq B_\nu$ (anisotropic, carries flux)	$I_\nu \neq B_\nu$
What sets populations?	—	collisions dominate	radiative rates compete with collisions

1.5 Where is the RTE used?

1. **Stellar atmospheres:** emergent spectra, $T_{\rm eff}$, $\log g$, abundances, limb darkening. Classic LTE territory (Lectures RT7–RT8 and SA5–SA6).
2. **The interstellar medium (ISM):** extinction and reddening by dust ($A_V = 1.086\,\tau_V$), 21-cm emission and absorption, molecular lines (CO), heating and cooling of clouds.
3. **H II regions and planetary nebulae:** recombination lines and free-free radio emission. These are optically thin to thick slabs (Lecture RT5) and strongly NLTE.
4. **Planetary atmospheres:** greenhouse effect, remote sensing, exoplanet transmission spectra.
5. **Accretion disks, AGN and quasars:** disk spectra, Compton scattering, broad-line regions.
6. **Cosmology:** the CMB is the best black body known ($T = 2.725$ K); Thomson scattering at recombination and reionisation; Lyman-α forest absorption.
7. **Stellar interiors:** the diffusion limit of the RTE gives the radiative energy flux and the **Rosseland mean opacity** (Lecture RT8).

1.6 How the course is built

This lecture opens **Course A: Radiative Transfer** (lectures RT1–RT10). It is followed by **Course B: Stellar Atmospheres** (lectures SA0–SA8), which applies the theory to stars and adds the physical processes and transition rates

in the style of Mihalas' *Stellar Atmospheres*. Cross-references use these labels: "Lecture RT7" is in this course, "Lecture SA5" in the stellar-atmospheres course.

#	Lecture	Key result
RT1	Introduction (this lecture)	RTE, TE/LTE/NLTE in one paragraph
RT2	Describing the radiation field	I_ν, J_ν, F_ν, K_ν; I_ν constant in vacuum
RT3	Black-body radiation	Planck, Wien, Stefan–Boltzmann, brightness temperature
RT4	Emission, absorption, scattering	α_ν, j_ν, cross-sections, Einstein coefficients, random walk
RT5	The RTE and its formal solution	$I = I_0 e^{-\tau} + S(1 - e^{-\tau})$; MIT toy models: thin H cloud, two-layer star, thermal inversion
RT6	How the RTE is formed	Boltzmann equation → general RTE → special forms; energy and momentum moments
RT7	Plane-parallel atmospheres	Schwarzschild–Milne, E_n, Eddington–Barbier
RT8	Moments, diffusion and scattering	Eddington approximation, Rosseland mean, $S(0) = \sqrt{\varepsilon} B$
RT9	Polarized light	Stokes parameters, polarized RTE, 11.7 % limb polarization, Zeeman and Hanle effects
RT10	Historical context of approximations and solutions	Schuster, Schwarzschild, Eddington, Milne, Hopf, Chandrasekhar → ALI, 3D
RT11	Final exam with full solutions	
RT12	Formula sheet	

Course B (Stellar Atmospheres), for orientation: SA0 overview · SA1 LTE thermodynamics (Saha–Boltzmann, Kirchhoff) · SA2 continuous opacity · SA3 transition rates and statistical equilibrium · SA4 line absorption and broadening · SA5 the gray LTE atmosphere · SA6 LTE spectral lines · SA7 TE, LTE and non-LTE · SA8 model atmospheres and the solar atmosphere.

Worked example 1.1 — Reading the RTE

A beam passes through a uniform slab of gas of thickness $L = 10^{12}$ m. The gas has $\alpha_\nu = 2 \times 10^{-12}$ m^{-1} and does not emit. What fraction of the beam emerges?

Solution. With $j_\nu = 0$, $dI_\nu/ds = -\alpha_\nu I_\nu$, so $I_\nu(L) = I_\nu(0)\, e^{-\alpha_\nu L}$. The optical depth is $\tau_\nu = \alpha_\nu L = 2$, and the transmitted fraction is $e^{-2} = 0.135$ (13.5 %). In magnitudes the extinction is $A = 2.5 \log_{10} e^{\tau} = 1.086\,\tau = 2.17$ mag.

Worked example 1.2 — Why does the Sun have a sharp edge?

The solar photosphere has a density scale height of about $H \approx 150$ km, and the opacity rises steeply with depth. Why does the solar limb look sharp even though the Sun is gaseous?

Solution. We see down to optical depth $\tau \approx 1$. Because $\alpha \propto \rho$ (and the opacity also rises with T), the optical depth changes from $\tau \approx 0.1$ to $\tau \approx 10$ within a few scale heights, i.e. a few hundred km. Seen from Earth, $1'' \approx 725$ km on the Sun, so the transition from transparent to opaque happens within less than $1''$, and the edge looks sharp. RT defines what "the surface" of a star means: the layer where $\tau \approx 2/3$ (Lecture SA5).

Common misconceptions

1. **"In LTE the intensity equals the Planck function."** No: in LTE the *source function* equals $B_\nu(T)$. The intensity is set by the RTE and carries information from many different depths.
2. **"Optical depth is a distance."** It is a *number of mean free paths*. One kilometre of air and one millimetre of a stellar interior can have very different optical depths.
3. **"Absorption lines always mean cold gas between us and the star."** In a stellar atmosphere the lines form because the temperature *decreases outward*; no separate cold cloud is needed (Lectures RT5 and SA6).
4. **"A star has a solid surface."** The "surface" is simply the layer from which photons can escape, at optical depth $\tau \approx 2/3$.

Lecture summary

- Radiative transfer follows a beam of light and adds up what matter removes and adds.
- The RTE, $dI_\nu/ds = -\alpha_\nu I_\nu + j_\nu$, is the mathematical form of this bookkeeping.
- Optical depth measures transparency; the source function measures what the gas emits per unit absorption.
- LTE is a statement about **matter** (local temperature controls excitation and ionisation), not about light.
- RT is used everywhere in astrophysics: stars, the ISM, nebulae, planets, accretion disks and cosmology.

Exam questions and answers — Lecture RT1

Q1. Write the RTE along a ray. Explain the physical meaning of each term and why the absorption term is proportional to I_ν but the emission term is not.

> **A1.** $dI_\nu/ds = -\alpha_\nu I_\nu + j_\nu$. Absorption removes a fixed *fraction* $\alpha_\nu\, ds$ of the photons already present in the beam, so the loss is $\propto I_\nu$. Spontaneous emission is produced by the matter (excited atoms decay at rate A_{21} whether or not there is light), so it does not depend on I_ν. (Stimulated emission *is* $\propto I_\nu$ and is therefore counted as negative absorption inside α_ν; see Lecture RT4.)

Q2. What is the difference between TE and LTE? "In LTE, $I_\nu = B_\nu$." True or false?

> **A2.** In TE, both matter and radiation are in equilibrium at one temperature: $I_\nu = J_\nu = B_\nu(T)$ is isotropic and there is no net flux. In LTE the *matter* is in equilibrium at the local temperature $T(\mathbf{r})$ (Maxwell, Boltzmann and

Saha distributions hold), which gives $j_\nu = \alpha_\nu B_\nu(T)$, i.e. $S_\nu = B_\nu(T)$. The *radiation field* may differ from B_ν. The statement is **false**: in LTE $S_\nu = B_\nu$, not $I_\nu = B_\nu$. At the surface of a star, for example, $I_\nu(\mu < 0) = 0$ (no incoming radiation) while $B_\nu \neq 0$.

Q3. Define optical depth. A medium has $\tau_\nu = 0.01$ at one frequency and $\tau_\nu = 100$ at another. Describe what an observer sees at each frequency.

A3. $\tau_\nu = \int \alpha_\nu \, ds$ is the path length measured in photon mean free paths. At $\tau_\nu = 0.01$ (optically thin), 99 % of the background light passes through; the medium adds $\approx \tau_\nu S_\nu$, so we see through it, and its own emission is weak and proportional to its column density. At $\tau_\nu = 100$ (optically thick), the background is blocked (e^{-100}), and we see only the medium's own surface layers, $I_\nu \to S_\nu$ evaluated near $\tau_\nu \approx 1$.

Lecture RT2 — Describing the Radiation Field: Intensity, Flux and Moments

> **Sources:** Hu, *Astro 305 Set 1* (observables, auxiliary quantities); Collins Ch. 9.2–9.3 (specific intensity, moments); Chandrasekhar Ch. I §1–6; MIT 8.901 Lec. 7.

Learning goals

1. Define the specific intensity I_ν and explain why it is the fundamental quantity.
2. Derive the mean intensity J_ν, flux F_ν, energy density u_ν and radiation pressure P_ν as angular moments of I_ν.
3. Prove that I_ν is constant along a ray in empty space.
4. Use these definitions to compute fluxes from stars.

Key concepts in plain words

Intensity versus flux: brightness versus net flow. Think of standing in a room lit by many lamps. *Intensity* answers "how bright does the room look when I point a narrow tube in one particular direction?" *Flux* answers "how much energy crosses this window per second, counting light going one way as positive and the other way as negative?" A single bright star seen through a narrow telescope gives an intensity; a light meter lying flat on a table measures a flux.

Solid angle — the two-dimensional angle. Just as an ordinary angle measures how much of a circle an object covers, a *solid angle* measures how much of the whole sky an object covers. The whole sky is 4π steradians (sr). The Sun, seen from Earth, covers only 6.8×10^{-5} sr. Intensity is "energy per steradian", so it tells you how bright each little patch of the sky is.

Why intensity does not fall with distance. When you walk away from a lamp, the energy reaching you falls as $1/r^2$, but the lamp also looks smaller, and its solid angle also falls as $1/r^2$. The ratio, the brightness per steradian, stays the same. This is why a photograph of the Sun's surface from Mars would be exactly as bright, pixel by pixel, as one taken from Earth; the Sun would only look smaller. This conservation law (I_ν = constant along a ray in vacuum) is why intensity, not flux, is the natural quantity for transfer.

Moments: replacing directions by averages. Following the intensity in every direction is complicated, so we often use averages over directions, called *moments*:

- the **mean intensity** J_ν is the simple average: how strongly a point is bathed in light from all sides (it decides how fast atoms there absorb);
- the **flux** F_ν is the average weighted by $\cos\theta$: the *net* flow of energy in one direction (it decides how much light leaves a star);
- the **K-integral** K_ν is weighted by $\cos^2\theta$: it measures the momentum carried, i.e. the radiation *pressure*.

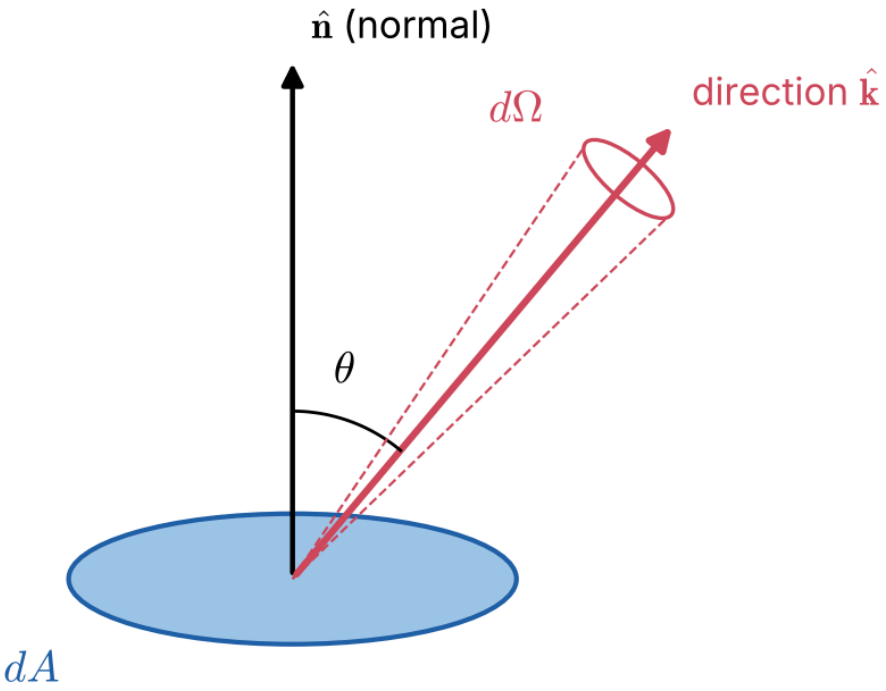
Isotropic light carries no flux. In a uniform fog, light arrives equally from all directions. Each point is strongly illuminated (J large) but as much energy flows left as right, so the net flux is zero. A star shines only because its radiation is slightly **anisotropic**: a little more light goes outward than inward.

Where the factor π comes from. If a surface glows with the same intensity I in every outward direction, the flux it emits is πI, not $2\pi I$, because light leaving at a slant crosses the surface less efficiently (factor $\cos\theta$). This is why a black body emits $F = \pi B = \sigma T^4$.

2.1 The four observables

A detector can measure four things about radiation (Hu): **energy flux**, **direction**, **frequency** and **polarisation**. Measuring the first three together defines the specific intensity. Polarisation adds the Stokes parameters Q, U, V; we use only I in this course.

2.2 Specific intensity



$$dE = I_\nu \cos\theta \, dA \, d\Omega \, d\nu \, dt$$

Figure 2.1 — Energy dE crosses an area dA whose normal $\hat{\mathbf{n}}$ makes angle θ with the direction of propagation $\hat{\mathbf{k}}$, into solid angle $d\Omega$, in frequency interval $d\nu$ and time dt.

$$\boxed{dE = I_\nu(\mathbf{r}, \hat{\mathbf{k}}, t)\ \cos\theta\ dA\ d\Omega\ d\nu\ dt}$$

Units: W m^{-2} Hz^{-1} sr^{-1} (cgs: erg s^{-1} cm^{-2} Hz^{-1} sr^{-1}). The factor $\cos\theta$ is the *projected* area seen by the beam.

Per unit wavelength: $I_\nu\,|d\nu| = I_\lambda\,|d\lambda|$. Since $\nu = c/\lambda$, $|d\nu/d\lambda| = c/\lambda^2$, so

$$I_\lambda = \frac{c}{\lambda^2} I_\nu, \qquad \nu I_\nu = \lambda I_\lambda.$$

Relation to the photon distribution function (Collins eq. 9.2.8). If $f(\mathbf{r}, \mathbf{p})$ is the photon phase-space density, then

$$I_\nu = \frac{h^4 \nu^3}{c^2}\, f.$$

Since f is conserved along a photon trajectory (Liouville's theorem), I_ν/ν^3 is invariant. This is why surface brightness dims as $(1+z)^{-4}$ in cosmology (Hu).

2.3 Constancy of I_ν along a ray in vacuum

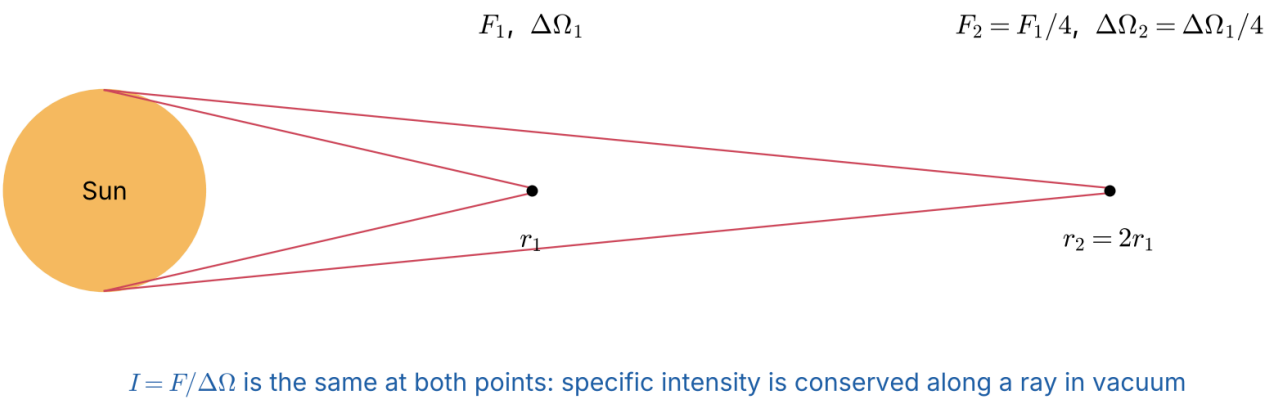


Figure 2.2 — The flux from the Sun falls as r^{-2}, but so does the solid angle it subtends. Their ratio, the intensity, is independent of distance.

Proof. Consider two small areas dA_1 and dA_2 separated by distance r along a ray, both perpendicular to it. Photons that pass through both areas cross dA_1 into solid angle $d\Omega_1 = dA_2/r^2$, and cross dA_2 from solid angle $d\Omega_2 = dA_1/r^2$. With no matter in between, energy is conserved:

$$I_{\nu,1}\, dA_1\, d\Omega_1\, d\nu\, dt = I_{\nu,2}\, dA_2\, d\Omega_2\, d\nu\, dt \;\Rightarrow\; I_{\nu,1}\frac{dA_1 dA_2}{r^2} = I_{\nu,2}\frac{dA_2 dA_1}{r^2} \;\Rightarrow\; I_{\nu,1} = I_{\nu,2}.$$

So $dI_\nu/ds = 0$ in vacuum. **Consequence:** the surface brightness of a resolved object does not depend on its distance (in a static, non-expanding space).

2.4 Angular moments of the intensity

Use spherical coordinates around a chosen axis: $d\Omega = \sin\theta\, d\theta\, d\phi = -d\mu\, d\phi$ with $\mu = \cos\theta$.

Moment	Definition	Plane-parallel form (I independent of ϕ)	Physical meaning
zeroth J_ν	$\dfrac{1}{4\pi}\oint I_\nu\, d\Omega$	$\frac{1}{2}\int_{-1}^{1} I_\nu\, d\mu$	mean intensity
first F_ν	$\oint I_\nu \cos\theta\, d\Omega$	$2\pi\int_{-1}^{1} I_\nu\, \mu\, d\mu$	net (physical) flux
first H_ν	$\dfrac{1}{4\pi}\oint I_\nu \cos\theta\, d\Omega$	$\frac{1}{2}\int_{-1}^{1} I_\nu\, \mu\, d\mu$	Eddington flux, $H_\nu = F_\nu/4\pi$
second K_ν	$\dfrac{1}{4\pi}\oint I_\nu \cos^2\theta\, d\Omega$	$\frac{1}{2}\int_{-1}^{1} I_\nu\, \mu^2\, d\mu$	radiation pressure, $P_\nu = 4\pi K_\nu/c$

> **Warning about flux definitions** (Collins §9.3b). Books use three different "fluxes": the physical flux F_ν (W m^{-2} Hz^{-1}), the Eddington flux $H_\nu = F_\nu/4\pi$, and the "astrophysical flux" $\mathcal{F}_\nu = F_\nu/\pi$. Always check which one a formula uses. **In this course, F is the physical flux** and $F = \sigma T_{\rm eff}^4$.

Energy density

Radiation crossing dA in time dt fills a cylinder of volume $dV = c\,dt\,dA$ (Hu). The energy in that volume travelling into $d\Omega$ is $dE = I_\nu\, dA\, d\Omega\, d\nu\, dt$, so the specific energy density per solid angle is I_ν/c. Integrating over angles:

$$u_\nu = \frac{1}{c}\oint I_\nu\, d\Omega = \frac{4\pi}{c} J_\nu, \qquad u = \int_0^\infty u_\nu\, d\nu .$$

Radiation pressure

A photon carries momentum E/c along $\hat{\mathbf{k}}$. The component normal to a surface is $(E/c)\cos\theta$, and the rate at which it crosses the surface carries another factor $\cos\theta$. For a surface that absorbs everything that hits it:

$$P_\nu = \frac{1}{c}\oint I_\nu \cos^2\theta\, d\Omega = \frac{4\pi}{c} K_\nu .$$

(A perfectly reflecting wall feels twice this from one side; Hu.)

The isotropic case

If $I_\nu = I$ is independent of direction:

$$J_\nu = I, \qquad F_\nu = 2\pi I\int_{-1}^{1}\mu\, d\mu = 0, \qquad K_\nu = \tfrac{1}{2} I\int_{-1}^{1}\mu^2 d\mu = \frac{I}{3} = \frac{J_\nu}{3}.$$

Therefore

$$\boxed{P = \tfrac{1}{3}u} \qquad \text{(equation of state of isotropic radiation)},$$

and $K_\nu = J_\nu/3$ is the basis of the **Eddington approximation** (Lecture RT8).

One-sided flux

The flux leaving a surface that radiates with the same intensity I in every outward direction ($0 \le \mu \le 1$), with nothing coming in:

$$F^{+} = 2\pi \int_0^1 I\,\mu\,d\mu = \pi I.$$

This is the origin of the ubiquitous factor π: a black body has $F = \pi B = \sigma T^4$.

2.5 Flux from a uniform sphere

A star of radius R with uniform surface intensity I (no limb darkening), seen from distance r: it subtends a cone of half-angle θ_c with $\sin\theta_c = R/r$.

$$F(r) = \int_0^{2\pi}\int_0^{\theta_c} I\cos\theta\sin\theta\,d\theta\,d\phi = \pi I \sin^2\theta_c = \pi I\left(\frac{R}{r}\right)^2.$$

At the surface ($r = R$), $F = \pi I$; far away, $F \propto r^{-2}$, while I itself is unchanged.

Worked example 2.1 — The Sun: from luminosity to intensity

Given $L_\odot = 3.828\times10^{26}$ W, $R_\odot = 6.957\times10^{8}$ m and 1 AU $= 1.496\times10^{11}$ m, find (a) the surface flux and $T_{\rm eff}$, (b) the mean intensity of the solar disk, (c) the solar constant, computed two ways.

(a) $F_\odot = L_\odot/(4\pi R_\odot^2) = 6.294\times10^{7}$ W m^{-2}. With $F = \sigma T_{\rm eff}^4$: $T_{\rm eff} = (F/\sigma)^{1/4} = 5772$ K.

(b) $\bar I = F/\pi = 2.003\times10^{7}$ W m^{-2} sr^{-1}.

(c) Solid angle of the Sun at 1 AU: $\Omega_\odot = \pi(R_\odot/\mathrm{AU})^2 = 6.794\times10^{-5}$ sr. Since I is conserved and the disk is small (so $\cos\theta \approx 1$), $F_\oplus = \bar I\,\Omega_\odot = 1361$ W m^{-2}. By the inverse-square law, $F_\odot(R_\odot/\mathrm{AU})^2 = 1361$ W m^{-2}, the same.

Worked example 2.2 — Radiation versus gas pressure at the centre of the Sun

At the solar centre, $T_c = 1.57\times10^{7}$ K and $P_{\rm gas} \approx 2.3\times10^{16}$ Pa. Find $P_{\rm rad}$.

Solution. In the interior the radiation is in TE, so it is isotropic and $P_{\rm rad} = u/3 = aT^4/3$, with $a = 4\sigma/c = 7.566\times10^{-16}$ J m^{-3} K^{-4}. Then $u = 4.6\times10^{13}$ J m^{-3} and $P_{\rm rad} = 1.5\times10^{13}$ Pa, so $P_{\rm rad}/P_{\rm gas} \approx 7\times10^{-4}$. Radiation pressure is negligible in the Sun but dominates in very massive stars ($M \gtrsim 50\,M_\odot$).

Worked example 2.3 — Moments of a linear (Eddington) intensity field

Suppose $I(\mu) = I_0 + I_1\mu$ for all $-1 \le \mu \le 1$. Compute J, H, K and show that $K = J/3$ exactly.

$$J = \tfrac12\int_{-1}^{1}(I_0 + I_1\mu)\,d\mu = I_0, \quad H = \tfrac12\int_{-1}^{1}(I_0 + I_1\mu)\mu\,d\mu = \frac{I_1}{3}, \quad K = \tfrac12\int_{-1}^{1}(I_0 + I_1\mu)\mu^2 d\mu =$$

So $K = J/3$ even though the field is **anisotropic** and carries a flux $F = 4\pi H = 4\pi I_1/3$. This is why the Eddington approximation works well deep in a star, where I is very nearly linear in μ.

Common misconceptions

1. **"Intensity falls as $1/r^2$."** Flux does; intensity (surface brightness) does not, in empty static space.
2. **"Zero flux means no light."** A perfectly isotropic radiation field can be extremely intense and still carry zero net flux, as deep inside a star.
3. **"Every book's flux is the same flux."** Watch for F (physical), $H = F/4\pi$ (Eddington) and $\mathcal{F} = F/\pi$ (astrophysical). Mixing them up gives errors of π or 4π.
4. **"The peak of a spectrum is a fixed property."** Its position depends on whether you plot per unit frequency or per unit wavelength.

Lecture summary

- I_ν is energy per time, area, frequency and solid angle: the brightness of one beam.
- I_ν is conserved along rays in vacuum; flux falls as $1/r^2$ only because objects look smaller.
- J_ν, F_ν and K_ν are the 0th, 1st and 2nd angular moments: illumination, net flow and pressure.
- Isotropic radiation has $F = 0$ and $P = u/3$; a surface of uniform intensity emits $F = \pi I$.

Exam questions and answers — Lecture RT2

Q1. Define I_ν, J_ν, F_ν and K_ν, give their units, and say which can be measured by a single pixel of a telescope pointed at a resolved source.

> **A1.** $I_\nu = dE/(\cos\theta\, dA\, d\Omega\, d\nu\, dt)$ [W m^{-2} Hz^{-1} sr^{-1}]; $J_\nu = \frac{1}{4\pi}\oint I_\nu d\Omega$ [same units]; $F_\nu = \oint I_\nu \cos\theta\, d\Omega$ [W m^{-2} Hz^{-1}]; $K_\nu = \frac{1}{4\pi}\oint I_\nu \cos^2\theta\, d\Omega$ [same as I_ν]. A pixel collects photons from a small solid angle, so it measures I_ν (the surface brightness). For an **unresolved** source the whole object falls in one pixel, and we measure F_ν.

Q2. Prove that I_ν is constant along a ray in vacuum. Hence explain why the Sun's disk looks equally bright from Earth and from Mars.

> **A2.** See §2.3: energy conservation between two areas dA_1, dA_2 at distance r gives $I_1 dA_1 (dA_2/r^2) = I_2 dA_2 (dA_1/r^2)$, so $I_1 = I_2$. From Mars the flux is lower by $(1/1.52)^2 = 0.43$, but the Sun's solid angle is smaller by the same factor. The surface brightness, i.e. I_ν, is the same.

Q3. A surface emits $I(\mu) = I_0$ for $\mu > 0$ and nothing for $\mu < 0$. Find J, F and K at the surface. What is J/F?

> **A3.** $J = \frac{1}{2}\int_0^1 I_0 d\mu = I_0/2$; $F = 2\pi\int_0^1 I_0 \mu\, d\mu = \pi I_0$; $K = \frac{1}{2}\int_0^1 I_0 \mu^2 d\mu = I_0/6$. So $J/F = 1/(2\pi)$, i.e. $J = 2H$ where $H = F/4\pi$. This is **Eddington's surface boundary condition** $J(0) =$

$2H(0)$, used in Lecture SA5.

Q4. A star of radius R and uniform surface intensity I is observed from distance $d \gg R$. Derive $F = \pi I (R/d)^2$, and hence $L = 4\pi R^2 \pi I$.

A4. $F = \int I \cos\theta \, d\Omega$ over the cone $\theta \le \theta_c$, $\sin\theta_c = R/d$: $F = 2\pi I \int_0^{\theta_c} \cos\theta \sin\theta \, d\theta = \pi I \sin^2\theta_c = \pi I R^2/d^2$. Then $L = 4\pi d^2 F = 4\pi R^2 (\pi I)$, consistent with a surface flux πI.

Lecture RT3 — Black-Body Radiation: the Planck Function

> **Sources:** MIT 8.901 Lec. 7 (Stefan–Boltzmann, Wien); Hu Set 1; Collins Ch. 9.1 (STE and photon statistics); Rybicki & Lightman Ch. 1 (standard reference).

Learning goals

1. Derive the Planck function from photon statistics.
2. Obtain its Rayleigh–Jeans and Wien limits, Wien's displacement law and the Stefan–Boltzmann law.
3. Define and use brightness, colour and effective temperatures.

Key concepts in plain words

What a black body is. A black body is an object that absorbs every photon that falls on it. The best laboratory example is a closed oven with a tiny hole: light entering the hole bounces around inside and is absorbed long before it can come out again, so the hole looks perfectly black when the oven is cold. When the oven is hot, the hole glows, and the light that escapes has a spectrum that depends **only on the temperature**, not on what the oven is made of. That universal spectrum is the Planck function $B_\nu(T)$.

Why the Planck curve has its shape. Two competing effects meet. (1) At higher frequencies there are many more possible "slots" (modes) for photons in a box: their number grows as ν^2. (2) Each photon of frequency ν costs energy $h\nu$, and thermal energy kT cannot easily pay for very expensive photons, so their occupation falls exponentially, as $e^{-h\nu/kT}$. Rising number of slots times falling occupation gives a curve that rises, peaks, and then drops steeply.

Colour tells temperature (Wien's law). The peak moves to shorter wavelengths as T increases: $\lambda_{\max} T = 2.9 \times 10^{-3}$ m K. Cool red stars (~ 3000 K) peak in the infrared; the Sun (5800 K) peaks in the green-blue; hot blue stars ($\sim 20\,000$ K) peak in the ultraviolet. This is why star colours are a thermometer.

Total power rises steeply with temperature (Stefan–Boltzmann). The total flux from a black body is σT^4. Doubling the temperature multiplies the emitted power by $2^4 = 16$. Small temperature differences therefore give large brightness differences, a fact used constantly in stellar astrophysics ($L = 4\pi R^2 \sigma T_{\rm eff}^4$).

Two useful limits. At low frequency (radio) the Planck function becomes the classical Rayleigh–Jeans law, which is simply proportional to T, so radio astronomers quote intensities as *brightness temperatures*. At high frequency (UV, X-rays) it becomes Wien's law, an exponential tail that is extremely sensitive to T.

Many different "temperatures". Astronomers use several temperatures because each answers a different question: *effective* temperature (which black body emits the same total flux?), *brightness* temperature (which black body has the same intensity at this frequency?), *colour* temperature (which black body has the same spectral shape?), *excitation* temperature (which temperature explains these level populations?). For a real star they differ; they coincide only for a true black body.

3.1 Black-body radiation

A **black body** is matter that absorbs all radiation falling on it. Radiation in an opaque cavity in thermodynamic equilibrium at temperature T (**Hohlraum** radiation) is isotropic, unpolarised and depends **only on** T. Its intensity is the **Planck function**

$$\boxed{B_\nu(T) = \frac{2h\nu^3}{c^2}\,\frac{1}{e^{h\nu/kT}-1}} \qquad \boxed{B_\lambda(T) = \frac{2hc^2}{\lambda^5}\,\frac{1}{e^{hc/\lambda kT}-1}}$$

Derivation (photon gas)

1. **Number of modes.** In a box of volume V the number of photon states with momentum in d^3p is $2\,V\,d^3p/h^3$ (the factor 2 counts polarisations). With $p = h\nu/c$ and $d^3p = 4\pi p^2 dp$, the number of modes per unit volume in $d\nu$ is $\dfrac{8\pi\nu^2}{c^3}d\nu$.
2. **Mean occupation number.** Photons are bosons with zero chemical potential: $\bar n = 1/(e^{h\nu/kT}-1)$.
3. **Energy density.** $u_\nu = \dfrac{8\pi\nu^2}{c^3}\,h\nu\,\bar n = \dfrac{8\pi h\nu^3}{c^3}\,\dfrac{1}{e^{h\nu/kT}-1}$.
4. **Intensity.** For isotropic radiation, $u_\nu = 4\pi I_\nu/c$ (Lecture RT2), so $I_\nu = cu_\nu/4\pi = B_\nu$. ∎

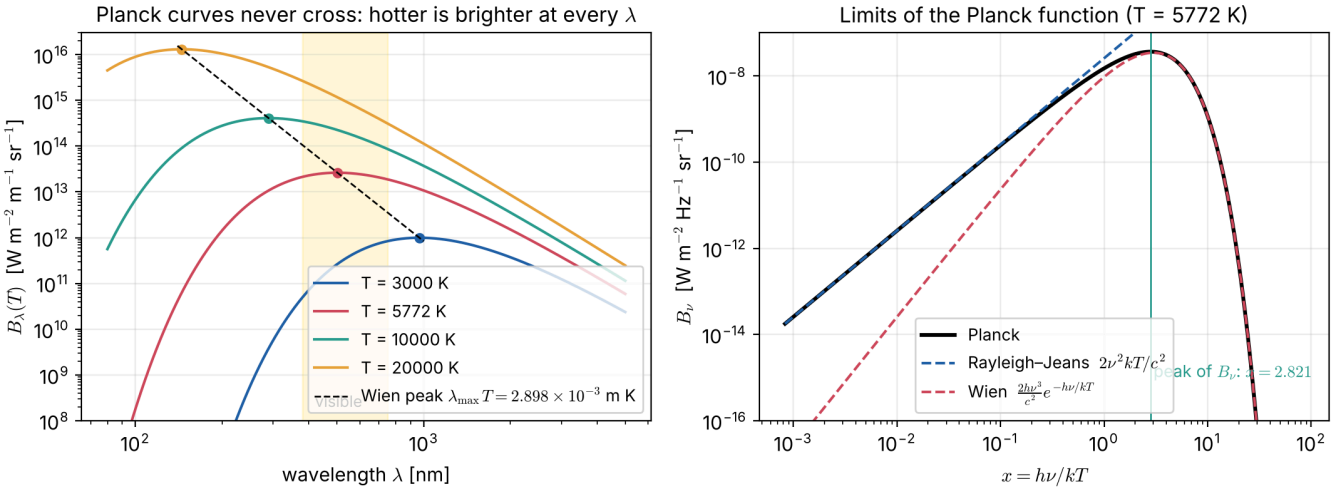


*Figure 3.1 — Left: B_λ for four temperatures, with the Wien peak (dots and dashed line). Hotter bodies are brighter at **every** wavelength, so Planck curves never cross. Right: the Rayleigh–Jeans ($h\nu \ll kT$) and Wien ($h\nu \gg kT$) limits.*

3.2 Limits

Rayleigh–Jeans ($x \equiv h\nu/kT \ll 1$, radio): $e^x - 1 \approx x$, so

$$B_\nu \approx \frac{2\nu^2 kT}{c^2} = \frac{2kT}{\lambda^2}.$$

This is linear in T and contains no h (classical). Integrated over all ν it diverges: the "ultraviolet catastrophe".

Wien ($x \gg 1$, UV and X-ray tail): $B_\nu \approx \dfrac{2h\nu^3}{c^2}e^{-h\nu/kT}$. The steep exponential makes the UV very sensitive to T.

3.3 Wien's displacement law

Set $\partial B_\nu/\partial\nu = 0$: with $x = h\nu/kT$ we need $3(1-e^{-x}) = x$, whose root is $x = 2.821$:

$$\nu_{\max} = 2.821\,\frac{kT}{h} = 5.879\times10^{10}\ \mathrm{Hz\,K^{-1}}\times T.$$

Set $\partial B_\lambda/\partial\lambda = 0$: $5(1-e^{-y}) = y$ with $y = hc/\lambda kT$, so $y = 4.965$:

$$\lambda_{\max} T = 2.898\times10^{-3}\ \mathrm{m\,K}.$$

Note that $\lambda_{\max} \neq c/\nu_{\max}$: the peak depends on the variable used, because $I_\lambda = (c/\lambda^2) I_\nu$ (Lecture RT2, §2.2).

3.4 Stefan–Boltzmann law

$$\int_0^\infty B_\nu\,d\nu = \frac{2h}{c^2}\left(\frac{kT}{h}\right)^4 \int_0^\infty \frac{x^3\,dx}{e^x-1} = \frac{2k^4T^4}{c^2h^3}\cdot\frac{\pi^4}{15} \equiv \frac{\sigma}{\pi}T^4,$$

with

$$\sigma = \frac{2\pi^5 k^4}{15c^2h^3} = 5.670\times10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}.$$

Quantity	Black-body value
Intensity $B = \int B_\nu d\nu$	$\sigma T^4/\pi$
Flux from a surface $F = \pi B$	σT^4
Energy density $u = 4\pi B/c$	aT^4, $a = 4\sigma/c = 7.566\times10^{-16}$ J m^{-3} K^{-4}
Pressure $P = u/3$	$aT^4/3$
Photon number density	$n_\gamma = 16\pi\zeta(3)(kT/hc)^3 = 2.03\times10^{7}\,T^3$ m^{-3}
Mean photon energy	$\langle h\nu\rangle = 2.70\,kT$

A useful derivative, needed for the diffusion approximation (Lecture RT8):

$$\frac{dB}{dT} = \frac{4\sigma T^3}{\pi}.$$

3.5 Characteristic temperatures

Name	Definition	Use
Brightness temperature $T_b(\nu)$	$I_\nu = B_\nu(T_b)$; in RJ: $T_b = c^2 I_\nu/(2k\nu^2)$	radio astronomy; T_b = actual T only for an optically thick thermal source
Colour temperature T_c	shape (slope) of the spectrum fitted by $B_\nu(T_c)$	photometric colours
Effective temperature $T_{\rm eff}$	$F = \sigma T_{\rm eff}^4$, i.e. $L = 4\pi R^2\sigma T_{\rm eff}^4$	stars; defines the surface flux
Excitation temperature $T_{\rm ex}$	$n_2/n_1 = (g_2/g_1)e^{-h\nu/kT_{\rm ex}}$	lines; $T_{\rm ex} = T$ only in LTE

For a real star these all differ. They coincide only for a true black body.

Worked example 3.1 — The cosmic microwave background

$T = 2.725$ K. Find (a) $\nu_{\max}$ of B_ν, (b) $\lambda_{\max}$ of B_λ, (c) the photon number and energy densities.

(a) $\nu_{\max} = 5.879 \times 10^{10} \times 2.725 = 1.60 \times 10^{11}$ Hz = 160 GHz. **(b)** $\lambda_{\max} = 2.898 \times 10^{-3}/2.725 = 1.06$ mm. Note that $c/\nu_{\max} = 1.87$ mm $\neq \lambda_{\max}$. **(c)** $n_\gamma = 2.03 \times 10^7 \times 2.725^3 = 4.1 \times 10^8$ m^{-3} = **411 photons cm^{-3}**; $u = aT^4 = 4.17 \times 10^{-14}$ J m^{-3}.

Worked example 3.2 — The Sun at 500 nm

Compute $B_\lambda(500\ \text{nm}, 5772\ \text{K})$ and decide whether the Wien approximation is adequate.

$x = hc/(\lambda k T) = 1.4388 \times 10^{-2}/(5 \times 10^{-7} \times 5772) = 4.985$, and $2hc^2/\lambda^5 = 3.811 \times 10^{15}$ W m^{-3} sr^{-1}, so

$$B_\lambda = \frac{3.811 \times 10^{15}}{e^{4.985} - 1} = 2.62 \times 10^{13}\ \text{W m}^{-3}\,\text{sr}^{-1}.$$

Wien drops the "-1": the relative error is $e^{-x} = 0.7\,\%$, so Wien is adequate in the visible for solar-type stars.

Worked example 3.3 — Brightness temperature of a radio source

At 1.4 GHz a source has $I_\nu = 2.0 \times 10^{-19}$ W m^{-2} Hz^{-1} sr^{-1}. Find T_b.

$h\nu/k = 0.067$ K, so the RJ limit is excellent for any $T \gg 0.07$ K:

$$T_b = \frac{c^2 I_\nu}{2k\nu^2} = \frac{(3 \times 10^8)^2 \times 2 \times 10^{-19}}{2 \times 1.381 \times 10^{-23} \times (1.4 \times 10^9)^2} = 332\ \text{K}.$$

If the source is an optically thin H II region with $T_e = 8000$ K, then $T_b = T_e \tau$ (Lecture RT5), so $\tau \approx 0.04$.

Worked example 3.4 — Inverting the Planck function

An observed intensity equals B_λ; solve for T.

$$T = \frac{hc}{\lambda k}\,\frac{1}{\ln\!\left(1 + \dfrac{2hc^2}{\lambda^5 B_\lambda}\right)}.$$

For $\lambda = 500$ nm and $B_\lambda = 4.0 \times 10^{13}$ W m^{-3} sr^{-1}: $T = 28\,776/\ln(1 + 95.3) = 6300$ K. We use this formula in Lecture RT7 to map out the solar temperature from limb darkening.

Common misconceptions

1. **"Black bodies look black."** Only when they are cold. The Sun is very nearly a black body.

2. **“$\lambda_{\max} = c/\nu_{\max}$.”** False: the peak of B_λ and the peak of B_ν are at different places.
3. **“Brightness temperature is the real temperature.”** Only for an optically thick, thermal source. Thin gas has $T_b \approx \tau T < T$; non-thermal sources (pulsars, masers) can have T_b far above any physical temperature.
4. **“Stars emit exact black-body spectra.”** They are approximately Planckian, but their spectra show lines, edges and depth-dependent temperature effects (Lecture SA5).

Lecture summary

- The Planck function is the spectrum of radiation in equilibrium; it depends only on T.
- Wien: $\lambda_{\max} T = 2.898 \times 10^{-3}$ m K. Stefan–Boltzmann: $F = \sigma T^4$, $u = aT^4$.
- Radio: Rayleigh–Jeans, $I_\nu \propto T_b$. UV: Wien tail, very sensitive to T.
- Different “temperatures” ($T_{\rm eff}$, T_b, T_c, $T_{\rm ex}$) answer different questions.

Exam questions and answers — Lecture RT3

Q1. Derive the Planck function from the density of photon states and Bose–Einstein statistics.

> **A1.** See §3.1. Modes per unit volume per $d\nu$: $8\pi\nu^2/c^3$. Mean energy per mode: $h\nu/(e^{h\nu/kT}-1)$. Therefore $u_\nu = (8\pi h\nu^3/c^3)/(e^{h\nu/kT}-1)$, and $B_\nu = cu_\nu/4\pi$.

Q2. Show that B_ν has its maximum at $h\nu = 2.821\,kT$. Is the peak of B_λ at $\lambda = c/\nu_{\max}$?

> **A2.** $B_\nu \propto x^3/(e^x-1)$; $d/dx = 0 \Rightarrow 3x^2(e^x-1) - x^3e^x = 0 \Rightarrow 3(1-e^{-x}) = x$, whose root is $x = 2.821$ (solve by iteration: $x_{n+1} = 3(1-e^{-x_n})$). For $B_\lambda \propto y^5/(e^y-1)$ the condition is $5(1-e^{-y}) = y$, $y = 4.965$. Since $4.965 \neq 2.821$, $\lambda_{\max} \neq c/\nu_{\max}$; the two differ by a factor 1.76.

Q3. Derive the Stefan–Boltzmann law, giving σ in terms of fundamental constants. (Use $\int_0^\infty x^3/(e^x-1)\,dx = \pi^4/15$.)

> **A3.** $F = \pi\int B_\nu d\nu = \pi\frac{2h}{c^2}\left(\frac{kT}{h}\right)^4\frac{\pi^4}{15} = \frac{2\pi^5k^4}{15c^2h^3}T^4 = \sigma T^4$.

Q4. Why is T_b of an optically thin source **less** than its kinetic temperature? Under what condition are they equal?

> **A4.** For an isothermal slab with no background, $I_\nu = B_\nu(T)(1-e^{-\tau_\nu})$ (Lecture RT5), so in RJ $T_b = T(1-e^{-\tau}) < T$, and $T_b \approx T\tau$ when $\tau \ll 1$. They are equal only when $\tau_\nu \gg 1$ (and the source is thermal, $S_\nu = B_\nu$).

Lecture RT4 — Emission, Absorption and Scattering

> **Sources:** Hu Set 1 (spontaneous emission, absorption, cross-section, scattering, random walk, multiple processes); MIT 8.901 Lec. 7 ($\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu$); Collins §9.2c (creation rate, redistribution function); Chandrasekhar Ch. I §§7–11.

Learning goals

1. Define j_ν, α_ν, κ_ν, σ_ν and the mean free path.
2. Derive the Einstein relations and the correction for stimulated emission.
3. Describe scattering as absorption followed by re-emission, and write the mixed source function.
4. Use the random walk to estimate photon escape times.

Key concepts in plain words

Absorption coefficient — probability per metre. α_ν is the probability that a photon is removed from the beam per unit length travelled. If $\alpha_\nu = 0.01$ m^{-1}, a photon has a 1 % chance of being stopped in each metre. Its inverse, the **mean free path** $\ell = 1/\alpha_\nu$, is the typical distance a photon travels before something happens to it; it plays the role of "visibility" in fog.

Cross-section — the effective size of a target. Each atom or electron acts like a little target of area σ_ν that photons can hit. This area is not a physical size: an electron has no measurable size, yet it has a Thomson cross-section of 6.65×10^{-29} m^2. Multiply the target area by the number of targets per volume and you get $\alpha_\nu = n\sigma_\nu$. Many tiny targets can make a medium very opaque.

Opacity — absorption per kilogram. Writing $\alpha_\nu = \rho\kappa_\nu$ separates "how much stuff there is" (ρ) from "how good each kilogram is at stopping light" (κ_ν, the opacity). Opacity depends strongly on temperature, composition and frequency, and that frequency dependence is what produces spectral features.

Emission coefficient. j_ν is the energy that matter emits per unit volume, time, frequency and solid angle. It does not depend on how much light is already present: hot, excited atoms emit spontaneously anyway.

Stimulated emission — photon cloning. A photon passing an excited atom can trigger it to emit a second photon identical to the first (same frequency, direction and phase). Because this process grows with the light already present, it acts like *negative absorption*. If more atoms are in the upper level than in the lower one (a population inversion), light is amplified instead of absorbed: this is a laser in the lab and a **maser** in space.

Absorption versus scattering — sponge versus pinball. In *true absorption* the photon's energy is given to the gas (it becomes heat) and the photon disappears; the gas may later emit a different photon according to its temperature. In *scattering* the photon only bounces off in a new direction, like a pinball, keeping its energy. Scattering does not heat the gas and does not "know" the gas temperature; it only redistributes the light that is already there. The number ε is the probability that an interaction is a true absorption (photon destroyed) rather than a scattering.

Random walk — why light leaves stars slowly. A photon in a scattering medium moves like a drunk person who takes steps in random directions: after N steps it is typically only $\sqrt{N}$ steps from the start. To escape from an optical depth

τ it needs about τ^2 scatterings. Light that would cross the Sun in 2 seconds takes thousands of years to diffuse out.

4.1 The four radiative processes

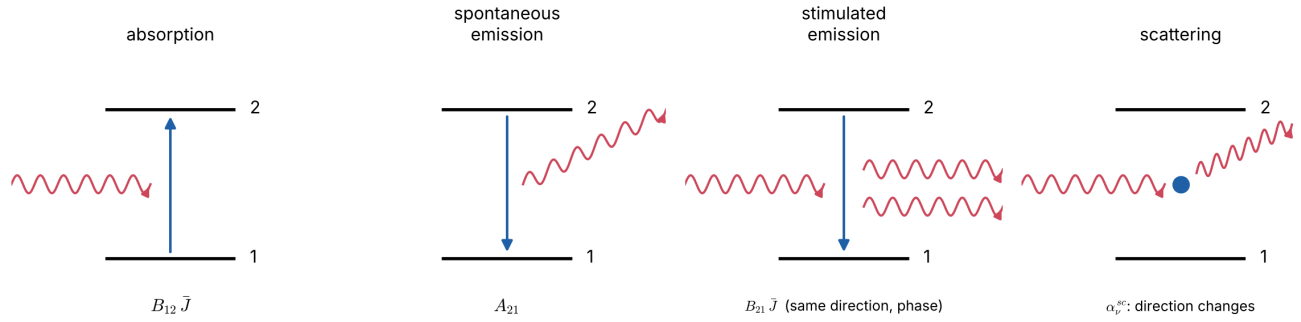


*Figure 4.1 — Absorption, spontaneous emission, stimulated emission and scattering. Stimulated emission produces a photon identical to the incoming one (same frequency, direction and polarisation), so mathematically it behaves like **negative absorption**.*

4.2 Emission coefficient

Let matter emit power $P_\nu\,d\nu$ per unit volume. If it emits isotropically, the energy into $d\Omega$ is

$$dE = j_\nu\,dV\,d\Omega\,d\nu\,dt, \qquad j_\nu = \frac{P_\nu}{4\pi}.$$

Along a path ds through a cylinder of cross-section dA ($dV = dA\,ds$), the beam gains $dI_\nu = j_\nu\,ds$. With emission only:

$$\frac{dI_\nu}{ds} = j_\nu \quad\Rightarrow\quad I_\nu(s) = I_\nu(s_0) + \int_{s_0}^{s} j_\nu\,ds'.$$

The emissivity per unit mass is $\epsilon_\nu = 4\pi j_\nu/\rho$ (Hu).

4.3 Absorption coefficient, cross-section and opacity

A beam loses a fraction $\alpha_\nu\,ds$ of its intensity over ds:

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu \quad\Rightarrow\quad I_\nu(s) = I_\nu(0)\,e^{-\int \alpha_\nu ds} = I_\nu(0)e^{-\tau_\nu}.$$

Particle picture (Hu). Absorbers of number density n each present a cross-section σ_ν. In a slab $dA \times ds$ they block an area $\sigma_\nu n\,dA\,ds$, so the fraction removed is $n\sigma_\nu ds$:

$$\boxed{\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu} \qquad \text{mean free path } \ell_\nu = \frac{1}{\alpha_\nu}.$$

An optical depth τ_ν therefore equals the number of mean free paths along the path.

Main opacity sources in stars

Process	Name	Frequency dependence	Where it matters
bound–bound	line absorption	narrow profile $\phi(\nu)$	spectral lines
bound–free	photo-ionisation	edges, $\propto \nu^{-3}$ above each edge	Balmer/Lyman jumps; H^- in the Sun
free–free	bremsstrahlung	$\propto \nu^{-3}$ (optical), ν^{-2} (radio)	hot stars, H II regions
electron scattering	Thomson	grey: $\sigma_T = 6.652 \times 10^{-29}$ m^2	O stars, interiors, accretion disks
H^-	b-f + f-f of the negative H ion	broad	**dominant continuum opacity in the solar photosphere**

For fully ionised gas, $\kappa_{\rm es} = \sigma_T/(\mu_e m_H) = 0.020(1+X)$ m^2 kg^{-1} $= 0.2(1+X)$ cm^2 g^{-1}.

4.4 Einstein coefficients

Consider a transition $1 \leftrightarrow 2$ with energy $h\nu_0 = E_2 - E_1$, statistical weights g_1, g_2, and line profile $\phi(\nu)$ normalised to $\int \phi\, d\nu = 1$. Let $\bar{J} = \int J_\nu \phi(\nu)\, d\nu$.

Process	Rate per unit volume
spontaneous emission	$n_2 A_{21}$
absorption	$n_1 B_{12} \bar{J}$
stimulated emission	$n_2 B_{21} \bar{J}$

Relations. In TE, $J_\nu = B_\nu(T)$, detailed balance holds ($n_1 B_{12}\bar{J} = n_2 A_{21} + n_2 B_{21}\bar{J}$), and the populations follow Boltzmann, $n_2/n_1 = (g_2/g_1) e^{-h\nu/kT}$. Solving for $\bar{J}$:

$$\bar{J} = \frac{A_{21}/B_{21}}{(g_1 B_{12}/g_2 B_{21}) e^{h\nu/kT} - 1}.$$

For this to equal $B_\nu(T)$ at **every** T we need

$$\boxed{g_1 B_{12} = g_2 B_{21}, \qquad A_{21} = \frac{2h\nu^3}{c^2} B_{21}}$$

These are **atomic** properties, so they hold whether or not the gas is in equilibrium.

Macroscopic coefficients. Each transition moves energy $h\nu$, spread over 4π sr and over the profile:

$$j_\nu = \frac{h\nu}{4\pi} n_2 A_{21} \phi(\nu), \qquad \alpha_\nu = \frac{h\nu}{4\pi}\left(n_1 B_{12} - n_2 B_{21}\right)\phi(\nu) = \frac{h\nu}{4\pi} n_1 B_{12}\left(1 - \frac{g_1 n_2}{g_2 n_1}\right)\phi(\nu).$$

The bracket is the **correction for stimulated emission**. If $g_1 n_2/g_2 n_1 > 1$ (a population inversion), then $\alpha_\nu < 0$ and the beam grows exponentially: a **maser**.

Line source function (any populations):

$$S_\nu = \frac{j_\nu}{\alpha_\nu} = \frac{2h\nu^3}{c^2}\,\frac{1}{\dfrac{g_2 n_1}{g_1 n_2} - 1}.$$

With Boltzmann populations at T this becomes exactly $B_\nu(T)$. This is the key LTE result of Lecture SA1.

4.5 Scattering

Scattering is absorption immediately followed by re-emission into a new direction (Hu). For **coherent** (same frequency), **isotropic** scattering with coefficient $\alpha_\nu^{\rm sc}$, the energy removed from all directions is redistributed evenly over 4π:

$$j_\nu^{\rm sc} = \alpha_\nu^{\rm sc} \oint I_\nu \frac{d\Omega}{4\pi} = \alpha_\nu^{\rm sc} J_\nu \quad \Rightarrow \quad S_\nu^{\rm sc} = J_\nu.$$

The RTE then becomes an **integro-differential equation**, because J_ν depends on I_ν in all directions:

$$\frac{dI_\nu}{ds} = -\alpha_\nu^{\rm sc}\left(I_\nu - \oint I_\nu \frac{d\Omega}{4\pi}\right).$$

Collins (eq. 9.2.27) gives the general form with a redistribution function $R(\nu', \nu, \Omega', \Omega)$, describing how photons are moved in frequency and angle.

Absorption plus scattering

The coefficients add (Hu): $\alpha_\nu = \alpha_\nu^a + \alpha_\nu^{\rm sc}$. The combined source function is

$$S_\nu = \frac{\alpha_\nu^a B_\nu + \alpha_\nu^{\rm sc} J_\nu}{\alpha_\nu^a + \alpha_\nu^{\rm sc}} = \varepsilon_\nu B_\nu + (1-\varepsilon_\nu) J_\nu, \qquad \varepsilon_\nu = \frac{\alpha_\nu^a}{\alpha_\nu^a + \alpha_\nu^{\rm sc}},$$

where $1-\varepsilon_\nu$ is the **single-scattering albedo** (the probability that an interaction is a scattering). This equation is the heart of Lecture RT8.

4.6 Random walk

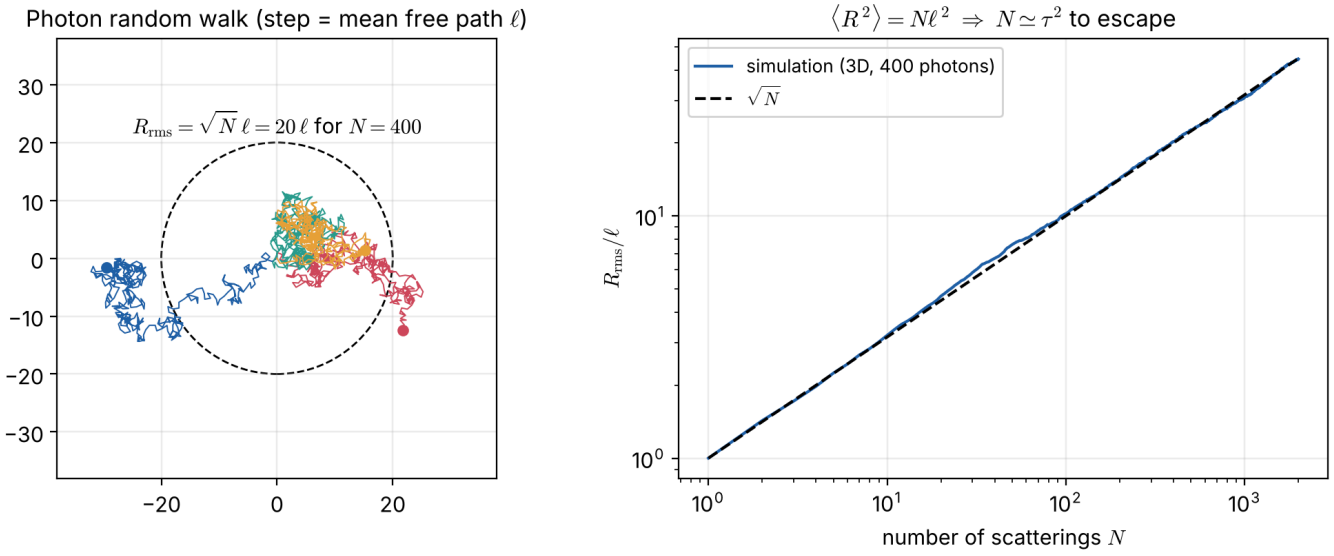


Figure 4.2 — Left: photon paths with unit steps; the dashed circle has radius $\sqrt{N}\ell$. Right: a 3D simulation confirms $R_{\rm rms} = \sqrt{N}\,\ell$.

With steps $\mathbf{r}_i$ of length ℓ in random directions, $\mathbf{R} = \sum \mathbf{r}_i$ and $\langle \mathbf{r}_i \cdot \mathbf{r}_j \rangle = \ell^2 \delta_{ij}$, so

$$\langle R^2 \rangle = N\ell^2 .$$

To escape from optical depth $\tau = R/\ell$ takes $N \approx \tau^2$ steps when $\tau \gg 1$. When $\tau \ll 1$, only a fraction τ of photons interact at all. A useful estimate is $N \approx \max(\tau, \tau^2)$ (Hu).

Thermalisation. If each interaction is an absorption with probability ε, a photon survives $N \approx 1/\varepsilon$ interactions before it is destroyed, travelling a net optical distance $\tau_{\rm th} \approx \sqrt{N} = 1/\sqrt{\varepsilon}$: the **thermalisation depth** (Lecture RT8).

Worked example 4.1 — How long does light take to leave the Sun?

Assume a uniform Sun of mean density $\bar{\rho}$, fully ionised with $X = 0.7$, and only Thomson scattering.

1. $\bar{\rho} = M_\odot / (\tfrac{4}{3}\pi R_\odot^3) = 1410$ kg m^{-3}.
2. $\mu_e = 2/(1+X) = 1.18$, so $n_e = \bar{\rho}/(\mu_e m_H) = 7.2 \times 10^{29}$ m^{-3}.
3. $\ell = 1/(n_e \sigma_T) = 2.1$ cm.
4. $\tau = R_\odot/\ell = 3.3 \times 10^{10}$, so $N = \tau^2 = 1.1 \times 10^{21}$ scatterings.
5. Total path $N\ell = R^2/\ell$; time $t = R_\odot^2/(\ell c) = 7.7 \times 10^{10}$ s $\approx$ **2400 yr**, compared with a light-crossing time of 2.3 s.

The real Sun is denser in the core and has higher (bound-free) opacity, so realistic estimates are 10^4–10^5 yr. The method is the point.

Worked example 4.2 — Stimulated emission at radio wavelengths

Evaluate $1 - e^{-h\nu/kT}$ (the LTE correction factor) for the 21-cm line at $T = 100$ K and for Hα in the Sun.

21 cm: $h\nu/kT = 6.8 \times 10^{-4}$, so the factor is 6.8×10^{-4}. Stimulated emission cancels **99.93 %** of the absorption, and the net opacity $\propto 1/T$. That is why 21-cm absorption measures the spin temperature. Hα (656 nm, 5772 K): $h\nu/kT = 3.8$, so the factor is 0.98; stimulated emission is a 2 % correction.

Worked example 4.3 — Source function of a two-level atom

Show that if the level populations are Boltzmann at temperature $T_{\rm ex}$, then $S_\nu = B_\nu(T_{\rm ex})$.

Insert $g_2 n_1/(g_1 n_2) = e^{h\nu/kT_{\rm ex}}$ into $S_\nu = (2h\nu^3/c^2)/[g_2 n_1/(g_1 n_2) - 1]$ to get $S_\nu = B_\nu(T_{\rm ex})$. **The line source function only measures the excitation temperature.** LTE is the special case $T_{\rm ex} = T_{\rm kin}$.

Common misconceptions

1. **"Scattering removes energy from the radiation."** It removes photons from one beam but puts them into others; the total energy is conserved.
2. **"Stimulated emission is a small correction that can always be ignored."** At radio wavelengths ($h\nu \ll kT$) it cancels almost all of the absorption.
3. **"Opacity is a fixed property of a material."** It depends on temperature, density (through ionisation and excitation) and, above all, frequency.
4. **"A photon from the Sun's core reaches us directly."** The energy random-walks out over thousands of years, being absorbed and re-emitted many times on the way.

Lecture summary

- $\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu$; mean free path $\ell = 1/\alpha_\nu$.
- Einstein relations link absorption, spontaneous and stimulated emission; they are atomic properties.
- Stimulated emission is negative absorption; population inversion gives masers.
- Scattering gives $S = J$; absorption plus scattering gives $S = \varepsilon B + (1-\varepsilon)J$.
- Random walk: $N \approx \tau^2$ scatterings to escape from optical depth τ.

Exam questions and answers — Lecture RT4

Q1. Derive the Einstein relations. Explain why they hold even when the gas is not in TE.

> **A1.** See §4.4. Detailed balance plus Boltzmann populations, required to reproduce $B_\nu(T)$ for all T, gives $g_1 B_{12} = g_2 B_{21}$ (high-T limit) and $A_{21} = (2h\nu^3/c^2)B_{21}$. The coefficients depend only on the atomic wavefunctions (matrix elements), not on the environment, so once found in TE they apply everywhere.

Q2. Write α_ν including stimulated emission. What happens physically when $n_2/g_2 > n_1/g_1$? Give an astrophysical example.

> **A2.** $\alpha_\nu = \frac{h\nu}{4\pi}\phi(\nu)(n_1 B_{12} - n_2 B_{21}) = \frac{h\nu}{4\pi}\phi\, n_1 B_{12}(1 - \frac{g_1 n_2}{g_2 n_1})$. With an inversion, $\alpha_\nu < 0$ and $\tau_\nu < 0$, so I_ν grows as $e^{|\tau|}$: maser amplification. Examples: OH (1665/1667 MHz) and H_2O (22 GHz) masers in star-

forming regions and the envelopes of AGB stars.

Q3. Derive $S_\nu = \varepsilon B_\nu + (1-\varepsilon) J_\nu$ and give the meaning of ε.

A3. $j = \alpha^{a} B + \alpha^{\mathrm{sc}} J$ (thermal emission by Kirchhoff, plus scattering) and $\alpha = \alpha^{a} + \alpha^{\mathrm{sc}}$. Divide to get $S = \varepsilon B + (1-\varepsilon) J$ with $\varepsilon = \alpha^{a}/(\alpha^{a} + \alpha^{\mathrm{sc}})$, the probability that an interaction destroys the photon (converts it into thermal energy of the gas).

Q4. A cloud has $\tau = 100$ for pure scattering. Roughly how many scatterings does a photon undergo before escaping, and how far does it travel in total?

A4. $N \approx \tau^2 = 10^4$; total path $N\ell = \tau^2 \ell = 100\, L$, where $L = \tau \ell$ is the size of the cloud. So the photon travels 100 times the cloud size, and the escape time is $100\, L/c$.

Lecture RT5 — The Radiative Transfer Equation and Its Formal Solution

> **Sources:** MIT 8.901 Lec. 7 §§7.1–7.3 (RTE built up in three steps, constant-S solution, formal solution, Kirchhoff's law for thick and thin lumps, thin H cloud, two-layer stellar atmosphere, thermal inversions); Hu Set 1 (optical depth, source function, formal solution, interpretation); Collins §9.2c–d (source function and its meaning).

Learning goals

1. Write the RTE in terms of τ_ν and S_ν and explain why "I_ν relaxes toward S_ν".
2. Derive the formal solution with an integrating factor.
3. Solve the uniform slab, including its optically thin and thick limits.
4. Explain Kirchhoff's three laws of spectroscopy with RT, including MIT's two-layer star and thermal inversions.

Key concepts in plain words

The RTE is a tug of war. Along a beam, two effects compete: the gas removes a fraction of the light (absorption), and the gas adds its own light (emission). If the beam is brighter than the gas's source function S_ν, absorption wins and the beam fades toward S_ν. If the beam is fainter, emission wins and the beam brightens toward S_ν. Either way, after about one optical depth the beam "forgets" where it came from and approaches S_ν.

The formal solution — looking down a foggy street of lamps. Imagine looking down a long, foggy street lined with lamps. The light reaching your eye is the sum of the light from every lamp, each one dimmed by the fog between it and you. Nearby lamps contribute strongly; distant lamps contribute only a little, because their light is attenuated by a factor $e^{-\tau}$. That is exactly the formal solution: $I = I_0 e^{-\tau} + \int S\, e^{-(\tau - t)} dt$. It also explains why we "see" only about one optical depth into any medium.

Optically thin versus optically thick. A *thin* medium ($\tau \ll 1$) is like a light mist: you see the background through it, and its own glow is weak and proportional to how much gas there is (each atom contributes independently). A *thick* medium ($\tau \gg 1$) is like a wall: you see only its surface layer, and its brightness is S_ν, independent of how much more gas lies behind.

Why lines appear — Kirchhoff's three laws explained. Atoms absorb strongly only at certain frequencies (the spectral lines). At those frequencies the gas is "thicker", so we see less deeply into it. If the background (or the deeper gas) is brighter than the gas in front, the line frequencies look darker: an *absorption line*. If there is no bright background, the gas glows mostly at the line frequencies: *emission lines*. If the gas is thick at all frequencies and at one temperature, we see a smooth continuum.

The MIT two-layer star. The simplest star that shows lines is a hot, thick base with a cooler, thin layer on top. In the continuum we see through the cool layer to the hot base; at the lines the cool layer blocks the view and we see its lower brightness. If the upper layer is *hotter* than the base (a **thermal inversion**), the same lines appear in emission. Thus line spectra are, at heart, thermometers of how temperature changes with height.

5.1 The equation, built up in three steps (MIT 8.901 §7.1)

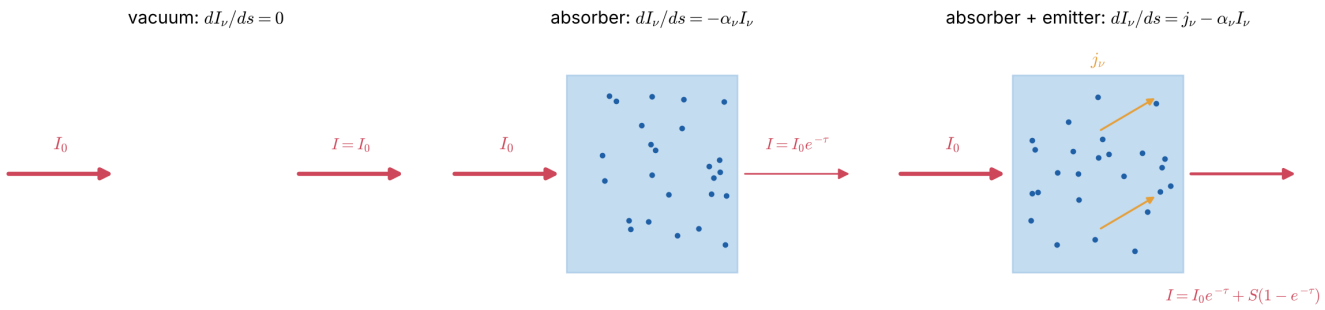


Figure 5.0 — After MIT 8.901 Lecture RT7, Fig. 9: radiation through (left) vacuum, $dI_\nu/ds = 0$; (centre) a purely absorbing medium, $dI_\nu/ds = -\alpha_\nu I_\nu$, so $I = I_0 e^{-\tau}$; (right) a medium that absorbs and emits, $dI_\nu/ds = j_\nu - \alpha_\nu I_\nu$.

MIT's notation: $\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu$ (extinction coefficient, m^{-1}) and j_ν (emissivity, $\mathrm{W\,m^{-3}\,Hz^{-1}\,sr^{-1}}$). Note that j_ν and I_ν have different units; their ratio $S_\nu = j_\nu/\alpha_\nu$ has the units of intensity. Combining Lecture RT4's gains and losses:

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu.$$

Divide by α_ν and use $d\tau_\nu = \alpha_\nu\, ds$ (optical depth measured **along the beam**) and $S_\nu = j_\nu/\alpha_\nu$:

$$\boxed{\frac{dI_\nu}{d\tau_\nu} = S_\nu - I_\nu}$$

Physical meaning. If $I_\nu > S_\nu$, then $dI_\nu/d\tau_\nu < 0$: intensity decreases. If $I_\nu < S_\nu$, it increases. **The intensity always relaxes toward the local source function, on a scale of one optical depth.** S_ν is the intensity the medium "wants" to have.

5.2 Formal solution

Multiply by the integrating factor e^{τ_ν} (MIT eq. 82–85):

$$\frac{d}{d\tau_\nu}\left(I_\nu e^{\tau_\nu}\right) = S_\nu e^{\tau_\nu}.$$

Integrate from 0 to τ_ν:

$$\boxed{I_\nu(\tau_\nu) = I_\nu(0)\, e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(t)\, e^{-(\tau_\nu - t)}\, dt}$$

Interpretation (Hu): what arrives at τ_ν is (i) the initial intensity attenuated by the whole column, plus (ii) the emission $S_\nu\, dt$ from every layer t, attenuated by the material between t and the observer, $e^{-(\tau_\nu - t)}$.

The solution is "formal" because S_ν may depend on I_ν itself (scattering, NLTE). It is then an integral equation, not an explicit answer. In **LTE without scattering**, $S_\nu = B_\nu(T)$ is known once $T(\tau)$ is known, and the formal solution **is** the solution.

5.3 The uniform (isothermal) slab

If S_ν is constant:

$$\boxed{I_\nu = I_\nu(0)\, e^{-\tau_\nu} + S_\nu \left(1 - e^{-\tau_\nu}\right)}$$

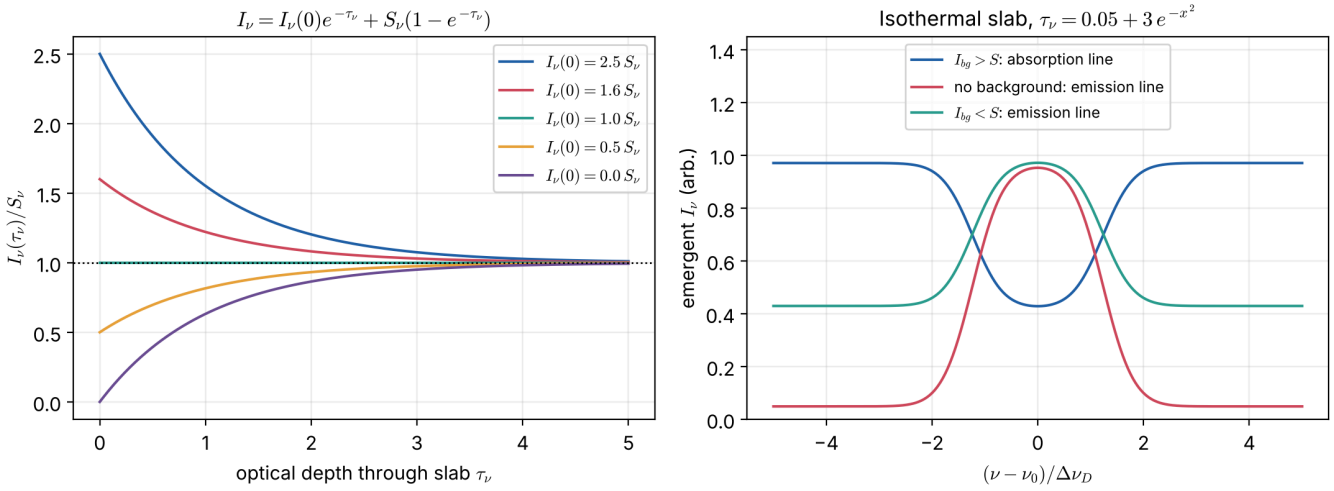


Figure 5.1 — Left: whatever the incident intensity, $I_\nu \to S_\nu$ after a few optical depths. Right: a slab with optical depth larger at line centre than in the wings produces an absorption line when the background is brighter than S, and an emission line when it is fainter.

Limit	Result	Meaning
$\tau_\nu \ll 1$ (optically thin)	$I_\nu \approx I_\nu(0)(1-\tau_\nu) + \tau_\nu S_\nu$	background passes; emission $\propto$ column density ($\tau S = j\,L$)
$\tau_\nu \gg 1$ (optically thick)	$I_\nu \to S_\nu$	memory of the background lost; we see the medium itself
thick, LTE	$I_\nu = B_\nu(T)$	an optically thick isothermal LTE body radiates as a black body

> Even in LTE, $I_\nu = B_\nu$ only inside an **optically thick, isothermal** body. A real star is not isothermal, so its emergent spectrum is not exactly Planckian.

Kirchhoff's laws of spectroscopy, explained

Write $\tau_\nu = \tau_c + \tau_0 \phi(\nu)$, with a weak continuum τ_c plus a line. Then:

1. **Hot, opaque body** ($\tau_\nu \gg 1$ at all ν): $I_\nu = B_\nu(T)$, a continuous spectrum.
2. **Hot, thin gas, no background** ($I_\nu(0) = 0$): $I_\nu = B_\nu\left(1 - e^{-\tau_\nu}\right)$ is largest where τ_ν is largest, so we see **emission lines**.
3. **Cooler gas in front of a hotter continuum** ($I_\nu(0) > S_\nu$): where τ_ν is large, I_ν is pulled *down* toward S_ν, giving **absorption lines**.

The real criterion is $I_{\text{background}}$ versus S_ν, not "cool versus hot" by itself. This idea carries over to stellar atmospheres in Lecture SA6.

5.4 MIT 8.901 §7.3: thin clouds, two-layer stars and thermal inversions

MIT's form of **Kirchhoff's law** says that a thermally emitting object in equilibrium has $S_\nu = B_\nu(T)$, but this does **not** mean that it emits a black-body spectrum. What emerges depends on τ_ν. Take two lumps of matter at the same T, with nothing behind them ($I_\nu(0) = 0$):

Lump	Emergent intensity	Spectrum
optically thick, $\tau_\nu \gg 1$	$I_\nu = S_\nu = B_\nu(T)$	black body
optically thin, $\tau_\nu \ll 1$	$I_\nu = S_\nu\,[1-(1-\tau_\nu)] = \tau_\nu B_\nu(T) = \alpha_\nu s\, B_\nu(T)$	black body × τ_ν: bright only where α_ν is large

(a) A hot, thin hydrogen cloud on a dark sky. α_ν is large only at the hydrogen lines, whose wavelengths follow the Rydberg formula

$$\frac{1}{\lambda_{\rm vac}} = R_{\rm H}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), \qquad R_{\rm H} = 1.0968 \times 10^7\ {\rm m}^{-1}$$

($n_1 = 1, 2, 3, 4$: Lyman, Balmer, Paschen, Brackett). Since $I_\nu = \tau_\nu B_\nu$, we see an **emission-line spectrum** (Figure 5.2a).

(b) The simplest stellar atmosphere (MIT Fig. 10): an optically thick interior at T_H under a cooler, thin layer at $T_C < T_H$. The thick base supplies $I_\nu(0) = B_\nu(T_H)$, and the layer gives

$$I_\nu = B_\nu(T_H)e^{-\tau_\nu} + B_\nu(T_C)\left(1 - e^{-\tau_\nu}\right) \overset{\tau_\nu \ll 1}{\approx} B_\nu(T_H) - \tau_\nu\left[B_\nu(T_H) - B_\nu(T_C)\right].$$

The **continuum** (small α_ν) shows $B_\nu(T_H)$ from deep down. At the **lines** (large α_ν) we see only down to $\tau_\nu \sim 1$, i.e. the cool layer, so $I_\nu \to B_\nu(T_C)$: **absorption lines** (Figure 5.2b). An isothermal atmosphere gives a pure black body whatever α_ν is.

(c) Thermal inversion. If T *increases* outward ($T_{\rm top} > T_{\rm base}$), every argument reverses and the same lines appear in **emission** above the continuum (Figure 5.2c). MIT notes that inversions are ubiquitous: the solar chromosphere, Solar System planets (stratospheres) and hot-Jupiter exoplanets.

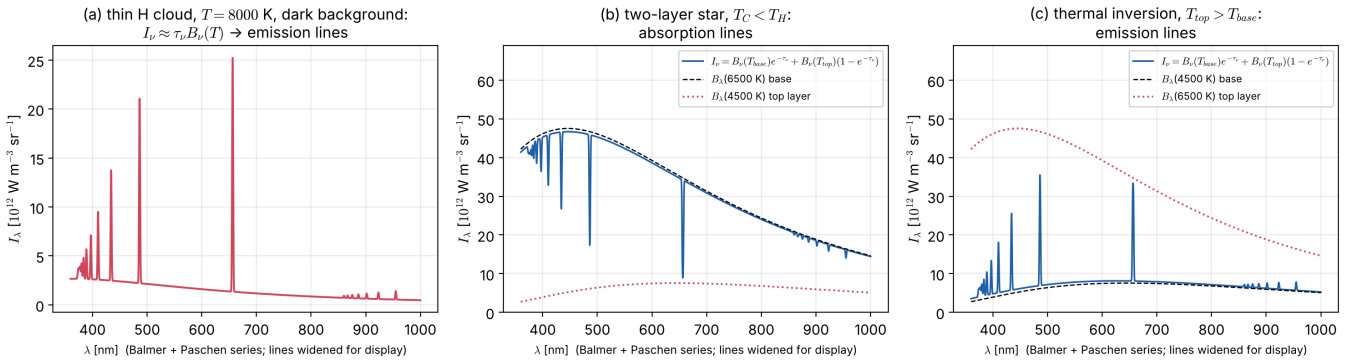


Figure 5.2 — The MIT 8.901 toy models, computed exactly with $I = I_0 e^{-\tau} + B(1 - e^{-\tau})$ for Balmer and Paschen lines placed by the Rydberg formula (line widths exaggerated for visibility). (a) Thin 8000 K cloud: emission lines. (b) 6500 K base under a 4500 K layer: absorption lines. (c) Inverted case: emission lines.

5.5 Radio spectrum of an H II region

For free–free absorption, $\tau_\nu \propto T_e^{-1.35}\nu^{-2.1}\int n_e^2\,ds$ (the integral is the emission measure). In the RJ limit, $I_\nu \propto T_b$:

$$T_b = T_e\left(1 - e^{-\tau_\nu}\right).$$

At low ν, $\tau \gg 1$, so $T_b = T_e$ and the flux $S_\nu \propto \nu^2 T_e$ (black body). At high ν, $\tau \ll 1$, so $T_b = T_e\tau_\nu \propto \nu^{-2.1}$ and the flux is $\propto \nu^{-0.1}$ (almost flat). The turnover at $\tau_\nu \approx 1$ measures the emission measure.

Worked example 5.1 — H II region brightness temperatures

$T_e = 8000$ K. *Find T_b for $\tau = 0.01, 0.1, 1, 3, 10$.*

τ	**0.01**	**0.1**	**1**	**3**	**10**
$T_b = T_e(1 - e^{-\tau})$ [K]	80	761	5057	7602	8000

Lesson: the brightness temperature equals the physical temperature only when the source is optically thick.

Worked example 5.2 — Absorption by a foreground cloud

A star with continuum intensity I_0 is seen through a cloud with $S = 0.3 I_0$. At line centre $\tau = 5$, in the wings $\tau = 0.1$. Find the line profile values and the central depth.

$I = I_0 e^{-\tau} + S(1 - e^{-\tau})$:

- wings ($\tau = 0.1$): $I = 0.933\,I_0$;
- $\tau = 1$: $I = 0.558\,I_0$;
- centre ($\tau = 5$): $I = 0.305\,I_0 \approx S$.

The line core is **saturated**: it cannot go below S, however large τ becomes. The central depth is $1 - 0.305/0.933 = 0.67$ relative to the local continuum.

Worked example 5.3 — Formal solution with a linearly increasing source function

Let $S(t) = a + bt$ along a path of total optical depth τ, with $I(0) = 0$. Find $I(\tau)$ and its thick limit.

$$I(\tau) = \int_0^\tau (a + bt) e^{-(\tau - t)}\,dt.$$

Put $u = \tau - t$: $I = \int_0^\tau [a + b(\tau - u)]e^{-u}du = (a + b\tau)(1 - e^{-\tau}) - b\left[1 - (1 + \tau)e^{-\tau}\right]$. For $\tau \to \infty$: $I \to a + b\tau - b = S(\tau - 1)$.

The emerging intensity equals the source function one optical depth back from the observer. This is the seed of the Eddington–Barbier relation (Lecture RT7).

Worked example 5.4 — The MIT two-layer star at Hα

Base $T_H = 6500$ K, upper layer $T_C = 4500$ K. At $\lambda = 656.3$ nm the layer has $\tau_c = 0.02$ in the continuum and $\tau = 2.02$ at the centre of Hα. (a) Find the line depth. (b) Compare with the linearised MIT formula, and say when the formula fails. (c) What is the deepest the line could ever be?

$B_\lambda(6500\ \mathrm{K}) = 3.474 \times 10^{13}$ and $B_\lambda(4500\ \mathrm{K}) = 7.551 \times 10^{12}$ W m^{-3} sr^{-1}.

(a) Continuum: $I_c = B_H e^{-0.02} + B_C(1 - e^{-0.02}) = 0.9845\, B_H$. Line: $I_l = B_H e^{-2.02} + B_C(1 - e^{-2.02}) = 0.321\, B_H$. Depth $= 1 - I_l/I_c = 0.67$.

(b) The linearised form $B_H - \tau(B_H - B_C)$ gives $0.984\, B_H$ for $\tau = 0.02$ (correct to 0.02 %) but $-0.57\, B_H$ for $\tau = 2$: **negative and meaningless**. The linear form holds only for $\tau_\nu \ll 1$; strong lines need the full exponential.

(c) For $\tau \to \infty$, $I_l \to B_C$, so the maximum depth is $1 - B_C/I_c = 0.78$. **The depth of a saturated line measures the temperature contrast of the atmosphere**, which is the same conclusion as the Milne–Eddington model of Lecture SA6.

Common misconceptions

1. **"The formal solution solves the transfer problem."** Only if S_ν is known in advance (e.g. LTE with known T). With scattering, S depends on I, and the formal solution becomes an integral equation.
2. **"An optically thick object always emits a black-body spectrum."** Only if it is also isothermal and $S = B$. A real star is thick but not isothermal.
3. **"Emission lines require a hot gas."** They require gas whose S_ν is larger than the background intensity, e.g. thin gas on a dark sky, or a temperature that rises outward.
4. **"The linearised two-layer formula works for every line."** It assumes $\tau \ll 1$; for strong lines it can even give negative intensities.

Lecture summary

- $dI_\nu/d\tau_\nu = S_\nu - I_\nu$: intensity relaxes toward the source function.
- Formal solution: background dimmed by $e^{-\tau}$ plus emission from each layer, each dimmed by the gas in front.
- Constant S: $I = I_0 e^{-\tau} + S(1 - e^{-\tau})$; thin ⇒ $I \approx I_0 + \tau(S - I_0)$, thick ⇒ $I \to S$.
- Absorption lines need background brighter than S; emission lines need the opposite.
- MIT toy models: thin cloud (emission lines), two-layer star (absorption lines), inversion (emission lines).

Exam questions and answers — Lecture RT5

Q1. Starting from $dI_\nu/ds = -\alpha_\nu I_\nu + j_\nu$, derive the formal solution.

> **A1.** With $d\tau = \alpha\, ds$ and $S = j/\alpha$: $dI/d\tau + I = S$. Multiply by e^τ: $d(Ie^\tau)/d\tau = Se^\tau$. Integrate: $I(\tau)e^\tau - I(0) = \int_0^\tau S(t)e^t dt$, so $I(\tau) = I(0)e^{-\tau} + \int_0^\tau S(t)e^{-(\tau - t)}dt$.

Q2. For a uniform slab with S constant, derive $I = I_0 e^{-\tau} + S(1 - e^{-\tau})$ and discuss the thin and thick limits.

A2. Take S out of the integral: $\int_0^\tau e^{-(\tau-t)}dt = 1 - e^{-\tau}$. Thin: $I \approx I_0 + \tau(S - I_0)$, so the change is linear in τ (the column density). Thick: $I \to S$, independent of I_0.

Q3. "Absorption lines form because the gas in a stellar atmosphere is cooler than the gas below it." Rephrase this precisely using I_ν and S_ν.

A3. At line frequencies τ_ν is large, so the emergent intensity is set by S_ν in higher layers. If S_ν decreases outward (in LTE, $S_\nu = B_\nu(T)$ with T decreasing outward), then line photons come from layers with smaller S_ν than continuum photons, and an absorption line results. If S_ν increases outward (a chromosphere, or NLTE effects), the line appears in emission.

Q4. (MIT 8.901 two-layer model.) A star has an optically thick base at T_H and a thin upper layer at T_C. Show that $I_\nu \approx B_\nu(T_H) - \tau_\nu[B_\nu(T_H) - B_\nu(T_C)]$ for $\tau_\nu \ll 1$. Explain why the spectrum shows absorption lines, and what changes in a thermal inversion.

A4. The base is thick, so $I_\nu(0) = B_\nu(T_H)$. The layer has $S_\nu = B_\nu(T_C)$ (LTE), so $I_\nu = B_\nu(T_H)e^{-\tau_\nu} + B_\nu(T_C)(1 - e^{-\tau_\nu})$, and expanding $e^{-\tau} \approx 1 - \tau$ gives the result. Where α_ν is large (lines), τ_ν is large and the intensity drops toward $B_\nu(T_C) < B_\nu(T_H)$: absorption lines. In a thermal inversion $T_C > T_H$, so $B_\nu(T_C) > B_\nu(T_H)$ and the same lines appear in emission (solar chromosphere, planetary stratospheres). For strong lines the linear form fails (it can even go negative), and the full exponential must be used (Worked example 5.4).

Lecture RT6 — How the Radiative Transfer Equation Is Formed (Part II: Deeper Foundations)

> **Sources:** Collins, *Fundamentals of Stellar Astrophysics*, Ch. 9.2 (photon phase-space density, eq. 9.2.1–9.2.27: Boltzmann equation → general RTE → creation rate, redistribution function, source function); Hu, *Astro 305 Set 1* (Liouville's theorem, I_ν/ν^3, Boltzmann equation); MIT 8.901 Lec. 7 §7.1 (building the RTE step by step); Chandrasekhar, *Radiative Transfer* (1950), Ch. I §§1–11; Rybicki & Lightman Ch. 1 and 4 (relativistic invariants).

Learning goals

1. Describe a radiation field by the photon distribution function $f(\mathbf{r}, \mathbf{p}, t)$ and relate it to I_ν.
2. Derive the radiative transfer equation from photon conservation (the Boltzmann equation) in a sequence of clear steps.
3. Write each physical gain and loss term (absorption, stimulated emission, spontaneous emission, scattering out and in).
4. Reduce the general equation to the forms used in practice: along a ray, plane-parallel, spherical, time-dependent, moving media.
5. Take energy and momentum moments to obtain the radiation energy equation and the radiation force.

Key concepts in plain words

Photons are particles in phase space. A gas of photons can be described exactly like a gas of atoms: by counting how many photons have position near $\mathbf{r}$ and momentum near $\mathbf{p}$ at time t. That count per unit phase-space volume is the distribution function f. Specific intensity is simply f multiplied by constants and by ν^3; it carries the same information in astronomers' units.

Transfer = conservation with sources and sinks. In empty space, photons stream in straight lines at speed c; nothing is created or destroyed, so f is constant along each photon's path (Liouville's theorem). Matter changes this by removing photons (absorption, scattering out of the beam) and adding photons (emission, scattering into the beam). The RTE is just "rate of change along the path = gains − losses", the same logic as a bank account.

Why the equation is hard. The losses are local and simple ($\propto I_\nu$). The gains from scattering, however, depend on the intensity arriving from *all other directions*, which depends on the whole medium. This turns a simple differential equation into an integro-differential equation, which is why most of the history of the subject (Lecture RT10) is about clever approximations and numerical methods.

One equation, many faces. Every form used in this course (along a ray, plane-parallel with μ, the formal solution, the moment equations) is the same physical statement written under particular assumptions: steady state, straight rays, layered geometry. Knowing *which assumption was made where* tells you when a formula may be used.

6.1 Step 1 — Describe the photons

Let $dN = f(\mathbf{r}, \mathbf{p}, t)\, d^3x\, d^3p$ be the number of photons in a phase-space volume. For photons $p = h\nu/c$, and in spherical momentum coordinates $d^3p = p^2 dp\, d\Omega = (h^3\nu^2/c^3)\, d\nu\, d\Omega$.

The photons crossing an area dA (normal at angle θ to their direction) in time dt fill the volume $d^3x = c\, dt\, dA \cos\theta$. Their energy is

$$dE = h\nu\, f\, d^3x\, d^3p = h\nu\, f\, (c\, dt\, dA \cos\theta) \left(\frac{h^3\nu^2}{c^3} d\nu\, d\Omega \right).$$

Comparing with the definition $dE = I_\nu \cos\theta\, dA\, d\Omega\, d\nu\, dt$ (Lecture RT2):

$$\boxed{I_\nu = \frac{h^4\nu^3}{c^2} f} \qquad \text{(Collins eq. 9.2.8)}$$

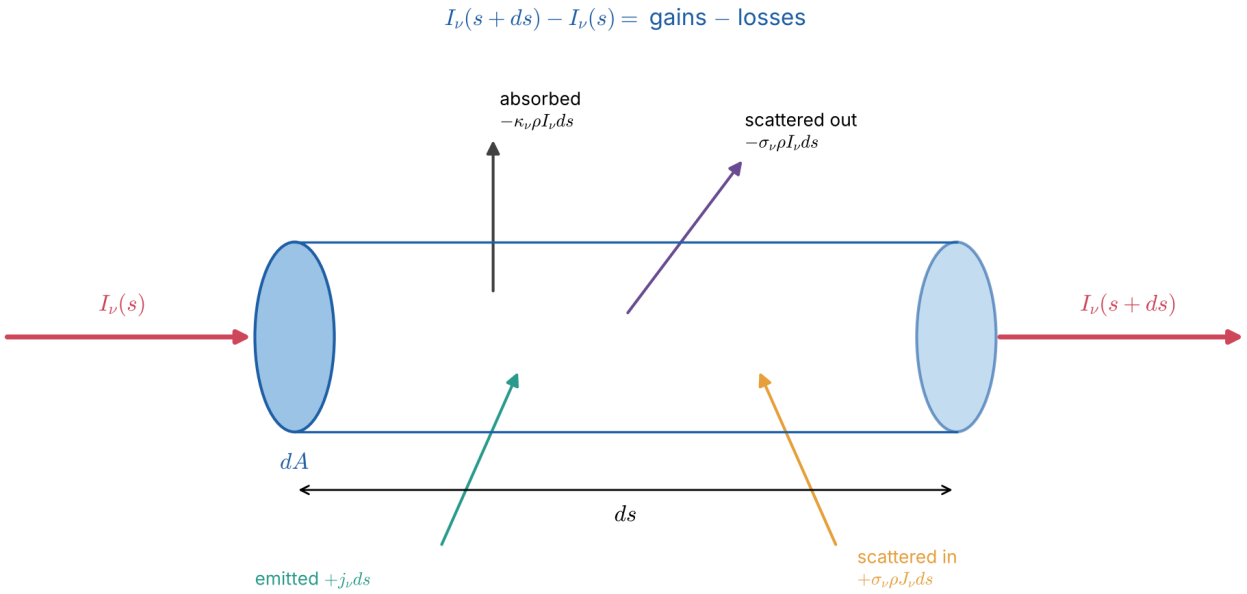


Figure 6.1 — Bookkeeping for a pencil of radiation crossing a small cylinder of cross-section dA and length ds. Losses: true absorption and scattering out of the beam. Gains: thermal emission and scattering into the beam from other directions.

6.2 Step 2 — Conservation of photons: the Boltzmann equation

The number of photons in a phase-space element changes only through "collisions" with matter:

$$\frac{\partial f}{\partial t} + \dot{\mathbf{r}} \cdot \nabla f + \dot{\mathbf{p}} \cdot \nabla_{\mathbf{p}} f = \left(\frac{\partial f}{\partial t} \right)_{\text{coll}} \qquad \text{(Collins eq. 9.2.1, 9.2.9)}$$

For photons $\dot{\mathbf{r}} = c\, \hat{\mathbf{n}}$. The term $\dot{\mathbf{p}}$ is non-zero only if photons are deflected or redshifted on their way (strong gravitational fields, a varying refractive index); Collins keeps it in eq. 9.2.11 for completeness. Without collisions the right-hand side is zero: f is constant along a photon path. This is **Liouville's theorem**, and it is the reason I_ν/ν^3 is conserved (Hu; Lecture RT2).

6.3 Step 3 — Convert to intensity

Multiply the Boltzmann equation by $h^4\nu^3/c^3$ and use Step 1. With $\dot{\mathbf{p}} = 0$:

$$\boxed{\frac{1}{c}\frac{\partial I_\nu}{\partial t} + \hat{\mathbf{n}}\cdot\nabla I_\nu = \frac{h^4\nu^3}{c^3}\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} \equiv (\text{gains}) - (\text{losses})}$$

The left-hand side is the rate of change of I_ν *following the photons*: $\frac{1}{c}\partial_t + \hat{\mathbf{n}}\cdot\nabla = \frac{d}{ds}$ along a ray.

6.4 Step 4 — The physical gain and loss terms

Process	Term in dI_ν/ds	Comment
true absorption	$-\kappa_\nu \rho\, I_\nu$	photon destroyed; energy goes into the gas
stimulated emission	$+\kappa_\nu^{\text{stim}} \rho\, I_\nu$	$\propto I_\nu$, so it is combined with absorption: $\kappa_\nu \to \kappa_\nu(1 - e^{-h\nu/kT})$ in LTE
spontaneous (thermal) emission	$+j_\nu$	independent of I_ν; in LTE $j_\nu = \kappa_\nu \rho B_\nu$ (Kirchhoff)
scattering out of the beam	$-\sigma_\nu \rho\, I_\nu$	photon redirected, not destroyed
scattering into the beam	$+\frac{\sigma_\nu \rho}{4\pi}\int_0^\infty \oint R(\nu', \nu; \hat{\mathbf{n}}', \hat{\mathbf{n}})\, I_{\nu'}(\hat{\mathbf{n}}')\, d\Omega'\, d\nu'$	R = redistribution function (Collins eq. 9.2.16, 9.2.20)

Putting everything together gives the **general equation of transfer** (Collins eq. 9.2.23):

$$\boxed{\frac{1}{c}\frac{\partial I_\nu}{\partial t} + \hat{\mathbf{n}}\cdot\nabla I_\nu = -(\kappa_\nu + \sigma_\nu)\rho\, I_\nu + j_\nu + \frac{\sigma_\nu \rho}{4\pi}\int_0^\infty \oint R\, I_{\nu'}\, d\Omega'\, d\nu'}$$

Special redistribution functions (Collins §9.2e): **coherent** scattering $R \propto \delta(\nu - \nu')$ (Thomson scattering of optical light); **complete redistribution** R independent of ν' (strong lines with frequent collisions); **isotropic** R independent of angle. For coherent isotropic scattering, the gain term becomes simply $\sigma_\nu \rho J_\nu$.

Defining the **extinction coefficient** $\chi_\nu = (\kappa_\nu + \sigma_\nu)\rho$ and the **source function**

$$S_\nu = \frac{j_\nu + \sigma_\nu \rho J_\nu}{\chi_\nu} = \varepsilon_\nu B_\nu + (1 - \varepsilon_\nu) J_\nu \quad (\text{LTE, coherent isotropic scattering}),$$

the static equation along a ray takes its familiar form, $dI_\nu/d\tau_\nu = S_\nu - I_\nu$ with $d\tau_\nu = \chi_\nu ds$.

6.5 Step 5 — Special forms of the equation

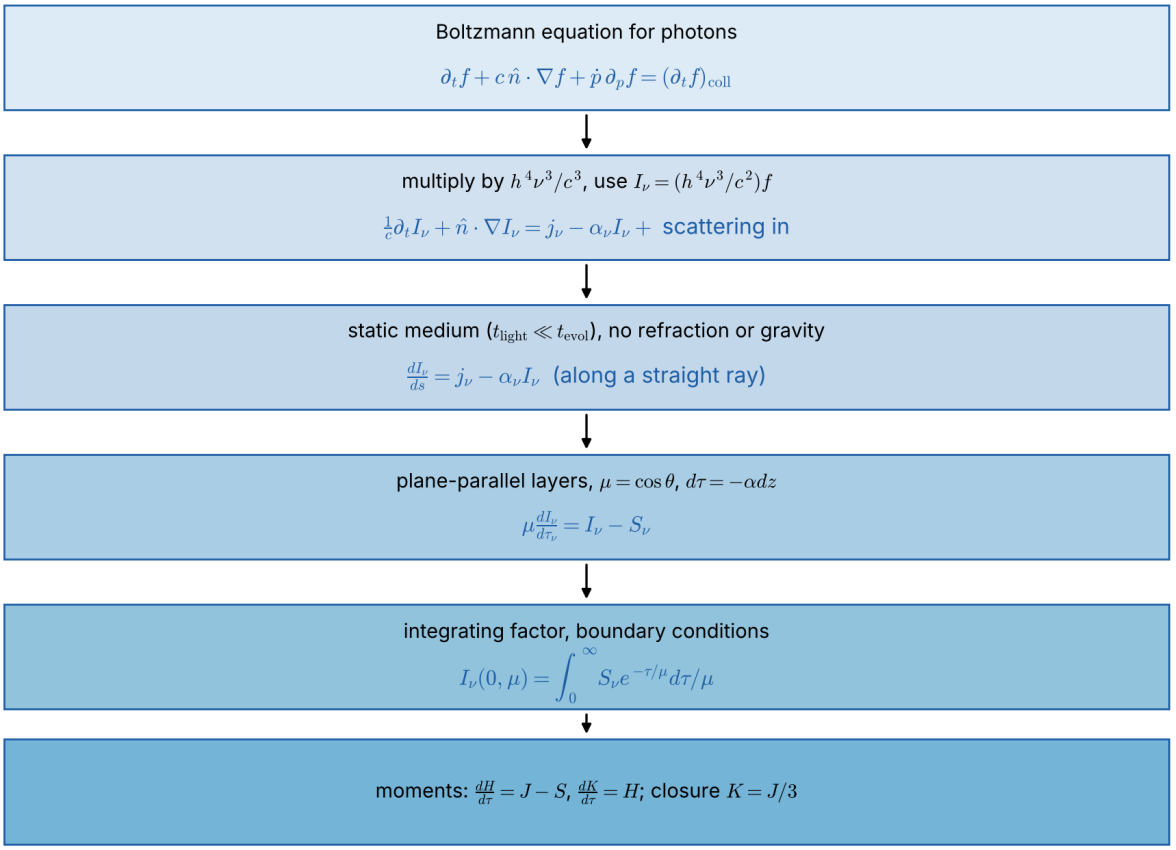


Figure 6.2 — Each simplification adds one assumption. Knowing which assumption was used tells you when a formula is valid.

Situation	Form of the RTE	Assumption added
general, time-dependent	$\frac{1}{c}\partial_t I_\nu + \hat{\mathbf{n}}\cdot\nabla I_\nu = \chi_\nu(S_\nu - I_\nu)$	none beyond straight rays
static medium	$\hat{\mathbf{n}}\cdot\nabla I_\nu = \chi_\nu(S_\nu - I_\nu)$	light-travel time ≪ evolution time
along one ray	$\frac{dI_\nu}{ds} = \chi_\nu(S_\nu - I_\nu)$	coordinate along the ray
3D Cartesian	$n_x\partial_x I + n_y\partial_y I + n_z\partial_z I = \chi(S - I)$	none (numerical 3D codes)
plane-parallel	$\mu\,\frac{dI_\nu}{d\tau_\nu} = I_\nu - S_\nu$	horizontally homogeneous layers; τ measured inward (Collins eq. 9.2.26)
spherical	$\mu\frac{\partial I_\nu}{\partial r} + \frac{1-\mu^2}{r}\frac{\partial I_\nu}{\partial \mu} = \chi_\nu(S_\nu - I_\nu)$	spherical symmetry (extended atmospheres, winds)
moving medium	write in the **comoving frame**, or use Doppler-shifted χ_ν, j_ν	velocities shift frequencies (Sobolev, Lecture SA8)

Why the spherical term appears. Along a straight ray with impact parameter p, the radius r and the angle to the radial direction both change: $r^2 = p^2 + z^2$ and $\mu = z/r$. Then $\frac{d}{dz} = \mu\frac{\partial}{\partial r} + \frac{1-\mu^2}{r}\frac{\partial}{\partial \mu}$. The second term vanishes when $r \to \infty$, recovering the plane-parallel form.

Relativistic invariants (Rybicki & Lightman). Between frames in relative motion, I_ν/ν^3, $\nu\,\chi_\nu$ and j_ν/ν^2 are invariant; these follow directly from f being a scalar. They give Doppler boosting of jets and the $(1+z)^{-4}$ surface-brightness dimming of Lecture RT2.

6.6 Step 6 — Energy and momentum of the radiation field

Integrate the general equation over all directions and frequencies (isotropic j_ν and χ_ν):

$$\text{energy:}\qquad \frac{\partial u}{\partial t} + \nabla\cdot\mathbf{F} = 4\pi\int_0^\infty (j_\nu - \kappa_\nu\rho J_\nu)\,d\nu$$

$$\text{momentum:}\qquad \frac{1}{c^2}\frac{\partial \mathbf{F}}{\partial t} + \nabla\cdot\mathsf{P} = -\frac{1}{c}\int_0^\infty \chi_\nu\,\mathbf{F}_\nu\,d\nu \equiv -\mathbf{f}_{\rm rad}$$

The first equation says the radiation gains energy where matter emits more than it absorbs. In a static, source-free atmosphere the right side is zero, giving **radiative equilibrium** and constant flux (Lecture RT8). The second gives the **radiation force** per unit volume, $\mathbf{f}_{\rm rad} = \frac{1}{c}\int \chi_\nu \mathbf{F}_\nu d\nu$, which drives the winds of hot stars and sets the Eddington luminosity.

Worked example 6.1 — Check: the black body from the photon distribution

In equilibrium, photons have occupation number $\bar n = 1/(e^{h\nu/kT}-1)$ per state and there are $2/h^3$ states per unit phase-space volume (two polarizations). Show that $I_\nu = B_\nu$.

$f = \dfrac{2}{h^3}\bar n$, so $I_\nu = \dfrac{h^4\nu^3}{c^2}\cdot\dfrac{2}{h^3}\cdot\dfrac{1}{e^{h\nu/kT}-1} = \dfrac{2h\nu^3}{c^2}\dfrac{1}{e^{h\nu/kT}-1} = B_\nu(T)$. ✓ The relation $I_\nu = (h^4\nu^3/c^2)f$ is consistent with Lecture RT3.

Worked example 6.2 — When is the time-dependent term needed?

Compare the photon escape time with the evolution time for (a) the solar photosphere and (b) a supernova 10 days after explosion ($v = 10^4$ km s^{-1}, $\tau \approx 100$).

a. The photosphere is about 500 km thick and $\tau \sim 1$, so photons cross it in $\sim 500\ \text{km}/c \approx 2$ ms. Granulation changes in about 5 minutes, 10^5 times slower, so the **static** RTE is excellent.
b. $R = vt = 10^7 \times 8.64\times10^5 = 8.6\times10^{12}$ m. The diffusion time is $t_d \approx \tau R/c = 100\times 8.6\times 10^{12}/3\times10^8 \approx 2.9\times10^6$ s $\approx$ 33 days, longer than the age. The **time-dependent** equation (or a diffusion treatment with time dependence) is required, which is why supernova light curves have a delayed peak.

Worked example 6.3 — Doppler boosting from the invariant I_ν/ν^3

A jet blob emits $I'_{\nu'} \propto \nu'^{-0.7}$ in its rest frame and moves toward us with Doppler factor $\mathcal{D} = 10$ ($\nu = \mathcal{D}\nu'$). How much brighter does it look at a fixed observed frequency?

$I_\nu/\nu^3 = I'_{\nu'}/\nu'^3 \Rightarrow I_\nu = \mathcal{D}^3 I'_{\nu'} = \mathcal{D}^3 I'_\nu\,(\nu'/\nu)^{-0.7} = \mathcal{D}^{3+0.7} I'_\nu$. So the blob is $10^{3.7} \approx 5000$ times brighter, which explains the extreme brightness of blazars.

Common misconceptions

1. **“The RTE is an empirical law.”** It follows rigorously from photon conservation (the Boltzmann equation); only the coefficients κ_ν, σ_ν, j_ν and R need microphysics.
2. **“Stimulated emission is part of the emission coefficient.”** Because it is proportional to I_ν, it is conventionally counted as negative absorption inside κ_ν.
3. **“The plane-parallel equation works for any star.”** It assumes the atmosphere is thin compared with the radius; extended atmospheres need the spherical form.
4. **“Scattering is just another kind of absorption.”** Scattering removes photons from one direction but adds them to others; it makes the equation non-local through the in-scattering integral.

Lecture summary

- Photons are described by $f(\mathbf{r}, \mathbf{p}, t)$; $I_\nu = (h^4\nu^3/c^2) f$.
- Photon conservation (Boltzmann) gives $\frac{1}{c}\partial_t I_\nu + \hat{\mathbf{n}} \cdot \nabla I_\nu$ = gains − losses.
- Gains and losses: absorption (with stimulated emission), spontaneous emission, scattering out, scattering in via $R(\nu', \nu; \hat{\mathbf{n}}', \hat{\mathbf{n}})$.
- Special forms follow from added assumptions: static, ray, plane-parallel, spherical, comoving frame.
- Moments give the radiation energy equation and the radiation force.

Exam questions and answers — Lecture RT6

Q1. Starting from the definition of f, show that $I_\nu = (h^4\nu^3/c^2) f$.

> **A1.** $dE = h\nu f \, d^3x \, d^3p$, with $d^3x = c\,dt\,dA\cos\theta$ and $d^3p = p^2 dp\, d\Omega = (h^3\nu^2/c^3) d\nu\, d\Omega$. So $dE = (h^4\nu^3/c^2) f \cos\theta \, dA\, dt\, d\nu\, d\Omega$; comparing with $dE = I_\nu \cos\theta\, dA\, d\Omega\, d\nu\, dt$ gives the result.

Q2. Write the general time-dependent RTE with coherent isotropic scattering and LTE thermal emission. Name each term.

> **A2.** $\frac{1}{c}\partial_t I_\nu + \hat{\mathbf{n}} \cdot \nabla I_\nu = -\kappa_\nu \rho I_\nu$ (absorption, corrected for stimulated emission) $-\sigma_\nu \rho I_\nu$ (scattering out) $+\kappa_\nu \rho B_\nu$ (thermal emission, Kirchhoff) $+\sigma_\nu \rho J_\nu$ (scattering in, coherent and isotropic). Equivalently $\chi_\nu (S_\nu - I_\nu)$ with $S_\nu = \varepsilon B_\nu + (1-\varepsilon) J_\nu$.

Q3. Derive the spherical form of the static RTE and explain the origin of the $(1-\mu^2)/r\, \partial I/\partial\mu$ term.

> **A3.** Along a straight ray with impact parameter p: $r^2 = p^2 + z^2$, $\mu = z/r$. Then $dr/dz = \mu$ and $d\mu/dz = (1-\mu^2)/r$. So $dI/dz = \mu\,\partial_r I + \frac{1-\mu^2}{r}\partial_\mu I = \chi(S - I)$. The second term exists because a straight ray changes its angle to the local radial direction as it moves; for $r \gg$ the atmosphere thickness it is negligible and we recover $\mu\, dI/dz$.

Q4. Integrate the static RTE over angle and frequency. Show that a static, source-free atmosphere has constant flux, and state what the momentum moment describes.

> **A4.** $\oint d\Omega$ of $\hat{\mathbf{n}} \cdot \nabla I_\nu$ gives $\nabla \cdot \mathbf{F}_\nu$; the right side gives $4\pi(j_\nu - \kappa_\nu \rho J_\nu)$ (scattering terms cancel). With no sources, energy absorbed = energy emitted, so $\int (j_\nu - \kappa_\nu \rho J_\nu) d\nu = 0$ and $\nabla \cdot \mathbf{F} = 0$; in plane-parallel geometry F = constant $= \sigma T_{\text{eff}}^4$. The first moment gives $\nabla \cdot \mathsf{P} = -\frac{1}{c}\int \chi_\nu \mathbf{F}_\nu d\nu$: the momentum transferred from radiation to matter, i.e. the radiation force.

Lecture RT7 — Plane-Parallel Atmospheres: Schwarzschild–Milne Equations and Eddington–Barbier

> **Sources:** Collins §9.2 (plane-parallel RTE, eq. 9.2.24–9.2.26, Fig. 9.3), §9.3 (moments J, H, K); Chandrasekhar Ch. I §§9–11 (the formal solution and the Λ, Φ, X operators); Hu Set 1 (formal solution); MIT 8.901 §7.2.

Learning goals

1. Write the RTE in plane-parallel geometry with τ measured **inward**.
2. Derive the formal solutions for outgoing and incoming intensity and the Schwarzschild–Milne integrals for J, H and K.
3. Prove the Eddington–Barbier relations and use them to invert limb darkening into a temperature structure.

Key concepts in plain words

Plane-parallel atmosphere — a thin skin on a big ball. The solar photosphere is only a few hundred km thick, while the Sun's radius is 700 000 km. Locally the atmosphere looks like a stack of flat, horizontal layers, just as the Earth looks flat when you stand on it. Every layer is uniform horizontally, so quantities depend only on depth.

Optical depth as the vertical coordinate. Instead of measuring depth in kilometres, we count how many photon mean free paths lie between a point and outer space: $\tau = 0$ at the top, increasing inward. This is natural because what we see depends on optical depth, not on kilometres. "$\tau = 1$" means "one mean free path below space": the typical depth from which photons escape.

Slanted rays and μ. A ray leaving at angle θ to the vertical crosses each layer obliquely, so it accumulates optical depth faster: a vertical optical depth τ corresponds to τ/μ along the ray ($\mu = \cos\theta$). Rays near the edge of the stellar disk (the limb) are almost horizontal, so they reach $\tau_{\text{ray}} = 1$ while still very high in the atmosphere.

The emergent intensity is a weighted average over depth. What leaves the surface in direction μ is the source function at all depths, weighted by $e^{-\tau/\mu}$ (the fraction that survives the trip out). Deep layers are hidden; shallow layers are seen. In practice the weight is concentrated within about one optical depth along the ray.

Eddington–Barbier — "we see down to $\tau \approx 1$". The weighted average turns out to be very close to the value of the source function at vertical depth $\tau = \mu$. At the centre of the disk ($\mu = 1$) we see down to $\tau = 1$; near the limb ($\mu \to 0$) we see only the top layers. The total flux comes from about $\tau = 2/3$. This is the single most useful rule for interpreting stellar spectra.

Limb darkening is a depth probe. Because the limb shows higher layers, and in the Sun the temperature decreases outward, the limb looks darker and redder. Measuring how the brightness changes from centre to limb therefore maps the temperature as a function of depth, without ever going there.

The Λ operator — "what light does this source produce?". If you know the source function everywhere, the operator Λ tells you the mean intensity J at any depth. It is the mathematical form of "light at one place comes from

everywhere else", which is why radiative transfer is non-local.

7.1 Geometry and sign conventions

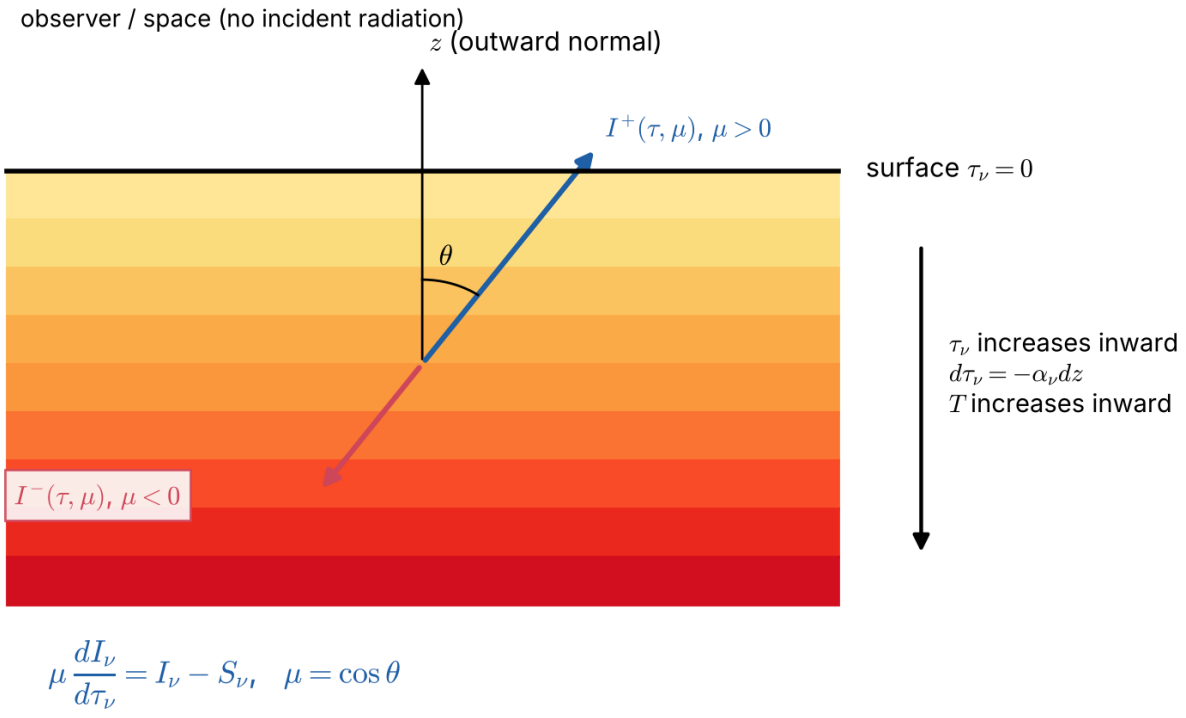


*Figure 7.1 — A plane-parallel atmosphere. z points outward, θ is measured from the outward normal, and $\mu=\cos\theta$. Outgoing rays have $\mu>0$, incoming rays $\mu<0$. The optical depth increases **inward**: $d\tau_\nu=-\alpha_\nu dz$.*

Along a ray, $ds = dz/\mu$. Then $dI_\nu/ds = \mu\, dI_\nu/dz = -\mu\,\alpha_\nu\, dI_\nu/d\tau_\nu$, and the RTE becomes (Collins eq. 9.2.26)

$$\boxed{\mu\,\frac{dI_\nu(\tau_\nu,\mu)}{d\tau_\nu} = I_\nu(\tau_\nu,\mu) - S_\nu(\tau_\nu)}$$

Note the sign change compared with Lecture RT5, because τ is now measured against the outgoing direction.

Boundary conditions for a semi-infinite stellar atmosphere:

- no incident radiation at the top: $I_\nu(0,\mu<0)=0$;
- the intensity does not grow faster than e^{τ} at depth: $\lim_{\tau\to\infty} I_\nu e^{-\tau/\mu}=0$.

7.2 Formal solution in a semi-infinite atmosphere

Multiply by $e^{-\tau/\mu}/\mu$: $\dfrac{d}{d\tau}\left(I e^{-\tau/\mu}\right) = -\dfrac{S}{\mu}e^{-\tau/\mu}$.

Outgoing ($0<\mu\le 1$), integrating from τ to ∞:

$$I_\nu^+(\tau,\mu) = \int_\tau^\infty S_\nu(t)\, e^{-(t-\tau)/\mu}\,\frac{dt}{\mu}.$$

Incoming ($\mu < 0$; write $\mu = -|\mu|$), integrating from 0 to τ with $I(0) = 0$:

$$I_\nu^-(\tau, \mu) = \int_0^\tau S_\nu(t)\, e^{-(\tau - t)/|\mu|}\, \frac{dt}{|\mu|}.$$

Emergent intensity (what we observe at disk position μ):

$$\boxed{I_\nu(0, \mu) = \int_0^\infty S_\nu(t)\, e^{-t/\mu}\, \frac{dt}{\mu}}$$

This is a **Laplace transform** of $S_\nu(\tau)$. Measuring $I_\nu(0, \mu)$ across the disk of the Sun (μ from 1 at centre to 0 at the limb) therefore probes $S_\nu(\tau)$ at different depths.

7.3 Exponential integrals and the Schwarzschild–Milne equations

$$E_n(x) = \int_1^\infty \frac{e^{-xt}}{t^n}\, dt = \int_0^1 e^{-x/\mu} \mu^{n-2}\, d\mu, \qquad \frac{dE_n}{dx} = -E_{n-1}, \quad E_n(0) = \frac{1}{n-1}\ (n > 1).$$

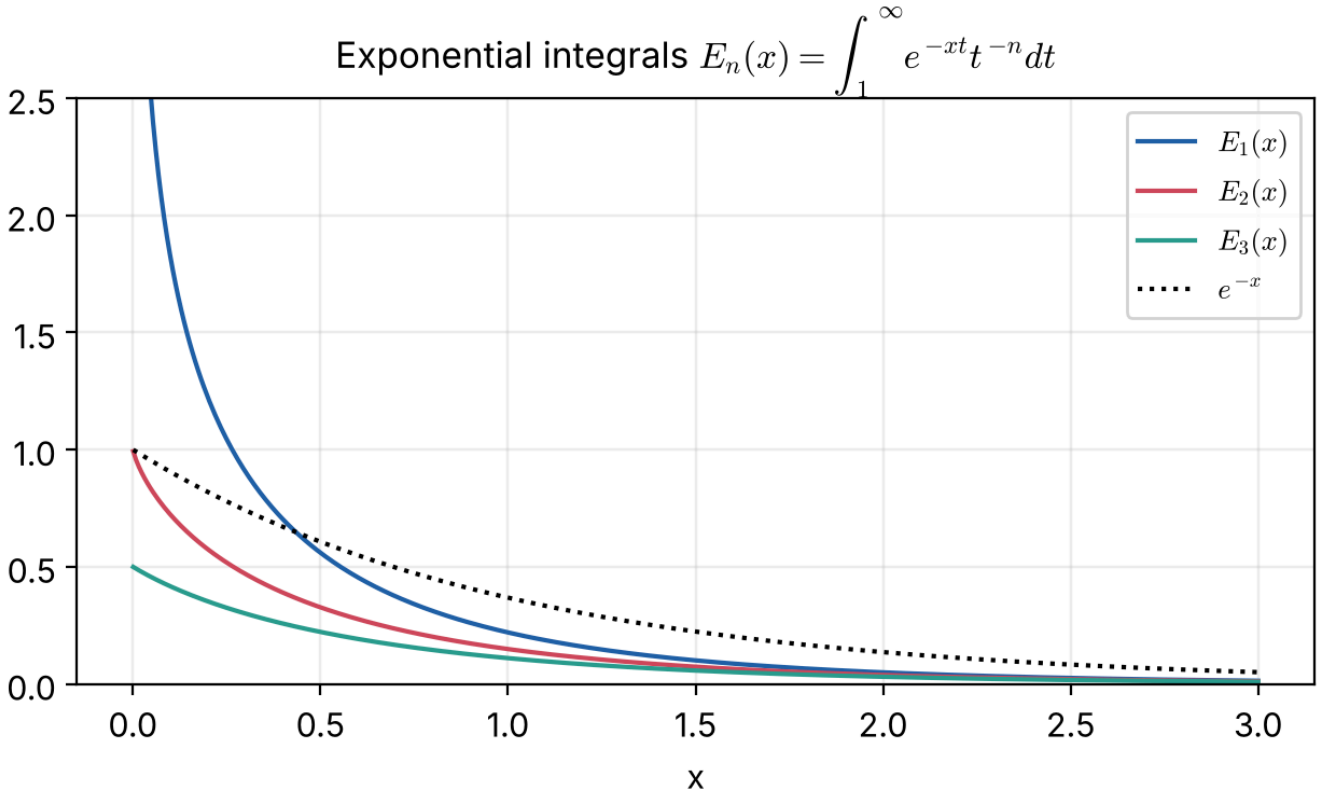


Figure 7.2 — E_1 diverges logarithmically at $x = 0$ (but is integrable); all E_n decay roughly like e^{-x}/x.

Insert the formal solution into the moment definitions and swap the order of integration:

$$J_\nu(\tau) = \tfrac{1}{2} \int_0^\infty S_\nu(t)\, E_1|t - \tau|\, dt \;\equiv\; \Lambda_\tau[S_\nu]$$

$$F_\nu(\tau) = 2\pi \left[\int_\tau^\infty S_\nu(t) E_2(t - \tau)\, dt - \int_0^\tau S_\nu(t) E_2(\tau - t)\, dt \right] \;\equiv\; \Phi_\tau[S_\nu]$$

$$K_\nu(\tau) = \tfrac{1}{2}\int_0^\infty S_\nu(t)\, E_3|t-\tau|\, dt \;\equiv\; X_\tau[S_\nu]$$

These are the **Schwarzschild–Milne equations**, and Λ is the **lambda operator** of Chandrasekhar.

Example: S constant. $J(0) = \frac{1}{2}S\int_0^\infty E_1(t)dt = \frac{1}{2}S\,E_2(0) = S/2$. At the surface only half the sky is filled, so $J = S/2$. Deep inside, $J \to S$.

7.4 The Eddington–Barbier relations

Expand the source function near the surface as $S_\nu(\tau) = a + b\tau$. Then

$$I_\nu(0,\mu) = \int_0^\infty (a+bt)e^{-t/\mu}\frac{dt}{\mu} = a + b\mu,$$

using $\int_0^\infty e^{-t/\mu}dt/\mu = 1$ and $\int_0^\infty te^{-t/\mu}dt/\mu = \mu$. So

$$\boxed{I_\nu(0,\mu) = S_\nu(\tau_\nu = \mu)}$$

exactly for a linear S, and approximately otherwise. For the surface flux:

$$F_\nu(0) = 2\pi\int_0^1 (a+b\mu)\mu\, d\mu = \pi\left(a + \tfrac{2}{3}b\right) \;\Rightarrow\; \boxed{F_\nu(0) = \pi S_\nu(\tau_\nu = 2/3)}$$

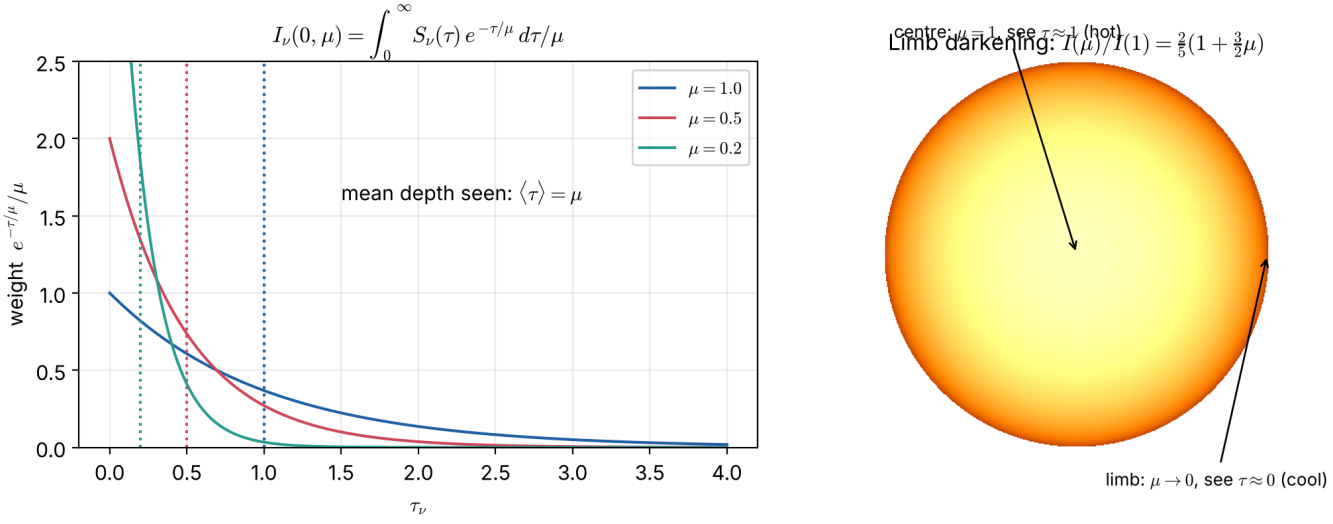


Figure 7.3 — Left: the weighting kernel $e^{-\tau/\mu}/\mu$ peaks at the surface but has mean depth $\langle\tau\rangle = \mu$. Right: limb darkening. At disk centre we see down to $\tau \approx 1$ (hot); at the limb, rays graze the atmosphere and we see $\tau \approx 0$ (cool).

Physical meaning:

- "We see down to $\tau \approx 1$ along the line of sight," i.e. to vertical depth $\tau \approx \mu$.
- Toward the limb we see higher, cooler layers. If S increases inward ($b > 0$), the limb is **darker**.
- The observed flux of a star reflects the conditions at $\tau \approx 2/3$, which is why the "photosphere" is defined there and why $T(\tau = 2/3) \approx T_{\rm eff}$ (Lecture SA5).

7.5 From limb darkening to temperature: an LTE inversion

In LTE, $S_\nu = B_\nu(T)$. Measuring $I_\nu(0,\mu)$ gives $S_\nu(\tau_\nu = \mu)$ by Eddington–Barbier, and inverting the Planck function then gives $T(\tau_\nu)$. This is how the first empirical models of the solar photosphere were built.

Worked example 7.1 — The temperature of the solar photosphere from limb darkening

At $\lambda = 500$ nm the solar limb darkening is approximately $I(\mu)/I(1) = 0.36 + 0.64\mu$, and the disk-centre intensity is $I(1) \approx 4.0 \times 10^{13}$ W m^{-3} sr^{-1}. Assume LTE. Find T at $\tau_{500} = 0,\ 0.5$ and 1.

Step 1 (Eddington–Barbier): $S(\tau) = I(1)\,(0.36 + 0.64\tau)$, so $S(0) = 0.36 I(1) = 1.44 \times 10^{13}$, $S(0.5) = 0.68 I(1) = 2.72 \times 10^{13}$, $S(1) = 4.0 \times 10^{13}$.

Step 2 (LTE): $S = B_\lambda(T)$, so $T = \dfrac{hc/\lambda k}{\ln(1 + 2hc^2/\lambda^5 S)}$ with $hc/\lambda k = 28\,776$ K and $2hc^2/\lambda^5 = 3.811 \times 10^{15}$:

τ_{500}	$S/I(1)$	T [K]
0	0.36	5155
0.5	0.68	5814
1	1.00	6300

Conclusion: the temperature **increases inward**, by roughly 1100 K over one optical depth. This is direct observational proof that the Sun is not isothermal, and it explains why the solar spectrum has absorption lines.

Worked example 7.2 — Emergent intensity for a quadratic source function

$S(\tau) = a + b\tau + c\tau^2$. Find $I(0,\mu)$ and the error of Eddington–Barbier.

$\int_0^\infty t^2 e^{-t/\mu} dt/\mu = 2\mu^2$, so $I(0,\mu) = a + b\mu + 2c\mu^2$, whereas $S(\mu) = a + b\mu + c\mu^2$. The error is $c\mu^2$, i.e. Eddington–Barbier is exact only for linear S.

Worked example 7.3 — Mean intensity at the surface for linear S

$S = a + b\tau$. Compute $J(0)$ and $H(0)$, and show that $J(0) \neq S(0)$.

$I^+(0,\mu) = a + b\mu$ and $I^-(0) = 0$, so $J(0) = \tfrac{1}{2}\int_0^1 (a + b\mu) d\mu = \tfrac{a}{2} + \tfrac{b}{4}$ and $H(0) = \tfrac{1}{2}\int_0^1 (a + b\mu)\mu\, d\mu = \tfrac{a}{4} + \tfrac{b}{6}$. With $b = 0$: $J(0) = a/2 = S(0)/2$. **The radiation field at the surface is not in equilibrium with the local source function**, a signature of photon escape.

Common misconceptions

1. **“Optical depth increases outward.”** In stellar atmospheres it is measured *inward* from the surface.

2. **“The limb is darker because it is farther away.”** The difference in distance is negligible; the limb is darker because we see higher, cooler layers there.
3. **“Eddington–Barbier is exact.”** It is exact only for a source function linear in τ.
4. **“The photosphere is at the same geometric height at all wavelengths.”** $\tau_\nu = 2/3$ occurs at different depths for different frequencies; line cores form higher than the continuum.

Lecture summary

- Plane-parallel RTE: $\mu\, dI/d\tau = I - S$, with τ inward and $\mu = \cos\theta$.
- Emergent intensity: $I(0,\mu) = \int_0^\infty S\, e^{-\tau/\mu} d\tau/\mu$ (a Laplace transform of S).
- Schwarzschild–Milne: $J = \Lambda[S]$ with the kernel $\frac{1}{2}E_1$; flux with E_2; K with E_3.
- Eddington–Barbier: $I(0,\mu) \approx S(\mu)$, $F \approx \pi S(2/3)$; limb darkening reveals $T(\tau)$.

Exam questions and answers — Lecture RT7

Q1. Derive the formal solution for the emergent intensity of a semi-infinite atmosphere.

> **A1.** $\frac{d}{d\tau}(Ie^{-\tau/\mu}) = -\frac{S}{\mu}e^{-\tau/\mu}$. Integrate from 0 to ∞ with $Ie^{-\tau/\mu} \to 0$: $-I(0,\mu) = -\int_0^\infty Se^{-t/\mu}dt/\mu$, so $I(0,\mu) = \int_0^\infty S(t)e^{-t/\mu}dt/\mu$.

Q2. Prove the Eddington–Barbier relations $I(0,\mu) = S(\tau=\mu)$ and $F(0) = \pi S(\tau = 2/3)$ for $S = a + b\tau$.

> **A2.** See §7.4: $I(0,\mu) = a + b\mu = S(\mu)$; $F = 2\pi\int_0^1 (a+b\mu)\mu d\mu = 2\pi(a/2 + b/3) = \pi(a + 2b/3) = \pi S(2/3)$.

Q3. Explain limb darkening physically. Under what condition would we see limb **brightening**?

> **A3.** At disk centre the line of sight is perpendicular to the layers, so $\tau = 1$ is reached at vertical depth $\tau_{\rm vert} = 1$. Near the limb the ray is oblique, and $\tau = 1$ along it occurs at $\tau_{\rm vert} = \mu$, in higher layers. If S (in LTE, T) decreases outward, the limb is darker. If S increases outward, as in the chromosphere at UV, EUV and radio wavelengths, the limb is brighter.

Q4. The solar limb darkening at some wavelength is $I(\mu)/I(1) = 0.3 + 0.7\mu$. In LTE and using Eddington–Barbier, at what optical depth does S equal half its value at $\tau = 1$?

> **A4.** $S(\tau)/S(1) = 0.3 + 0.7\tau = 0.5 \Rightarrow \tau = 0.29$.

Lecture RT8 — Moment Equations, the Diffusion and Eddington Approximations, and Scattering Atmospheres

> **Sources:** Collins §9.3–9.4 (moments; radiative equilibrium eq. 9.4.3–9.4.5; diffusion approximation eq. 9.4.7–9.4.9; Eddington approximation eq. 9.4.12–9.4.13); Hu Set 1 (scattering, multiple processes, single-scattering albedo); Chandrasekhar Ch. I §§11–14 and Ch. III (isotropic scattering).

Learning goals

1. Derive the zeroth and first moment equations and the condition of radiative equilibrium.
2. Close the moment hierarchy with the Eddington approximation and its surface boundary condition.
3. Derive the diffusion approximation and the Rosseland mean opacity.
4. Solve an isothermal atmosphere with scattering, obtain the $\sqrt{\varepsilon}$ law and the thermalisation depth, and understand why Λ-iteration fails.

Key concepts in plain words

Moments trade detail for simplicity. Instead of following $I(\mu)$ in every direction, we follow a few averages (J, H, K). The price is that each equation for one moment involves the next one, so we always have one equation too few. This is the **closure problem**, exactly as in fluid dynamics, where the momentum equation needs an equation of state to close it. In RT, the closure is an assumption about the angular shape of the radiation, such as $K = J/3$.

Radiative equilibrium — every parcel is a perfect thermostat. In a static atmosphere with no energy sources, each small piece of gas must emit exactly as much energy as it absorbs; otherwise it would heat up or cool down. The equation $\int \kappa_\nu (J_\nu - B_\nu) d\nu = 0$ says precisely that. It is this condition that **fixes the temperature** at each depth. Scattering drops out because it only redirects photons and neither gives energy to the gas nor takes it.

Diffusion — radiation behaves like heat conduction deep inside. Deep in a star, photons are absorbed after very short distances and the radiation is almost the same in every direction, with a tiny excess flowing outward. The net flow is then proportional to the temperature gradient, $F \propto -dT/dz$, just like heat flowing down a metal rod. The "conductivity" is inversely proportional to the opacity.

Rosseland mean — traffic takes the easy lanes. When the opacity varies with frequency, the energy flows mainly through the frequencies where the gas is most transparent, as cars on a motorway crowd into the fastest lanes. Mathematically this is a harmonic mean, like adding electrical resistors in parallel: the low-resistance (low-opacity) paths dominate.

The Eddington approximation. Eddington assumed that the radiation is "almost isotropic" everywhere, so $K = J/3$ even near the surface, and that the outgoing light at the surface is uniform over angles, so $J(0) = 2H(0)$. Both assumptions are a little wrong, but their errors partly cancel, and the results are usually accurate to a few per cent.

Thermalisation — how long a photon survives. With scattering, a photon may bounce many times before being truly absorbed. If the destruction probability per event is ε, a photon survives about $1/\varepsilon$ scatterings, during which it wanders about $1/\sqrt{\varepsilon}$ optical depths. Only below this **thermalisation depth** does the gas "lock" the radiation to its own

temperature ($S \to B$). Above it, photons leak out, J falls below B, and so does S, all the way to $S(0) = \sqrt{\varepsilon} B$ at the surface. A useful image is water in a sieve: the deeper the water, the longer it stays.

Why simple iteration fails. Λ-iteration is like passing a message down a long line of people, where each person only talks to their neighbours: the news that "the surface is leaking" travels only one optical depth per iteration. With a thermalisation depth of 1000, you need about a million iterations. Modern methods (direct solution, ALI) send the message along the whole line at once.

8.1 Moments of the RTE

Take $\mu\, dI_\nu/d\tau_\nu = I_\nu - S_\nu$ (S_ν isotropic) and apply $\frac{1}{2}\int_{-1}^{1} d\mu$ and $\frac{1}{2}\int_{-1}^{1} \mu\, d\mu$:

$$\boxed{\frac{dH_\nu}{d\tau_\nu} = J_\nu - S_\nu} \qquad\qquad \boxed{\frac{dK_\nu}{d\tau_\nu} = H_\nu}$$

Each equation brings in a higher moment ($J \to H \to K \to \ldots$): the **closure problem**, as in the hydrodynamic moments of the Boltzmann equation (Collins §9.4).

Radiative equilibrium

If radiation carries all the energy and there are no sources, the total flux is constant: $dF/dz = 0$ (Collins eq. 9.4.5). Integrating the zeroth moment over frequency:

$$\frac{dH}{d\tau} \;\propto\; \int_0^\infty \alpha_\nu \left(J_\nu - S_\nu\right) d\nu = 0.$$

In LTE ($S_\nu = B_\nu$ plus coherent scattering, which cancels out because scattering conserves energy):

$$\boxed{\int_0^\infty \kappa_\nu\, J_\nu\, d\nu = \int_0^\infty \kappa_\nu\, B_\nu(T)\, d\nu} \qquad \text{(energy absorbed = energy emitted)}$$

This condition **determines** $T(\tau)$. It is the extra equation that turns the RTE into a model atmosphere.

8.2 The diffusion approximation (deep interior)

At large depth, $S_\nu = B_\nu$ varies slowly on the scale of a mean free path. Expanding $B_\nu(t)$ in a Taylor series about τ in the formal solution gives

$$I_\nu(\tau, \mu) = B_\nu + \mu \frac{dB_\nu}{d\tau_\nu} + \mu^2 \frac{d^2 B_\nu}{d\tau_\nu^2} + \ldots$$

The radiation is almost isotropic with a small linear anisotropy. Then (cf. Lecture RT2, Worked example 2.3)

$$J_\nu \approx B_\nu, \qquad H_\nu = \frac{1}{3}\frac{dB_\nu}{d\tau_\nu}, \qquad K_\nu = \frac{B_\nu}{3} \approx \frac{J_\nu}{3}.$$

The flux is therefore

$$F_\nu = 4\pi H_\nu = \frac{4\pi}{3}\frac{dB_\nu}{d\tau_\nu} = -\frac{4\pi}{3}\frac{1}{\alpha_\nu}\frac{\partial B_\nu}{\partial T}\frac{dT}{dz}.$$

Radiation **diffuses** down the temperature gradient, like heat conduction.

Rosseland mean opacity

Integrate over frequency:

$$F = -\frac{4\pi}{3}\frac{dT}{dz}\int_0^\infty \frac{1}{\alpha_\nu}\frac{\partial B_\nu}{\partial T}d\nu \equiv -\frac{4\pi}{3}\frac{1}{\alpha_R}\frac{dB}{dT}\frac{dT}{dz},$$

which defines the **Rosseland mean**

$$\boxed{\frac{1}{\alpha_R} = \frac{\displaystyle\int_0^\infty \frac{1}{\alpha_\nu}\frac{\partial B_\nu}{\partial T}d\nu}{\displaystyle\int_0^\infty \frac{\partial B_\nu}{\partial T}d\nu}} \qquad \Rightarrow \qquad \boxed{F = -\frac{16\sigma T^3}{3\kappa_R \rho}\frac{dT}{dz}}$$

using $dB/dT = 4\sigma T^3/\pi$. It is a **harmonic** mean: energy flows through the frequency "windows" where the opacity is low (Hu: "energy escapes in the channel with the lowest opacity"). This is the radiative-transfer equation of **stellar interiors**.

8.3 The Eddington approximation (whole atmosphere)

Eddington assumed $K_\nu = J_\nu/3$ **everywhere**, not only at depth (Collins eq. 9.4.8). At the surface we also need a boundary condition. Assume the outgoing intensity is isotropic, $I(0,\mu>0) = I_0$, with $I(0,\mu<0) = 0$. Then $J(0) = I_0/2$ and $H(0) = I_0/4$ (Lecture RT2, Q3), so

$$\boxed{J(0) = 2H(0)} \qquad \text{(Eddington boundary condition; Collins eq. 9.4.13)}$$

The two errors partly cancel, and the approximation is typically good to about 10 % (Collins), and to a few per cent for the temperature of a gray atmosphere (Lecture SA5: $T(0)$ is $0.841\,T_{\rm eff}$ versus the exact $0.811\,T_{\rm eff}$).

Combined, the moment equations become a **second-order ODE**:

$$\frac{1}{3}\frac{d^2 J_\nu}{d\tau_\nu^2} = J_\nu - S_\nu.$$

8.4 Atmosphere with scattering: thermalisation

Take the source function of Lecture RT4,

$$S = \varepsilon B + (1-\varepsilon)J, \qquad J - S = \varepsilon(J - B),$$

in an **isothermal** atmosphere (B = constant), with ε constant.

Eddington solution

$\frac{1}{3}J'' = \varepsilon(J - B)$, so $J = B + Ce^{-\sqrt{3\varepsilon}\,\tau}$ (the growing solution is excluded). The boundary condition $J(0) = 2H(0) = \frac{2}{3}J'(0)$ gives $B + C = -\frac{2}{3}\sqrt{3\varepsilon}\,C$, hence

$$J(\tau) = B\left[1 - \frac{e^{-\sqrt{3\varepsilon}\,\tau}}{1 + 2\sqrt{\varepsilon/3}}\right], \qquad S(0) = B\left[\varepsilon + (1-\varepsilon)\frac{2\sqrt{\varepsilon/3}}{1 + 2\sqrt{\varepsilon/3}}\right] \approx \frac{2}{\sqrt{3}}\sqrt{\varepsilon}\,B = 1.15\sqrt{\varepsilon}\,I$$

Exact result

The exact solution of the integral equation $S = \varepsilon B + (1-\varepsilon)\Lambda[S]$ (Chandrasekhar's H-function theory) gives the famous

$$\boxed{S(0) = \sqrt{\varepsilon}\,B}$$

This is one of the most important results in radiative transfer.

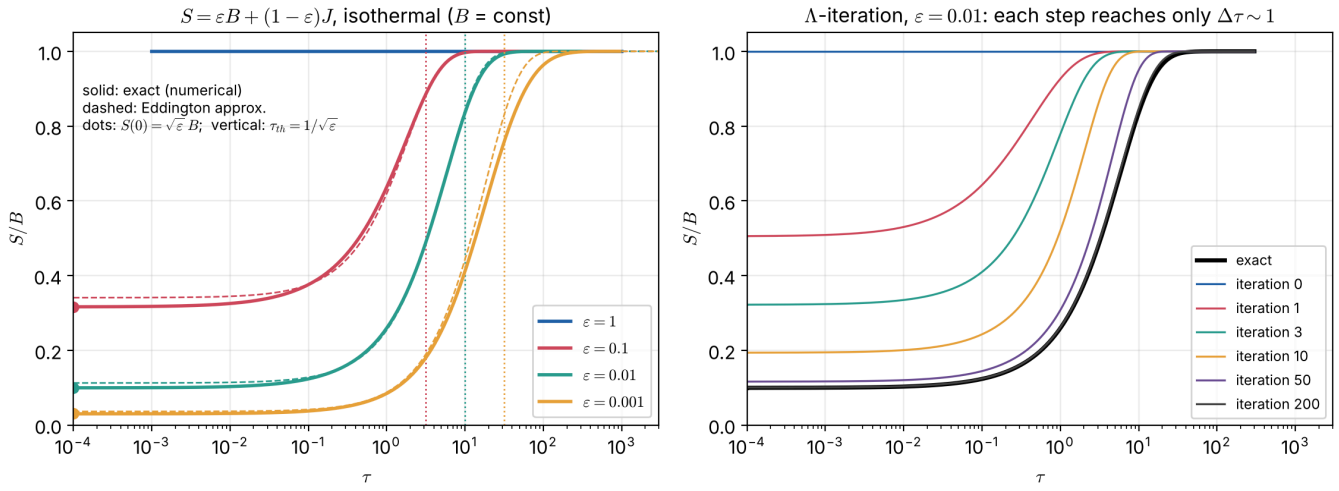


Figure 8.1 — Left: exact numerical solutions (solid; computed with `scripts/lte_solvers.py`) and the Eddington approximation (dashed) for an isothermal atmosphere with $S = \varepsilon B + (1-\varepsilon)J$. Even though B is constant, S drops far below B near the surface, and $S \to B$ only below the thermalisation depth $\tau_{\rm th} \sim 1/\sqrt{\varepsilon}$ (dotted lines). Right: Λ-iteration for $\varepsilon = 0.01$. Each iteration propagates information only about one optical depth; after 200 iterations it is still not converged.

Interpretation.

- A photon is destroyed with probability ε per interaction, so it survives $\sim 1/\varepsilon$ scatterings and random-walks a distance $\Delta\tau \sim \sqrt{1/\varepsilon}$. That is the **thermalisation length**.
- Within $\tau_{\rm th}$ of the surface, photons "know" about the boundary and leak out. J drops below B, so $S < B$: **NLTE behaviour even when the gas is isothermal and in LTE populations.**
- For strong resonance lines $\varepsilon \sim 10^{-4}$–10^{-8}, so $\tau_{\rm th} \sim 10^{2}$–$10^{4}$. Thermalisation happens very deep, and line cores are much darker than LTE predicts.

Why Λ-iteration fails

The obvious scheme $S^{(n+1)} = \varepsilon B + (1-\varepsilon)\Lambda[S^{(n)}]$ converges at rate $(1-\varepsilon)$ per step. Each step only couples points within $\Delta\tau \sim 1$. For $\varepsilon \ll 1$ it can appear converged (tiny changes) while still being badly wrong (Figure 8.1,

right). Modern codes use **accelerated Λ-iteration (ALI)**, or solve the linear system directly, as our script does: $(\mathbf{1} - (1-\varepsilon)\mathbf{\Lambda})\mathbf{S} = \varepsilon\mathbf{B}$.

Numerical solution: how the script works

1. Choose a grid $\tau_0 = 0 < \tau_1 < \cdots < \tau_N$ (logarithmic).
2. Represent $S(\tau)$ as piecewise linear between grid points.
3. Integrate $J_i = \frac{1}{2}\int S(t) E_1|t-\tau_i|dt$ analytically on each segment, using $\int E_1 = -E_2$ and $\int x E_1\,dx = -xE_2 - E_3$. This gives a matrix Λ_{ik} with $J_i = \sum_k \Lambda_{ik} S_k$.
4. Solve the linear system for S_k.

The results reproduce $S(0)/B = 0.316,\ 0.099$ for $\varepsilon = 0.1,\ 0.01$, against exact values of 0.316 and 0.100.

Worked example 8.1 — Rosseland mean of a two-band opacity

Suppose the weight $\partial B_\nu/\partial T$ is split equally between two bands with opacities $\kappa_1 = 1$ and $\kappa_2 = 100$ (arbitrary units). Compare the Rosseland mean with the straight (Planck-like) mean.

$$\frac{1}{\kappa_R} = \frac{1}{2}\left(\frac{1}{1} + \frac{1}{100}\right) = 0.505 \;\Rightarrow\; \kappa_R = 1.98; \qquad \kappa_{\text{Planck}} = \tfrac{1}{2}(1+100) = 50.5.$$

The flux flows through the transparent band. Adding many strong lines (line blanketing) raises κ_R only modestly, unless they fill the windows.

Worked example 8.2 — Thermalisation of a resonance line

A resonance line in a stellar chromosphere has $\varepsilon = 10^{-4}$. Estimate (a) $S(0)/B$, (b) the thermalisation depth, (c) the number of scatterings before destruction.

a. $S(0) = \sqrt{\varepsilon}\,B = 0.01\,B$. (b) $\tau_{\text{th}} \approx 1/\sqrt{\varepsilon} = 100$ (Eddington: $1/\sqrt{3\varepsilon} = 58$). (c) $N \approx 1/\varepsilon = 10^4$. **Consequence:** an LTE calculation ($S = B$) would predict a line core 100 times brighter than the NLTE one. This is why strong lines like Ca II H & K and Mg II h & k need NLTE treatment.

Worked example 8.3 — Diffusion flux and the solar interior

At $r \approx 0.5 R_\odot$ in the Sun, $T \approx 3.4\times10^6$ K, $\rho \approx 1.3\times10^3$ kg m^{-3} and $\kappa_R \approx 1$ m^2 kg^{-1}, and the luminosity passing through is $L_\odot$. Find dT/dr.

$$F = L_\odot/(4\pi r^2) = 3.828\times10^{26}/(4\pi(3.48\times10^8)^2) = 2.5\times10^8\ \text{W m}^{-2}.$$

$$\left|\frac{dT}{dr}\right| = \frac{3\kappa_R \rho F}{16\sigma T^3} = \frac{3\times1\times1.3\times10^3\times2.5\times10^8}{16\times5.67\times10^{-8}\times(3.4\times10^6)^3} = 2.7\times10^{-2}\ \text{K m}^{-1}.$$

That is about 27 K per km, or roughly $T/R_\odot \times 5$: the right order of magnitude for the solar interior.

Common misconceptions

1. **“Scattering heats the gas.”** Coherent scattering conserves photon energy; it drops out of radiative equilibrium.
2. **“The Rosseland mean is an ordinary average of the opacity.”** It is a weighted *harmonic* mean, dominated by the most transparent frequencies.
3. **“An isothermal atmosphere has $S = B$ everywhere.”** With scattering, S falls to $\sqrt{\varepsilon}B$ at the surface even though T is constant.
4. **“If the solution changes very little between iterations, it has converged.”** In Λ-iteration tiny changes per step can hide large remaining errors.

Lecture summary

- Moments: $dH/d\tau = J - S$, $dK/d\tau = H$; a closure ($K = J/3$) is needed.
- Radiative equilibrium $\int \kappa_\nu (J_\nu - B_\nu) d\nu = 0$ fixes $T(\tau)$; scattering drops out.
- Deep inside: diffusion, $F = -\frac{16\sigma T^3}{3\kappa_R \rho}\frac{dT}{dz}$ with the Rosseland mean.
- Scattering atmospheres: $S(0) = \sqrt{\varepsilon}B$, thermalisation depth $\sim 1/\sqrt{\varepsilon}$; Λ-iteration converges slowly.

Exam questions and answers — Lecture RT8

Q1. State radiative equilibrium in integral form. Why do scattering terms drop out?

> **A1.** $\int \alpha_\nu (J_\nu - S_\nu) d\nu = 0$, equivalently $dF/dz = 0$. For coherent scattering the scattering part of $\alpha_\nu (J_\nu - S_\nu)$ is $\alpha_\nu^{\text{sc}}(J_\nu - J_\nu) = 0$: scattering redirects photons but neither creates nor destroys energy (Collins §9.4). The condition becomes $\int \kappa_\nu^a (J_\nu - B_\nu) d\nu = 0$.

Q2. Derive the diffusion approximation and the Rosseland mean opacity. Why is it a harmonic mean?

> **A2.** See §8.2. Deep inside, $I \approx B + \mu\, dB/d\tau$, so $F_\nu = \frac{4\pi}{3}\frac{dB_\nu}{d\tau_\nu} = -\frac{4\pi}{3\alpha_\nu}\frac{\partial B_\nu}{\partial T}\frac{dT}{dz}$. Summing the fluxes over frequencies adds the $1/\alpha_\nu$ (“conductances”, like resistors in parallel), hence the harmonic mean weighted by $\partial B_\nu/\partial T$.

Q3. For an isothermal atmosphere with $S = \varepsilon B + (1-\varepsilon)J$, solve in the Eddington approximation for $J(\tau)$. Explain why $S \neq B$ near the surface although the atmosphere is isothermal.

> **A3.** $\frac{1}{3}J'' = \varepsilon(J - B) \Rightarrow J = B + Ce^{-\sqrt{3\varepsilon}\tau}$. With $J(0) = \frac{2}{3}J'(0)$: $C = -B/(1 + 2\sqrt{\varepsilon/3})$. Near the surface photons escape, so $J < B$; scattering makes S depend on J, so S is also less than B. Only below $\tau \sim 1/\sqrt{\varepsilon}$ do photons get destroyed (thermalised) before they can escape, and there $S \to B$.

Q4. Why does Λ-iteration converge so slowly when $\varepsilon \ll 1$? Name a better approach.

A4. The Λ operator couples points only over $\Delta\tau \sim 1$, so each iteration spreads information about the boundary by about one optical depth. The thermalisation depth is $\sim 1/\sqrt{\varepsilon}$, and the error decays as $(1-\varepsilon)^n$, so $\gtrsim 1/\varepsilon$ iterations are needed. Remedies: solve the linear system directly (Feautrier method, or matrix inversion as in our script), or use accelerated Λ-iteration (ALI) with an approximate local operator.

Lecture RT9 — Polarized Light and Polarized Radiative Transfer

> **Sources:** Hu, *Astro 305 Set 1* (Stokes parameters as differences of filtered intensities); Chandrasekhar, *Radiative Transfer* (1950), Ch. I §15–17 (Stokes parameters, Rayleigh phase matrix) and Ch. X §68 (exact laws of darkening in the two states of polarization; Table XXIV); Landi Degl'Innocenti & Landolfi, *Polarization in Spectral Lines* (2004); Stenflo, *Solar Magnetic Fields* (1994).

Learning goals

1. Define the Stokes parameters I, Q, U, V and the degree and angle of polarization.
2. Write the polarized transfer equation in matrix form and name its terms.
3. Explain polarization by scattering (Thomson, Rayleigh) and Chandrasekhar's result for electron-scattering atmospheres.
4. Explain how the Zeeman and Hanle effects let us measure magnetic fields, and estimate Zeeman splittings.

Key concepts in plain words

Light is a transverse wave. The electric field oscillates perpendicular to the direction of travel. If its oscillation has a preferred orientation, the light is *polarized*. Natural thermal emission is unpolarized because countless independent emitters oscillate in random directions. Polarization appears when something breaks the symmetry: a scattering geometry, a magnetic field, aligned dust grains, or relativistic electrons spiralling in a field.

Four numbers describe any beam. Hu's picture is the clearest: put filters in front of a detector. I is the total intensity. Q is the difference between light passing a horizontal and a vertical polarizer. U is the same difference for polarizers rotated by 45°. V is the difference between right- and left-circular filters. These four Stokes parameters all have units of intensity, so they can be transported exactly like I_ν.

Why scattering polarizes. An electron hit by light oscillates along the incoming electric field and re-radiates like a small antenna, which emits nothing along its axis of oscillation. Light scattered at 90° therefore contains only one direction of oscillation: it is 100 % linearly polarized. This is why the blue sky is polarized and why the limb of a hot star should be slightly polarized.

Magnetic fields leave fingerprints. In a magnetic field, atomic levels split (the Zeeman effect) into components with definite polarization. A field along the line of sight produces *circular* polarization (V); a field across the line of sight produces *linear* polarization (Q, U). Measuring the Stokes profiles of spectral lines is how solar magnetograms are made. Very weak fields instead modify scattering polarization (the Hanle effect).

9.1 Stokes parameters

For a monochromatic wave $E_x = a_1\cos(\omega t - \delta_1)$, $E_y = a_2\cos(\omega t - \delta_2)$, with $\delta = \delta_2 - \delta_1$ (up to a common constant):

$$I = a_1^2 + a_2^2, \quad Q = a_1^2 - a_2^2, \quad U = 2a_1a_2\cos\delta, \quad V = 2a_1a_2\sin\delta.$$

Operationally (Hu): $I = I_1 + I_2, Q = I_1 - I_2, U = I_{1'} - I_{2'}$ (axes rotated by 45°), $V = I_+ - I_-$.

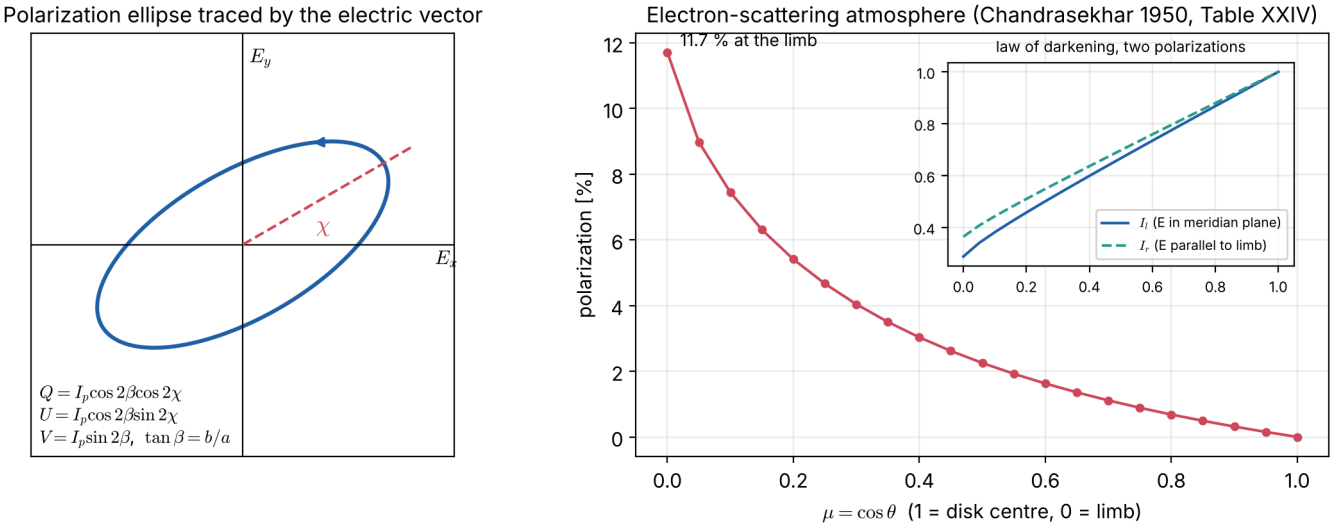


Figure 9.1 — Left: the electric vector traces an ellipse with orientation χ and ellipticity $\tan\beta = b/a$. Right: Chandrasekhar's exact result for a semi-infinite electron-scattering atmosphere. The polarization rises from 0 at disk centre to 11.7 % at the limb (data: Chandrasekhar 1950, Table XXIV). Inset: the limb darkening differs for the two polarization states.

Quantity	Definition
degree of polarization	$p = \sqrt{Q^2+U^2+V^2}\,/\,I \le 1$
degree of linear polarization	$p_L = \sqrt{Q^2+U^2}\,/\,I$
polarization angle	$\chi = \tfrac{1}{2}\arctan(U/Q)$
circular polarization	$p_V = V/I$
fully polarized light	$I^2 = Q^2+U^2+V^2$

Rotation of axes by an angle ϕ: $Q' = Q\cos 2\phi + U\sin 2\phi$, $U' = -Q\sin 2\phi + U\cos 2\phi$, with I and V unchanged. The factor 2ϕ means linear polarization is a "spin-2" quantity, the basis of the E- and B-mode decomposition of CMB polarization.

9.2 The polarized transfer equation

Write the Stokes vector $\mathbf{I} = (I, Q, U, V)^T$. The transfer equation becomes a matrix equation:

$$\boxed{\frac{d\mathbf{I}_\nu}{ds} = -\mathsf{K}_\nu\,\mathbf{I}_\nu + \boldsymbol{\epsilon}_\nu} \qquad \mathsf{K} = \begin{pmatrix} \eta_I & \eta_Q & \eta_U & \eta_V \\ \eta_Q & \eta_I & \rho_V & -\rho_U \\ \eta_U & -\rho_V & \eta_I & \rho_Q \\ \eta_V & \rho_U & -\rho_Q & \eta_I \end{pmatrix}$$

- η_I: ordinary absorption. $\eta_{Q,U,V}$: **dichroism**, meaning different absorption for different polarization states.
- $\rho_{Q,U,V}$: **magneto-optical (Faraday) effects**, rotating the polarization plane and converting linear into circular polarization.
- $\boldsymbol{\epsilon}$: emission vector. In LTE $\boldsymbol{\epsilon} = \mathsf{K}\,(B_\nu, 0, 0, 0)^T$, the polarized form of Kirchhoff's law.
- For an unmagnetized, isotropic medium $\mathsf{K} = \alpha_\nu \mathbf{1}$, and the equations for I, Q, U, V decouple.
- **Scattering** adds a gain term with the **phase matrix** $\mathsf{P}(\hat{\mathbf{n}}', \hat{\mathbf{n}})$ (Chandrasekhar's Rayleigh phase matrix for Thomson and Rayleigh scattering).
- With Zeeman-split absorption profiles these are the **Unno–Rachkovsky equations** (Unno 1956; Rachkovsky 1962). Sign conventions differ between authors (see Landi Degl'Innocenti & Landolfi 2004).

9.3 Polarization by scattering

For Thomson or Rayleigh scattering of unpolarized light through a scattering angle Θ:

$$I_{\rm sc} \propto \frac{1 + \cos^2\Theta}{2}, \qquad p_L = \frac{1 - \cos^2\Theta}{1 + \cos^2\Theta} = \frac{\sin^2\Theta}{1 + \cos^2\Theta}.$$

So the light is unpolarized in the forward and backward directions and 100 % polarized at $\Theta = 90^\circ$.

Chandrasekhar's electron-scattering atmosphere (1946). In a semi-infinite atmosphere where Thomson scattering dominates (hot O and B stars), the radiation emerging near the limb has travelled mostly parallel to the surface before its last scattering. It is therefore polarized, with the electric vector **parallel to the limb**. The exact solution of the polarized transfer equation gives:

μ	0 (limb)	0.1	0.2	0.4	0.6	0.8	1 (centre)
degree of polarization	11.71 %	7.45 %	5.41 %	3.03 %	1.63 %	0.68 %	0

For an unresolved spherical star the polarization cancels by symmetry. It becomes observable when the symmetry is broken: in eclipsing binaries (where it is measured during eclipses), rapidly rotating oblate stars, or disks.

9.4 Magnetic fields: the Zeeman and Hanle effects

Zeeman splitting. A field B splits a line by

$$\Delta\lambda_B = 4.67 \times 10^{-13}\, g_{\rm eff}\, \lambda^2\, B,$$

with λ and $\Delta\lambda_B$ in ångström (Å) and B in gauss (G), where $g_{\rm eff}$ is the effective Landé factor. The σ components are circularly polarized when viewed along the field and linearly polarized when viewed across it; the π component is linearly polarized.

Weak-field approximation (splitting much smaller than the Doppler width): $V(\lambda) \approx -\Delta\lambda_B \cos\gamma \, \dfrac{\partial I}{\partial \lambda}$, where γ is the angle between field and line of sight. V is antisymmetric about line centre and proportional to the longitudinal field, which is the principle of **magnetographs** (e.g. SDO/HMI).

Hanle effect. Weak fields (roughly 1–100 G, depending on the line) change the polarization of *scattered* line radiation (depolarization and rotation). The linearly polarized spectrum of the solar limb, the "second solar spectrum" (Stenflo & Keller 1997), is a diagnostic of turbulent magnetic fields that the Zeeman effect cannot detect.

9.5 Where polarized transfer is used

Field	Mechanism	Example
solar physics	Zeeman, Hanle	sunspot and quiet-Sun magnetic maps; Milne–Eddington inversions of Stokes profiles
hot stars, binaries, disks	Thomson scattering	eclipse polarization; Be-star disk polarization
interstellar medium	dichroic extinction by aligned dust (Hiltner and Hall, 1949)	Galactic magnetic-field direction
radio galaxies, pulsars, blazars	synchrotron emission	up to ~70 % linear polarization; Faraday rotation measures
cosmology	Thomson scattering of a quadrupole anisotropy	CMB E-modes (first detected by DASI in 2002); B-modes
planets and exoplanets	Rayleigh scattering	sky polarization; reflected light of planets

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Worked example 9.1 — Stokes parameters from filter measurements

Through six filters a detector measures (in units of the total): horizontal 0.70, vertical 0.30, +45° 0.60, −45° 0.40, right-circular 0.50, left-circular 0.50. Find I, Q, U, V, p and χ.

$I = 0.70 + 0.30 = 1.00$, $Q = 0.40$, $U = 0.20$, $V = 0$. So $p = \sqrt{0.40^2 + 0.20^2} = 0.447$ (44.7 % linearly polarized) and $\chi = \frac{1}{2}\arctan(0.20/0.40) = 13.3^\circ$.

Worked example 9.2 — Polarization of scattered light

Unpolarized sunlight is Thomson-scattered by electrons. Find the degree of polarization at $\Theta = 90^\circ$, 60° and 30°.

$p = \sin^2\Theta/(1+\cos^2\Theta)$: 90° → $1/1 = 100\,\%$; 60° → $0.75/1.25 = 60\,\%$; 30° → $0.25/1.75 = 14\,\%$. The same law gives the polarization pattern of the blue sky (Rayleigh scattering) and of the solar corona (the K-corona, Thomson scattering by free electrons).

Worked example 9.3 — Is a line split in a sunspot?

The Fe I line at 6302.5 Å has $g_{\rm eff} = 2.5$. Compute the Zeeman splitting for $B = 3000$ G (sunspot umbra) and $B = 100$ G (plage), and compare with the Doppler width (about 28 mÅ at 5800 K, Lecture SA6).

$\Delta\lambda_B = 4.67 \times 10^{-13} \times 2.5 \times (6302.5)^2 \times B = 4.64 \times 10^{-5}\, B$ Å.

- 3000 G: $\Delta\lambda_B = 0.139$ Å $= 139$ mÅ, 5 times the Doppler width. The line is visibly split (strong-field regime).
- 100 G: $\Delta\lambda_B = 4.6$ mÅ, much smaller than the Doppler width. The line is not split, but $V \propto \partial I/\partial\lambda$ still measures the field (weak-field regime).

Worked example 9.4 — Polarization of synchrotron radiation

For electrons with a power-law energy distribution $N(E) \propto E^{-p}$ in a uniform field, the maximum linear polarization is $\Pi = (p+1)/(p+7/3)$. Evaluate it for $p = 2.5$.

$\Pi = 3.5/4.833 = 0.72$, i.e. 72 %. Observed values are lower (typically 10–30 %) because the field is tangled and Faraday rotation varies along the line of sight. The ratio of observed to maximum polarization measures the order of the magnetic field.

Common misconceptions

1. **“Thermal emission is polarized.”** Isotropic thermal emission in LTE is unpolarized; polarization requires an asymmetry (scattering geometry, magnetic field, aligned grains).
2. **“Q and U are properties of the source alone.”** They depend on the chosen reference axes; only p and, after fixing axes, χ have direct physical meaning.
3. **“A star’s limb polarization is easy to see.”** For an unresolved round star it averages to zero; it is measured only when the symmetry is broken (eclipses, disks, rotation).
4. **“The Zeeman effect only works for strong fields.”** Even when the splitting is much smaller than the line width, circular polarization measures the longitudinal field (weak-field approximation).

Lecture summary

- Stokes vector (I, Q, U, V): total, two linear polarizations (0°/90° and ±45°), circular.
- Polarized RTE: $d\mathbf{I}/ds = -\mathsf{K}\mathbf{I} + \boldsymbol{\epsilon}$, with absorption, dichroism and magneto-optical terms; the Unno–Rachkovsky equations.
- Thomson and Rayleigh scattering: $p = \sin^2\Theta/(1+\cos^2\Theta)$; an electron-scattering atmosphere has 11.7 % polarization at the limb (Chandrasekhar).
- Zeeman: $\Delta\lambda_B = 4.67\times10^{-13} g \lambda^2 B$; V measures the longitudinal field. Hanle effect: weak fields.

Exam questions and answers — Lecture RT9

Q1. Define the four Stokes parameters operationally. Show that for completely polarized light $I^2 = Q^2 + U^2 + V^2$.

> **A1.** $I = I_1 + I_2$, $Q = I_1 - I_2$ (0°/90° polarizers), $U = I_{1'} - I_{2'}$ (±45°), $V = I_+ - I_-$ (circular). For a fully polarized wave, $I = a_1^2 + a_2^2$, $Q = a_1^2 - a_2^2$, $U = 2a_1a_2\cos\delta$, $V = 2a_1a_2\sin\delta$, so $Q^2 + U^2 + V^2 = (a_1^2 - a_2^2)^2 + 4a_1^2a_2^2 = (a_1^2 + a_2^2)^2 = I^2$.

Q2. Why is light scattered by an electron at 90° completely polarized? What is the degree of polarization at 45°?

> **A2.** The electron oscillates along the incident electric field, which is perpendicular to the incoming direction. A dipole does not radiate along its axis, so at 90° only the component of oscillation perpendicular to the scattering plane is seen: 100 % linear polarization. At 45°: $p = \sin^2 45^\circ/(1+\cos^2 45^\circ) = 0.5/1.5 = 33\,\%$.

Q3. Describe Chandrasekhar's result for the polarization of an electron-scattering atmosphere. Why is it hard to observe, and how can it be detected?

> **A3.** In a semi-infinite pure Thomson-scattering atmosphere, the emergent polarization rises from 0 at disk centre to 11.7 % at the limb, with the electric vector parallel to the limb. For a spherical, unresolved star the contributions from opposite limbs cancel. It becomes observable when symmetry is broken: during eclipses of binaries, in rotationally flattened stars, or in circumstellar disks.

Q4. What is the Zeeman splitting of a line with $g = 1.5$ at 5000 Å in a 1000 G field? In which regime is it, and how would you measure the field?

> **A4.** $\Delta\lambda_B = 4.67 \times 10^{-13} \times 1.5 \times (5000)^2 \times 1000 = 0.0175$ Å $= 17.5$ mÅ. This is smaller than a typical Doppler width (about 20–30 mÅ), so it is in the weak-field regime and the line is not visibly split. Measure the circular polarization: $V \approx -\Delta\lambda_B \cos\gamma \, \partial I/\partial\lambda$ gives the longitudinal component $B\cos\gamma$.

Lecture RT10 — Historical Context: How the Approximations and Solutions Were Found

> **Sources:** Chandrasekhar, *Radiative Transfer* (1950), Preface and bibliographical notes of Chs. II–V (Schuster, Schwarzschild, Milne, Eddington, Hopf, Bronstein, Wiener–Hopf, Ambartsumian); Collins Ch. 9 and its supplemental reading (Eddington 1916; Mihalas 1978; Sobolev 1963; Mustel 1958); A670 Lecture RT1 (historical background); Hubeny & Mihalas, *Theory of Stellar Atmospheres* (2014), for the numerical era.

Learning goals

1. Place the main approximations of this course (two-stream, Eddington, diffusion, Milne–Eddington, Eddington–Barbier) in their historical setting.
2. Know who found the exact solutions of the classical problems, and by which methods.
3. Name the main numerical methods of the computer era and the problem each one solved.
4. Compare the accuracy of the historical approximations on the gray atmosphere.

Key concepts in plain words

Why approximations came first. Until the 1960s there were no computers able to solve integro-differential equations. Astronomers needed answers they could compute with pencil, paper and tables. They invented approximations that replace the full angular dependence of the radiation by one or two numbers (two streams, or the Eddington closure $K = J/3$). Remarkably, these simple ideas already explained limb darkening, the temperature structure of the Sun and the formation of absorption lines.

Exact solutions as benchmarks. Between 1930 and 1950, mathematicians and astrophysicists (Hopf, Wiener, Ambartsumian, Chandrasekhar) solved a few idealised problems *exactly*: the gray atmosphere, isotropic scattering, the polarization of scattered light. These solutions, such as the Hopf function and the H-functions, are still used to test every new numerical code, exactly as we did with our own solver in Lecture SA5.

The numerical era: solving the hard problems. Real atmospheres need thousands of frequencies, many atomic levels and NLTE populations that depend on the radiation field. From the 1960s onwards, new algorithms turned this into problems computers could solve: Feautrier's method, complete linearization, accelerated Λ-iteration, short characteristics, Monte Carlo. Today's 3D radiation-hydrodynamic simulations build on all of them.

Old approximations never die. Two-stream and Eddington approximations are still used in climate models, exoplanet atmospheres and supernova codes. The Milne–Eddington model is the basis of solar magnetic-field measurements. Understanding the history explains why these "old" formulas appear in modern papers.

10.1 Before transfer theory (1729–1917)

Year	Who	Contribution
1729, 1760, 1852	Bouguer, Lambert, Beer	exponential attenuation of light, $I = I_0 e^{-\tau}$
1859–1860	Kirchhoff (with Bunsen)	emission/absorption ratio is universal; spectral analysis; dark solar lines explained
1900	Planck	the universal function $B_\nu(T)$
1917	Einstein	A and B coefficients; stimulated emission
1920	Saha	ionisation equation, which made the spectral sequence a temperature sequence

10.2 The birth of stellar radiative transfer (1905–1930)

- **Schuster (1905)**, "Radiation through a foggy atmosphere": the first transfer problem with scattering, treated with **two streams** (one up, one down). He explained how a scattering layer can produce absorption or emission lines.
- **Karl Schwarzschild (1906)**: showed that the solar photosphere is in **radiative equilibrium**, not convective (adiabatic) equilibrium. With a two-stream approximation he found $T^4 = \frac{1}{2}T_{\text{eff}}^4(1 + 2\tau)$ and the limb-darkening law $I(\mu)/I(1) = (1 + 2\mu)/3$, close to the observed solar limb darkening. In 1914 he wrote the theory of scattering lines.
- **Rosseland (1924)**: the diffusion limit and the **Rosseland mean** opacity (Lecture RT8).
- **Milne (1921)**: formulated the transfer problem as **integral equations** (the Schwarzschild–Milne equations and the "Milne problem" of Lecture SA5), and with Eddington the line-formation model now called **Milne–Eddington** (Lecture SA6).
- **Eddington (1916, 1926)**: radiative equilibrium of whole stars; the **Eddington approximation** $K = J/3$ with $J(0) = 2H(0)$ (Lecture RT8), which gave $T^4 = \frac{3}{4}T_{\text{eff}}^4(\tau + \frac{2}{3})$.
- **Unsöld (1928–1938)**: quantitative abundance analysis; the solar atmosphere is mostly hydrogen. His book *Physik der Sternatmosphären* (1938) became the standard text.

10.3 The era of exact solutions (1930–1960)

- **Hopf (1930–1934)** and **Bronstein (1929)**: exact solution of the Milne problem; Hopf–Bronstein relation $J(0) = \sqrt{3}H$; Hopf function $q(\tau)$ with $q(\infty) = 0.7104$.
- **Wiener & Hopf (1931)**: the Wiener–Hopf technique for integral equations on a half-line, first motivated by this problem.
- **Wildt (1939)**: the negative hydrogen ion H^- is the main continuum opacity of the Sun. Chandrasekhar then computed its cross-sections (1940s).
- **Barbier (1943)**: the relation $I(0, \mu) \approx S(\tau = \mu)$, now the **Eddington–Barbier relation** (Lecture RT7).
- **Ambartsumian (1943)**: **principles of invariance**. Adding a thin layer to a semi-infinite atmosphere cannot change its reflected light, which gives equations for the emergent radiation without solving for the interior.
- **Chandrasekhar (1944–1950)**: the **method of discrete ordinates** (Gaussian quadrature in angle), the H-**functions**, the exact law of darkening, the polarization of electron-scattering atmospheres (11.7 % at the limb,

1946) and the sunlit-sky (Rayleigh) problem. Collected in *Radiative Transfer* (1950).

- **Kourganoff (1952)**, *Basic Methods in Transfer Problems*; **Sobolev (1947, 1957, 1963)**: the **Sobolev approximation** for rapidly expanding envelopes and his *Treatise on Radiative Transfer*.
- **R. N. Thomas (1957–1965)**: NLTE line source functions, the two-level atom, thermalisation length (Lectures RT8 and SA7).

10.4 The numerical era (1960–today)

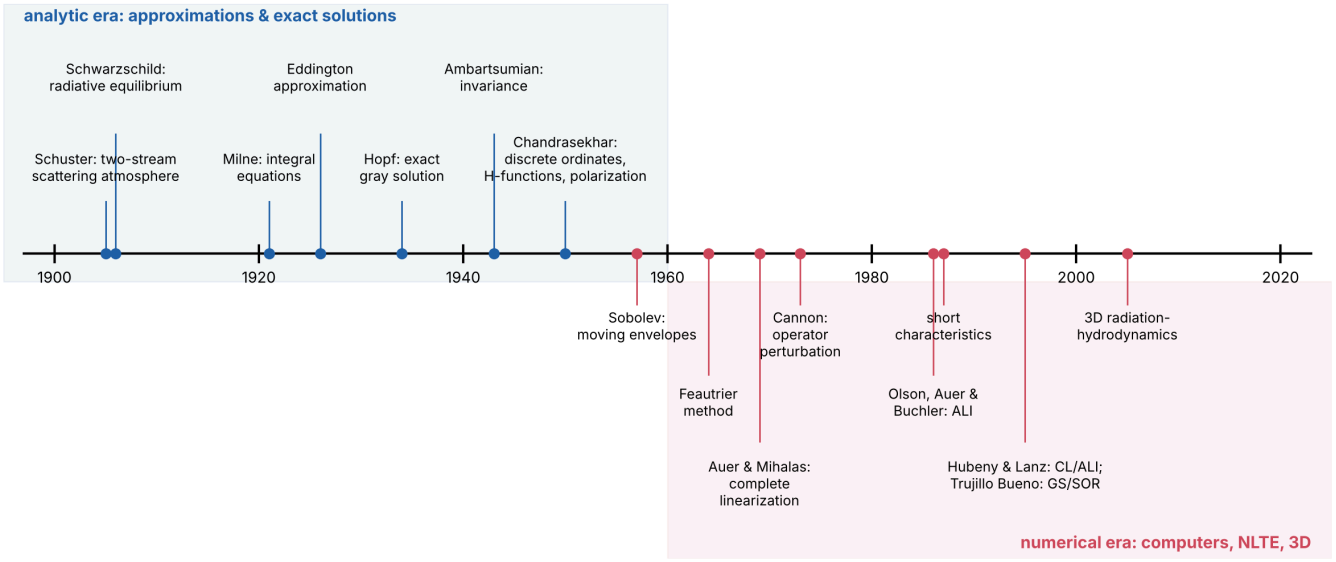


Figure 10.1 — Two eras. Above the axis: approximations and exact solutions. Below: numerical methods. Dates are approximate and refer to the first major publication.

Year	Method	Problem it solved
1964	**Feautrier** method (second-order difference equations)	stable solution of the RTE with scattering
1969	**Complete linearization** (Auer & Mihalas)	first self-consistent NLTE model atmospheres
1970	**Variable Eddington factors** (Auer & Mihalas)	closes the moment equations accurately
1970	**ATLAS** code (Kurucz)	LTE line-blanketed model atmosphere grids
1971–1973	**Rybicki method**; **operator perturbation** (Cannon)	efficient scattering problems; seed of ALI
1975	**MARCS** (Gustafsson et al.)	models of cool stars with molecules
1981–1986	**Accelerated Λ-iteration** (Scharmer; Werner & Husfeld; Olson, Auer & Buchler)	fast NLTE solutions with an approximate Λ operator
1987	**Short characteristics** (Olson & Kunasz)	multi-dimensional formal solutions
1991	**MALI** (Rybicki & Hummer)	multi-level NLTE atoms
1995	**Hybrid CL/ALI** (Hubeny & Lanz); **Gauss–Seidel/SOR** (Trujillo Bueno & Fabiani Bendicho)	NLTE metal-line-blanketed models; faster convergence
1982–today	**3D radiation-hydrodynamics** (Nordlund; Stein & Nordlund; Asplund et al.)	convection and granulation; revised solar abundances
1960s–today	**Monte Carlo** radiative transfer	complex geometry: dust, disks, winds, polarization

10.5 How good were the historical approximations?

Applied to the same gray atmosphere:

Approximation	$T^4/T_{\rm eff}^4$	$T(0)/T_{\rm eff}$	limb darkening $I(\mu)/I(1)$	$I(0)/I(1)$
Schwarzschild (two-stream, $\mu = \pm\frac{1}{2}$)	$\tau + \frac{1}{2}$	0.841	$(1+2\mu)/3$	0.333
Two-stream with $\mu = \pm 1/\sqrt{3}$	$\frac{3}{4}(\tau + 0.577)$	0.811	$(1+\sqrt{3}\mu)/(1+\sqrt{3})$	0.366
Eddington	$\frac{3}{4}(\tau + \frac{2}{3})$	0.841	$(2+3\mu)/5$	0.400
Exact (Hopf, Chandrasekhar)	$\frac{3}{4}(\tau + q(\tau))$	**0.811**	$H(\mu)/H(1)$	**0.344**

All the approximations are within about 15 % of the exact answer, accurate enough to have guided astrophysics for half a century.

Worked example 10.1 — Reproducing Schwarzschild's 1906 result

Use two streams at $\mu = \pm\frac{1}{2}$ (I^+ outward, I^- inward) in a gray LTE atmosphere in radiative equilibrium.

With these streams: $J = \frac{1}{2}(I^+ + I^-)$, $H = \frac{1}{4}(I^+ - I^-)$, $K = \frac{1}{8}(I^+ + I^-) = J/4$. Radiative equilibrium gives H = const and $dK/d\tau = H$, so $J = 4H\tau + C$. At the surface $I^- = 0$: $J(0) = I^+/2$ and $H(0) = I^+/4$, so $J(0) = 2H$ and $C = 2H$. Then $J = 4H(\tau + \frac{1}{2})$, and with $J = \sigma T^4/\pi$, $H = \sigma T_{\text{eff}}^4/4\pi$:

$$T^4 = T_{\text{eff}}^4 \left(\tau + \tfrac{1}{2}\right) = \tfrac{1}{2} T_{\text{eff}}^4 (1 + 2\tau).$$

The emergent intensity, by Eddington–Barbier (exact for linear S), is $\propto 1 + 2\mu$, which is Schwarzschild's limb-darkening law.

Worked example 10.2 — Which approximation is closest at the limb?

Using the table, compute the error of $I(0)/I(1)$ for each approximation.

Schwarzschild: $(0.333 - 0.344)/0.344 = -3.2\,\%$. Two-stream with $\mu = 1/\sqrt{3}$: $+6.4\,\%$. Eddington: $+16\,\%$. The oldest approximation is the most accurate at the limb, although Eddington's is better on average and for the temperature deep in the atmosphere ($q(\infty) = 0.710$ against Eddington's 0.667).

Common misconceptions

1. **"The Eddington approximation is obsolete."** Variable-Eddington-factor methods, climate models and many astrophysical codes still use it as a closure.
2. **"Exact solutions exist for realistic atmospheres."** They exist only for idealised problems (gray, isotropic scattering); realistic problems are numerical.
3. **"Numerical methods replaced the theory."** Numerical codes are tested against the exact solutions (Hopf, H-functions), as our own solver was.
4. **"Radiative transfer started with computers."** The key physical ideas (radiative equilibrium, limb darkening, line formation, scattering) were all established before 1930.

Lecture summary

- 1905–1930: Schuster and Schwarzschild (two streams), Milne (integral equations), Eddington ($K = J/3$), Rosseland (diffusion).
- 1930–1960: exact solutions by Hopf, Wiener–Hopf, Ambartsumian and Chandrasekhar; Sobolev's approximation for moving media.
- 1960–today: Feautrier, complete linearization, ALI/MALI, short characteristics, Monte Carlo, 3D radiation-hydrodynamics.
- The historical approximations are within about 15 % of the exact gray solution, and they are still used today.

Exam questions and answers — Lecture RT10

Q1. What did Karl Schwarzschild show in 1906, and why was it important?

> **A1.** He showed that the solar photosphere is in radiative, not adiabatic (convective), equilibrium. Using a two-stream approximation he derived $T^4 = \frac{1}{2}T_{\rm eff}^4(1+2\tau)$ and the limb-darkening law $(1+2\mu)/3$, which matched solar observations. This established radiative equilibrium and the transfer equation as the basis of stellar-atmosphere theory.

Q2. Explain the difference between the two-stream approximation and the Eddington approximation.

> **A2.** The two-stream approximation replaces the full angular distribution by two beams at fixed angles $\mu = \pm\mu_0$, solving two coupled ordinary differential equations. The Eddington approximation keeps the full angular range but closes the moment equations with $K = J/3$ (exact for a field linear in μ) and uses the surface condition $J(0) = 2H(0)$. For $\mu_0 = 1/\sqrt{3}$ the two-stream method also has $K = J/3$, but its surface condition is $J(0) = \sqrt{3}H$.

Q3. Name three contributions of Chandrasekhar to radiative transfer.

> **A3.** (1) The method of discrete ordinates (Gaussian quadrature in angle) with explicit solutions in every order. (2) The H-functions and the exact laws of darkening for isotropic and Rayleigh scattering. (3) The polarized transfer equation and the prediction of 11.7 % limb polarization in electron-scattering atmospheres; he also solved the sunlit-sky (Rayleigh scattering) problem.

Q4. Which numerical method first made self-consistent NLTE model atmospheres possible, and which later family of methods made them fast?

> **A4.** Complete linearization (Auer & Mihalas 1969) first produced self-consistent NLTE models. Accelerated Λ-iteration (operator perturbation: Cannon 1973; Scharmer 1981; Olson, Auer & Buchler 1986), its multi-level form MALI (Rybicki & Hummer 1991) and the hybrid CL/ALI scheme (Hubeny & Lanz 1995) made them fast enough for many atomic levels and lines.

Final Examination — Course A: Radiative Transfer (with complete solutions)

Format suggestion: 3 hours. Part A (20 %): short answers. Part B (50 %): answer any 6 of the 8 problems. Part C (20 %): RT6, RT9 and RT10 topics. Part E (10 %): essay. Solutions follow each question; numbers were checked with `scripts/worked_examples.py` and `scripts/lte_solvers.py`. Constants: $h = 6.626 \times 10^{-34}$ J s, $k = 1.381 \times 10^{-23}$ J K^{-1}, $c = 2.998 \times 10^{8}$ m s^{-1}, $\sigma = 5.670 \times 10^{-8}$ W m^{-2} K^{-4}, $m_e = 9.109 \times 10^{-31}$ kg, $m_H = 1.674 \times 10^{-27}$ kg, $r_e = 2.818 \times 10^{-15}$ m, $\sigma_T = 6.652 \times 10^{-29}$ m^2, $R_\odot = 6.957 \times 10^{8}$ m, 1 pc $= 3.086 \times 10^{16}$ m, 1 eV $= 1.602 \times 10^{-19}$ J. $\int_0^\infty E_n(t)\,dt = \dfrac{1}{n}$, $\int_0^\infty t\,E_n(t)\,dt = \dfrac{1}{n+1}$, $E_n(0) = \dfrac{1}{n-1}$.

Part A — Short answers (2 marks each)

A1. What is conserved along a ray in vacuum: F_ν, I_ν or J_ν?

> I_ν (more generally I_ν/ν^3).

A2. Give the source function for coherent isotropic scattering plus thermal absorption.

> $S_\nu = \varepsilon B_\nu + (1-\varepsilon) J_\nu$, with $\varepsilon = \kappa^a/(\kappa^a + \sigma^{\rm sc})$.

A3. At what optical depth does the emergent flux “originate” according to Eddington–Barbier?

> $\tau_\nu = 2/3$: $F_\nu(0) \approx \pi S_\nu(2/3)$.

A4. Why is the Rosseland mean a harmonic mean?

> The flux in each frequency band is $\propto 1/\kappa_\nu$ (like conductances in parallel), so the total is dominated by low-opacity windows.

A5. Write the Eddington closure and the Eddington surface condition.

> $K = J/3$; $J(0) = 2H(0)$.

A6. What does the factor $1 - e^{-h\nu/kT}$ represent?

> The LTE correction for stimulated emission (negative absorption).

A7. For an isothermal scattering atmosphere, what is $S(0)/B$ exactly?

> $\sqrt{\varepsilon}$.

A8. What is an optically thin source's brightness temperature in terms of T and τ?

> $T_b = T(1 - e^{-\tau}) \approx T\tau$ (RJ, no background).

A9. Two stars with equal $T_{\rm eff}$ but different radii: which quantity is the same, the flux at Earth or the surface brightness?

> The surface brightness.

Part B — Problems

B1. Intensity and flux (8 marks)

An A-type star has $R = 1.7 R_\odot$ and $T_{\rm eff} = 9900$ K and lies at $d = 2.64$ pc. (a) Find its surface flux and luminosity in $L_\odot$ ($L_\odot = 3.828 \times 10^{26}$ W). (b) Find the bolometric flux at Earth. (c) Show that its mean surface brightness is the same at the star and at Earth.

> **Solution.** (a) $F_s = \sigma T^4 = 5.67 \times 10^{-8} \times (9900)^4 = 5.45 \times 10^{8}$ W m^{-2}; $L = 4\pi R^2 F_s = 25\, L_\odot$. (b) $F_\oplus = F_s (R/d)^2$, with $R = 1.18 \times 10^{9}$ m and $d = 8.15 \times 10^{16}$ m, so $(R/d)^2 = 2.1 \times 10^{-16}$ and $F_\oplus = 1.15 \times 10^{-7}$ W m^{-2}. (c) The mean intensity is $\bar I = F_s/\pi$. At Earth, $\bar I = F_\oplus / \Omega$ with $\Omega = \pi (R/d)^2$, giving $F_s (R/d)^2 / (\pi (R/d)^2) = F_s/\pi$, the same. Intensity is conserved along rays.

B2. Planck function (8 marks)

a. Show that in the RJ limit $I_\nu = (2k\nu^2/c^2) T_b$. (b) A compact source has $I_\nu = 10^{-18}$ W m^{-2} Hz^{-1} sr^{-1} at 5 GHz. Find T_b. (c) If the source is an H II region with $T_e = 10^4$ K, what is its optical depth at 5 GHz? (d) Is it thick or thin at 0.5 GHz if $\tau \propto \nu^{-2.1}$?

> **Solution.** (a) $e^x - 1 \approx x$ gives $B_\nu \approx 2h\nu^3/(c^2 x) = 2\nu^2 kT/c^2$; define T_b by $I_\nu = B_\nu(T_b)$. (b) $T_b = c^2 I_\nu / (2k\nu^2) = (8.99 \times 10^{16} \times 10^{-18}) / (2 \times 1.381 \times 10^{-23} \times 2.5 \times 10^{19}) = 130$ K. (c) $T_b = T_e(1 - e^{-\tau}) \Rightarrow 1 - e^{-\tau} = 0.013 \Rightarrow \tau = 0.013$. (d) $\tau(0.5\ \text{GHz}) = 0.013 \times 10^{2.1} = 1.6$: marginally thick, so $T_b = 10^4 (1 - e^{-1.6}) = 8000$ K.

B3. Formal solution in a plane-parallel atmosphere (8 marks)

$S(\tau) = a + b\tau + c\tau^2$ in a semi-infinite atmosphere. (a) Find $I(0,\mu)$. (b) Find $F(0)$. (c) Compare with the Eddington–Barbier values and state when they are exact.

> **Solution.** (a) $I(0,\mu) = \int_0^\infty (a + bt + ct^2) e^{-t/\mu} dt/\mu = a + b\mu + 2c\mu^2$. (b) $F = 2\pi \int_0^1 (a + b\mu + 2c\mu^2)\mu\, d\mu = 2\pi(a/2 + b/3 + c/2) = \pi(a + \tfrac{2}{3}b + c)$. (c) E–B: $I \approx S(\mu) = a + b\mu + c\mu^2$ (error $c\mu^2$); $F \approx \pi S(2/3) = \pi(a + \tfrac{2}{3}b + \tfrac{4}{9}c)$ (error $\tfrac{5}{9}\pi c$). Both are exact when $c = 0$ (linear S).

B4. Testing the Eddington solution with the Λ operator (8 marks)

For the Eddington gray solution $S(\tau) = 3H(\tau + \tfrac{2}{3})$, compute $J(0) = \Lambda_0[S] = \tfrac{1}{2}\int_0^\infty S(t) E_1(t)\, dt$. Compare with $S(0)$ and with the exact (Hopf–Bronstein) value $J(0) = \sqrt{3}H$. What does this tell you?

> **Solution.** $J(0) = \tfrac{3H}{2}\left[\int_0^\infty t E_1 dt + \tfrac{2}{3}\int_0^\infty E_1 dt\right] = \tfrac{3H}{2}\left[\tfrac{1}{2} + \tfrac{2}{3}\right] = \tfrac{7}{4}H = 1.750\,H$. But $S(0) = 2H$, so $\Lambda[S] \neq S$: the Eddington solution does **not** satisfy Milne's equation exactly ($J \neq S$ at the surface). The exact value $\sqrt{3}H = 1.732H$ is close to the result of one Λ-iteration (1.750H), which shows that Λ-iteration improves the solution near the surface, where it couples to the boundary.

B5. Two-stream (Schuster–Schwarzschild) gray atmosphere (10 marks)

Represent the radiation field by two streams, I^+ at $\mu = +1/\sqrt{3}$ and I^- at $\mu = -1/\sqrt{3}$, so that $J = \tfrac{1}{2}(I^+ + I^-)$, $H = \tfrac{1}{2\sqrt{3}}(I^+ - I^-)$, $K = \tfrac{1}{6}(I^+ + I^-)$. (a) Show that $K = J/3$ exactly. (b) With $I^-(0) = 0$, show $J(0) = \sqrt{3}H$. (c) Hence derive $T^4 = \tfrac{3}{4}T_{\rm eff}^4(\tau + 1/\sqrt{3})$ and compare $T(0)$ with the exact result.

> **Solution.** (a) $K = \tfrac{1}{2}\sum \mu_i^2 I_i = \tfrac{1}{2}\cdot\tfrac{1}{3}(I^+ + I^-) = J/3$. (b) $J(0) = I^+/2$ and $H(0) = I^+/(2\sqrt{3})$, so $J(0) = \sqrt{3}H(0)$. (c) As in B6: $J = 3H(\tau + C)$ with $3HC = \sqrt{3}H \Rightarrow C = 1/\sqrt{3}$. So $T^4 = \tfrac{3}{4}T_{\rm eff}^4(\tau + 0.577)$ and $T(0) = (\sqrt{3}/4)^{1/4} T_{\rm eff} = 0.811\,T_{\rm eff}$, which is **exactly the true surface value**. This is Chandrasekhar's "first approximation" of the discrete-ordinate method; it gives $J(0)$ exactly in every order (Chandrasekhar eq. 64), while $q(\infty)$ improves with the order.

B6. Limb darkening and LTE temperature structure (8 marks)

At $\lambda = 500$ nm, the observed solar limb darkening is $I(\mu) = I(1)(0.36 + 0.64\mu)$ with $I(1) = 4.0 \times 10^{13}$ W m^{-3} sr^{-1}. Assuming LTE and the Eddington–Barbier relation, find $T(\tau_{500} = 0)$ and $T(\tau_{500} = 1)$. Comment.

> **Solution.** $S(\tau) = I(1)(0.36 + 0.64\tau)$, and LTE gives $S = B_\lambda(T)$, so $T = (hc/\lambda k)/\ln(1 + 2hc^2/(\lambda^5 B))$, with $hc/\lambda k = 28\,776$ K and $2hc^2/\lambda^5 = 3.811 \times 10^{15}$. $\tau = 0$: $B = 1.44 \times 10^{13}$, so $T = 28776/\ln(265.7) = 5155$ K. $\tau = 1$: $B = 4.0 \times 10^{13}$, so $T = 28776/\ln(96.3) = 6300$ K. The temperature rises inward by about 1100 K per unit optical depth, which is why the Sun shows limb darkening and absorption lines. The values bracket $T_{\rm eff} = 5772$ K, consistent with the gray model (Lecture SA5).

B7. Scattering and thermalisation (10 marks)

An isothermal atmosphere has $S = \varepsilon B + (1-\varepsilon) J$. (a) In the Eddington approximation, derive the differential equation for J and solve it with the appropriate boundary conditions. (b) For $\varepsilon = 10^{-2}$, find $S(0)/B$ (Eddington) and compare with the exact value. (c) Estimate the thermalisation depth and the number of scatterings a photon undergoes before destruction. (d) Why does Λ-iteration starting from $S = B$ converge so slowly?

> **Solution.** (a) $dH/d\tau = J - S = \varepsilon(J - B)$ and $H = \tfrac{1}{3} dJ/d\tau$, so $\tfrac{1}{3} J'' = \varepsilon (J - B)$. Then $J = B + C e^{-\sqrt{3\varepsilon}\tau}$; with $J(0) = \tfrac{2}{3} J'(0)$, $C = -B/(1 + 2\sqrt{\varepsilon/3})$. (b) $\sqrt{\varepsilon/3} = 0.0577$, so $J(0)/B = 1 - 1/1.1155 = 0.1035$ and $S(0)/B = 0.01 + 0.99 \times 0.1035 = 0.1125$. The exact value is $\sqrt{\varepsilon} = 0.100$; Eddington overestimates by 12 %. (c) $\tau_{\rm th} \sim 1/\sqrt{\varepsilon} = 10$ (Eddington: $1/\sqrt{3\varepsilon} = 5.8$); $N \sim 1/\varepsilon = 100$ scatterings. (d) Each iteration transmits information only about $\Delta\tau \sim 1$, and the error decreases by a factor $(1-\varepsilon) = 0.99$ per step, so hundreds of iterations are needed (see Figure 8.1). Our solver gives $S(0)$ errors of 95 % after 10 iterations, 17 % after 50, 5 % after 100 and 0.9 % after 200; yet beyond 100 iterations each step changes $S(0)$ by less than 0.1 %, so a naive convergence test would stop far too early.

B8. Diffusion and Rosseland mean (8 marks)

a. From $I \approx B + \mu\, dB/d\tau$, derive $F = \frac{4\pi}{3} \frac{dB}{d\tau}$. (b) A gas has three frequency bands carrying 50 %, 30 % and 20 % of $\partial B/\partial T$, with $\kappa = 1$, 10 and 1000 (arbitrary units). Find the Rosseland and Planck means. (c) Which mean controls the energy flux in a stellar interior, and why?

> **Solution.** (a) $F = 2\pi \int_{-1}^{1} (B + \mu B') \mu \, d\mu = 2\pi B' \cdot \tfrac{2}{3} = \tfrac{4\pi}{3} B'$. (b) $1/\kappa_R = 0.5/1 + 0.3/10 + 0.2/1000 = 0.5302 \Rightarrow \kappa_R = 1.89$. $\kappa_P = 0.5 + 3 + 200 = 203.5$. (c) The Rosseland mean, because the flux is $\propto \sum (1/\kappa_\nu)$: radiation leaks through the most transparent band, so $\kappa_R \approx 2\times$ the window opacity, while the Planck mean is dominated by the most opaque band.

Part C — Formation of the RTE, polarization, history (4 marks each)

C1. (Lecture RT6) Show how the term $\frac{1-\mu^2}{r}\frac{\partial I}{\partial \mu}$ arises in the spherical RTE, and state when it may be neglected.

> **Solution.** Along a straight ray with impact parameter p: $r^2 = p^2 + z^2$, $\mu = z/r$, so $dr/dz = \mu$ and $d\mu/dz = (1-\mu^2)/r$. Then $dI/dz = \mu\, \partial_r I + \frac{1-\mu^2}{r} \partial_\mu I$. The term describes the change of a ray's angle to the local vertical. It is negligible when the atmosphere is thin compared with r (about $10H \ll R$), e.g. the Sun.

C2. (Lecture RT10) Schwarzschild (1906) used two streams at $\mu = \pm \tfrac{1}{2}$ and found $T^4 = \tfrac{1}{2} T_{\rm eff}^4 (1 + 2\tau)$. Derive the corresponding limb-darkening law and compare $I(0)/I(1)$ with the exact gray value 0.344.

> **Solution.** $S \propto 1 + 2\tau$ is linear, so Eddington–Barbier is exact: $I(0,\mu) \propto 1 + 2\mu$ and $I(\mu)/I(1) = (1 + 2\mu)/3$. At the limb, $I(0)/I(1) = 1/3 = 0.333$, only 3 % below the exact 0.344 (Eddington's

approximation gives 0.400).

C3. (Lecture RT9) Light is measured through filters: $I_1 = 0.55$, $I_2 = 0.45$, $I_{1'} = 0.50$, $I_{2'} = 0.50$, $I_+ = 0.52$, $I_- = 0.48$ (units of the total). Find p, p_L and p_V. Which physical process could produce circular polarization in a stellar line?

> **Solution.** $I = 1$, $Q = 0.10$, $U = 0$, $V = 0.04$. Then $p = \sqrt{0.01 + 0.0016} = 0.108$, $p_L = 0.10$, $p_V = 0.04$. Circular polarization in a spectral line comes from the Zeeman effect with a magnetic field component along the line of sight ($V \propto B \cos\gamma \, \partial I/\partial\lambda$ in the weak-field limit).

Part E — Essay (10 marks)

E1. Starting from photon conservation, explain how the radiative transfer equation is derived, which assumptions lead to its plane-parallel static form, and how the main historical approximations (two-stream, Eddington, diffusion) and exact methods (Hopf, Chandrasekhar) solve it.

> **Model answer outline.** (1) Photon distribution f, $I_\nu = (h^4\nu^3/c^2) f$; Boltzmann equation; gains and losses (absorption with stimulated emission, emission, scattering out and in with redistribution function). (2) Assumptions: static, straight rays, plane-parallel, so $\mu \, dI/d\tau = I - S$. (3) Formal solution and Schwarzschild–Milne equations; scattering makes the problem an integral equation. (4) Approximations: two-stream (Schuster, Schwarzschild 1906: $T^4 = \tfrac{1}{2} T_{\rm eff}^4 (1 + 2\tau)$), Eddington $K = J/3$, $J(0) = 2H$, diffusion and Rosseland mean deep inside. (5) Exact: Hopf $q(\tau)$, Chandrasekhar's discrete ordinates and H-functions (limb darkening 0.344 at the limb). (6) Numerical era: Feautrier, Λ-iteration and its failure, ALI, short characteristics, Monte Carlo.

Formula Sheet — Course A: Radiative Transfer

Radiation field (Lecture RT2)

Specific intensity	$dE = I_\nu \cos\theta \, dA \, d\Omega \, d\nu \, dt$; I_ν constant along rays in vacuum; I_ν/ν^3 invariant
Mean intensity	$J_\nu = \frac{1}{4\pi}\oint I_\nu d\Omega = \frac{1}{2}\int_{-1}^{1} I_\nu d\mu$
Flux	$F_\nu = \oint I_\nu \cos\theta \, d\Omega = 2\pi \int_{-1}^{1} I_\nu \mu \, d\mu$; $H_\nu = F_\nu/4\pi$
K-integral, pressure	$K_\nu = \frac{1}{2}\int_{-1}^{1} I_\nu \mu^2 d\mu$; $P_\nu = 4\pi K_\nu/c$
Energy density	$u_\nu = 4\pi J_\nu/c$
Isotropic field	$F = 0, K = J/3, P = u/3$
Uniform hemisphere	$F = \pi I$
Sphere seen from d	$F = \pi I (R/d)^2$
$\nu \leftrightarrow \lambda$	$I_\lambda = (c/\lambda^2) I_\nu$

Black body (Lecture RT3)

$$B_\nu = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/kT}-1}, \quad B_\lambda = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda kT}-1}$$

- RJ: $B_\nu \approx 2\nu^2 kT/c^2$ · Wien: $B_\nu \approx \frac{2h\nu^3}{c^2} e^{-h\nu/kT}$ · $h\nu_{\max} = 2.821kT \cdot \lambda_{\max} T = 2.898 \times 10^{-3}$ m K
- $\int B_\nu d\nu = \sigma T^4/\pi \cdot F = \sigma T^4 \cdot u = aT^4, a = 4\sigma/c \cdot dB/dT = 4\sigma T^3/\pi \cdot n_\gamma = 2.03 \times 10^7 T^3$ m^{-3}
- T_b: $I_\nu = B_\nu(T_b)$, RJ $T_b = c^2 I_\nu/2k\nu^2 \cdot L = 4\pi R^2 \sigma T_{\text{eff}}^4$

Coefficients (Lecture RT4)

- $\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu \cdot \ell = 1/\alpha_\nu \cdot \tau_\nu = \int \alpha_\nu ds \cdot S_\nu = j_\nu/\alpha_\nu \cdot \kappa_{\text{es}} = 0.02(1+X)$ m^2 kg^{-1}
- Einstein: $g_1 B_{12} = g_2 B_{21}$, $A_{21} = \frac{2h\nu^3}{c^2} B_{21}$
- $j_\nu = \frac{h\nu}{4\pi} n_2 A_{21} \phi$, $\alpha_\nu = \frac{h\nu}{4\pi}\phi(n_1 B_{12} - n_2 B_{21})$, $S = \frac{2h\nu^3/c^2}{g_2 n_1/g_1 n_2 - 1}$
- Scattering + absorption: $S = \varepsilon B + (1-\varepsilon)J$ · Random walk: $R = \sqrt{N}\ell$, $N \approx \max(\tau, \tau^2)$

Transfer equation (Lectures RT5, RT7)

- $\frac{dI_\nu}{d\tau_\nu} = S_\nu - I_\nu$ (τ along the ray) · $I = I_0 e^{-\tau} + \int_0^\tau S(t) e^{-(\tau - t)} dt$ · slab: $I = I_0 e^{-\tau} + S(1 - e^{-\tau})$
- Plane-parallel (τ inward): $\mu \frac{dI}{d\tau} = I - S$ · $I(0, \mu) = \int_0^\infty S \, e^{-t/\mu} dt/\mu$

- $J = \frac{1}{2}\int_0^\infty S\,E_1|t-\tau|dt \cdot F = 2\pi\left[\int_\tau^\infty SE_2(t-\tau)dt - \int_0^\tau SE_2(\tau-t)dt\right] \cdot K = \frac{1}{2}\int SE_3|t-\tau|dt$
- $E_n(x) = \int_1^\infty e^{-xt}t^{-n}dt$, $E_n' = -E_{n-1}$, $E_n(0) = 1/(n-1)$, $\int_0^\infty E_n = 1/n$, $\int_0^\infty tE_n = 1/(n+1)$
- **Eddington–Barbier:** $I(0,\mu) \approx S(\tau=\mu)$, $F(0) \approx \pi S(\tau = 2/3)$ (exact for linear S)

Moments, diffusion, scattering (Lecture RT8)

- $\dfrac{dH}{d\tau} = J - S \cdot \dfrac{dK}{d\tau} = H$ · RE: $\int \kappa_\nu(J_\nu - B_\nu)d\nu = 0$, $F = \sigma T_{\text{eff}}^4$ = const
- Eddington: $K = J/3$, $J(0) = 2H(0) \Rightarrow \frac{1}{3}J'' = J - S$
- Diffusion: $F_\nu = \frac{4\pi}{3}\frac{dB_\nu}{d\tau_\nu} \cdot \dfrac{1}{\kappa_R} = \dfrac{\int \kappa_\nu^{-1}\partial_T B_\nu d\nu}{\int \partial_T B_\nu d\nu} \cdot F = -\dfrac{16\sigma T^3}{3\kappa_R\rho}\dfrac{dT}{dz}$
- Isothermal scattering: $S(0) = \sqrt{\varepsilon}\,B$ (exact) · Eddington $J = B\left[1 - \frac{e^{-\sqrt{3\varepsilon}\tau}}{1+2\sqrt{\varepsilon/3}}\right]$ · $\tau_{\text{th}} \sim 1/\sqrt{\varepsilon}$

Forming the RTE (Lecture RT6)

- $I_\nu = (h^4\nu^3/c^2)f$ · Boltzmann: $\partial_t f + c\hat{\mathbf{n}}\cdot\nabla f = (\partial_t f)_{\text{coll}}$
- General: $\frac{1}{c}\partial_t I_\nu + \hat{\mathbf{n}}\cdot\nabla I_\nu = -(\kappa_\nu + \sigma_\nu)\rho I_\nu + j_\nu + \frac{\sigma_\nu\rho}{4\pi}\iint R I_{\nu'}d\Omega' d\nu'$
- Spherical: $\mu\partial_r I + \frac{1-\mu^2}{r}\partial_\mu I = \chi(S - I)$ · invariants I_ν/ν^3, $\nu\chi_\nu$, j_ν/ν^2
- Energy: $\partial_t u + \nabla\cdot\mathbf{F} = 4\pi\int(j_\nu - \kappa_\nu\rho J_\nu)d\nu$ · force: $\mathbf{f}_{\text{rad}} = \frac{1}{c}\int \chi_\nu \mathbf{F}_\nu d\nu$

Historical approximations (Lecture RT10)

- Schwarzschild ($\mu = \pm\frac{1}{2}$): $T^4 = \frac{1}{2}T_{\text{eff}}^4(1+2\tau)$, $I(\mu)/I(1) = (1+2\mu)/3$
- Two-stream ($\mu = \pm 1/\sqrt{3}$): $T^4 = \frac{3}{4}T_{\text{eff}}^4(\tau + 1/\sqrt{3})$ · Eddington: $\frac{3}{4}T_{\text{eff}}^4(\tau + \frac{2}{3})$ · exact: $\frac{3}{4}T_{\text{eff}}^4(\tau + q)$

Polarization (Lecture RT9)

- $I = I_1 + I_2, Q = I_1 - I_2, U = I_{1'} - I_{2'}, V = I_+ - I_-$ · $p = \sqrt{Q^2+U^2+V^2}/I$ · $\chi = \frac{1}{2}\arctan(U/Q)$
- $d\mathbf{I}/ds = -\mathsf{K}\mathbf{I} + \boldsymbol{\epsilon}$ (Unno–Rachkovsky with Zeeman profiles)
- Thomson/Rayleigh: $p = \sin^2\Theta/(1+\cos^2\Theta)$ · electron-scattering atmosphere: $p(\mu=0) = 11.7\,\%$
- Zeeman: $\Delta\lambda_B = 4.67\times10^{-13} g\lambda^2 B$ (λ, Δλ in Å; B in G) · weak field: $V \approx -\Delta\lambda_B\cos\gamma\,\partial I/\partial\lambda$